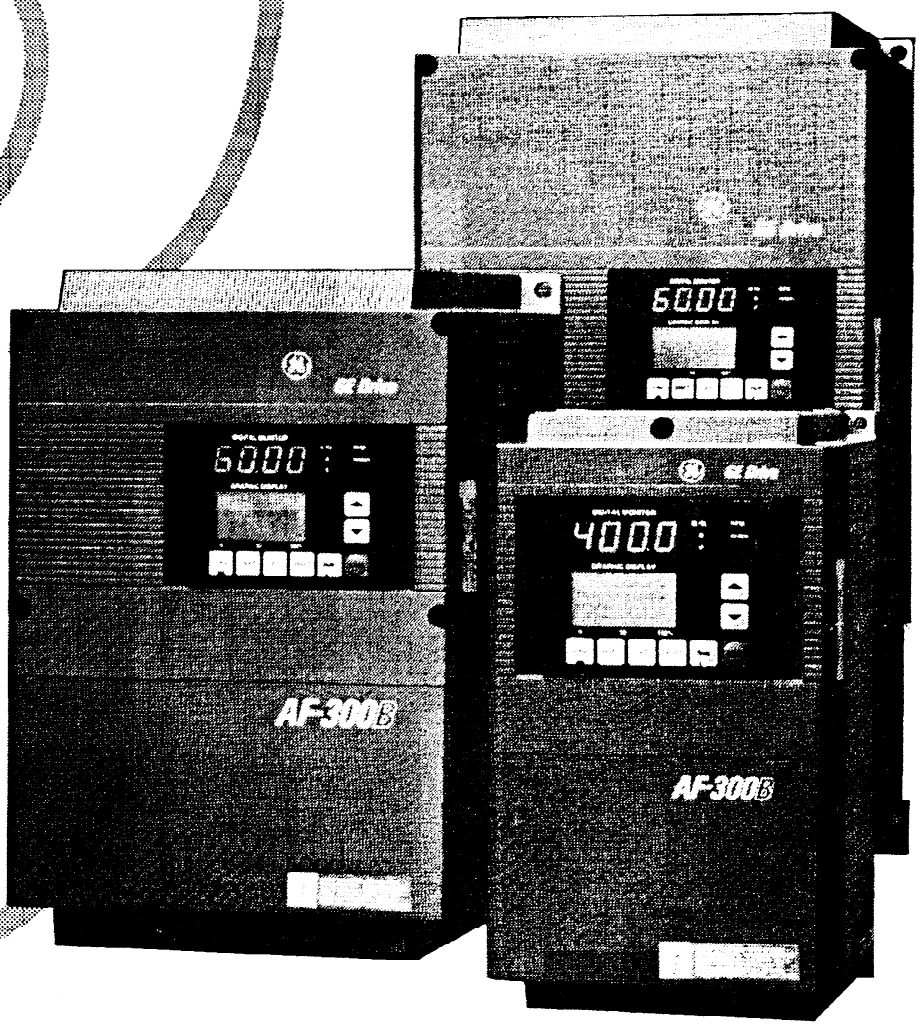
GE Drives

AF-300B™

1/2-30 HP

Instructions



TM Trademark of General Electric Company, U.S.A.

These instructions do not purport to cover all details or variations in equipment to provide for every possible contingency to be met in connection with installation operation maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchasers purposes, the matter should be referred to General Electric Company

SUPPLEMENT #1

FOR INSTRUCTION MANUAL GEK-85715A

These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems, Salem, Virginia U.S.A.

This supplement contains additional information, and corrected information, needed when installing and operating the AF-300B series of inverters. Refer to the information of this supplement in addition to, or instead of, that contained in GEK-85715A for the figures, tables, and paragraphs listed in this supplement.

Table 3: Application of Wiring and Equipment

The recommended wire sizes shown in Table 3 are minimum sizes. Due to variations in installation, check that the recommended wire sizes are adequate for that particular installation. All inverter installations must meet the applicable national (NEC, CEC) and local electrical codes. It is the system engineer's responsibility to be certain that the inverter installation complies with all applicable electrical codes. (If a system engineer is not involved, it is the user's responsibility.) Only adequately trained persons should install any programmable inverter.

Figure 4-28. Basic Connection Diagram

The 3-phase circuit breaker shown in this figure is incorrectly illustrated as three breakers instead of a single 3-pole breaker. The START/STOP switch, STOP switch position, and control terminal connection shield symbol are also shown incorrectly in this figure. Refer to figure 1 on page 2 of this supplement for the corrected figure.

Table 9: Function Code Settings
Table 10: Function Code Descriptions

Function Code 43, Electronic Thermal Overload Protection, that is built in to the AF-300B series of inverters is factory set as Operational. If the user changes this to Non-Operational, conventional overload protection for the motor and wiring must then be provided by the user. Typical wiring for an overload relay is shown in figure 1 on page 2 of this supplement. All inverter installations must meet the applicable national (NEC, CEC) and local electrical codes. It is the system engineer's responsibility to be certain that the inverter installation complies with all applicable electrical codes. (If a system engineer is not involved, it is the user's responsibility.)

Table 16: AC Line Fuses and Fuseholders

Due variations in installation when mounting fuseholders and fuses external to the inverter, check that the recommended fuse sizes are correct for that particular installation. All inverter installations must meet the applicable national (NEC, CEC) and local electrical codes. It is the system engineer's responsibility to be certain that the inverter installation complies with all applicable electrical codes. (If a system engineer is not involved, it is the user's responsibility.)

(10/14/94)

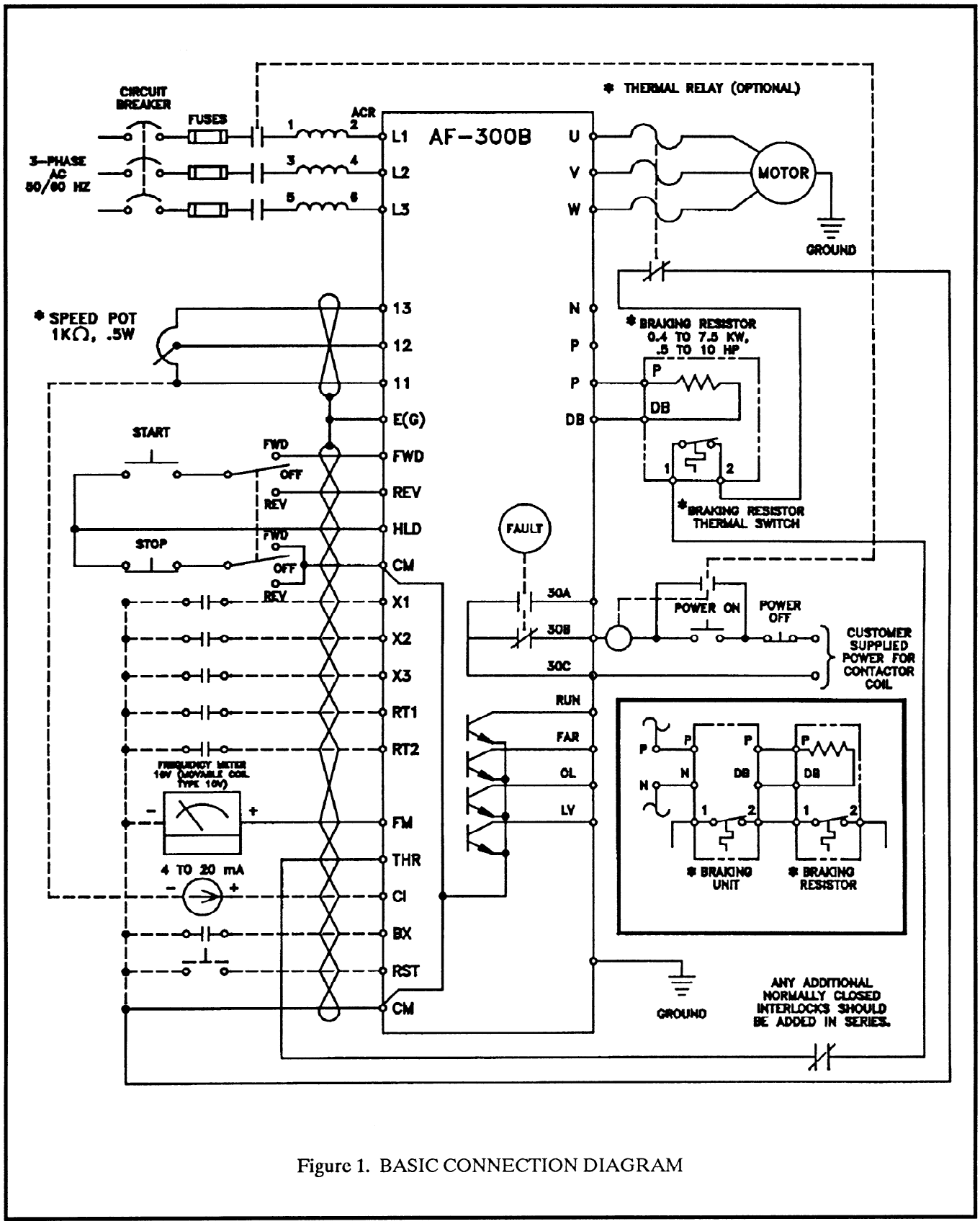


Figure 1. BASIC CONNECTION DIAGRAM

AF-300B Inverter
1/2 to 30 Horsepower
200-230 and 380-460 VAC

GEK-85715A

Issue Date: August 1994

These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that not covered sufficiently for the purchaser's purpose, the matter should be referred to GE Drive Systems, Salem, Virginia, U.S.A.

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NOTES:

1. SAFETY PRECAUTIONS

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions on them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.

WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.

CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.

NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - MECHANICAL MOTION HAZARD: Inverter systems cause mechanical motion. It is the responsibility of the user to insure that any such motion does not result in an unsafe condition. Factory provided interlocks and operating limits should not be bypassed or modified.

WARNING - FIRE HAZARD: Use only dynamic braking resistors that are provided by GE Drive Systems. These are special resistors that have a thermal fuse built in that opens if the resistor gets too hot. This fuse should only open if the DB transistor in the inverter or DB unit fails short circuited. The resistor will neet to be replaced if this happens.

WARNING - ELECTRICAL SHOCK AND BURN HAZARD: When using instruments such as oscilloscopes to work on live equipment, the oscilloscope's chassis should be grounded and a differential amplifier input should be used. Care should be used in the selection of probes and leads and in the adjustment of the oscilloscope so that accurate readings may be made. See instrument manufacturers instruction book for proper operation and adjustments to the instrument.

WARNING - STRAIN HAZARD: Improper lifting practices can cause serious or fatal injury. Lift only with adequate equipment and trained personnel.

WARNING - FIRE AND EXPLOSION HAZARD: Fires or explosions might result from mounting inverters in hazardous areas such as locations where flammable or combustible vapors or dusts are present. Inverters should be installed away from hazardous areas, even if used with motors suitable for use in these locations.

WARNING - ELECTRICAL SHOCK HAZARD: All motor bases and equipment enclosure housings should be grounded in accordance with the National Electric Code or equivalent.

WARNING - MOTOR OVERSPEED HAZARD: With 400 Hz inverter output possible, the inverter could cause the motor to run up to 6 - 7 times its base speed. Never operate the motor above its top mechanical speed or a catastrophic failure may occur. Consult factory for any application exceeding 120 Hz.

WARNING - MOTOR OVERSPEED HAZARD: Bias is operational when Function Code is 1601 or 1602. When the inverter does not have a RUN command, the reference frequency will flash on the display. When Bias is operational and the speed reference is zero, the display will flash zero. When a RUN command is given, the motor will run at the Bias setting even though zero was flashing on the display.

WARNING - PERSONAL INJURY HAZARD: To prevent personal injury or equipment damage caused by equipment malfunction, only adequately trained persons should modify any programmable inverter.

CAUTION: Do not connect power supply voltage that exceeds the standard specification voltage fluctuation permissible. If excessive voltage is applied to the inverter, damage to the internal components will result.

CAUTION: Do not connect power supply to the output terminals (U, V, W). Connect power supply only to the power terminals (L1, L2, L3).

CAUTION: Do not connect power supply to the braking resistor connection terminals (P, DB). Never short-circuit between P-N or P-DB terminals, and do not connect any resistance with a resistance value less than the standard application braking resistor.

CAUTION: Do not connect power supply to the control circuit terminals (except 30A, B, C).

CAUTION: For RUN and STOP, use the FWD-CM (forward) and REV-CM (reverse) terminals. Avoid using a contactor (ON/OFF) installed on the line side of the inverter for RUN and STOP.

CAUTION: Do not use a switch on the output side of the inverter for ON/OFF operation.

CAUTION: The AC Supply Power Transformer KVA rating should be greater than 1.5 times the HP rating of the inverter, but not greater than 500 KVA. If an AC Power Supply Transformer greater than 500 KVA is to be used, install a line reactor (optional; must be purchased seperately) on the line side of the inverter. A Line Reactor is also required if an SCR Type controller and/or Power Factor Correction Capacitors are used on the same power lines.)

CAUTION: Do not connect filter capacitors on the output side of the inverter.

CAUTION: Do not operate the inverter without a ground wire connected.

CAUTION: If the inverter's Fault Alarm is activated, consult the TROUBLESHOOTING section of this instruction book, and after correcting the problem, resume operation. Do not reset the alarm automatically by external sequence, etc.

CAUTION: Do not perform a megger test between the inverter terminals or on the control circuit terminals.

CAUTION: Install a suppressor in parallel with any relays, solenoid type coils, brake coils, etc. that are close to the inverter or wired to the inverter.

CAUTION: Motor Thermal Overload protection must be provided, either by motor thermoswitch, motor overload relay, or inverter electronic thermal overload.

CAUTION: Because the ambient temperature greatly affects inverter life and reliability, do not install the inverter in any location that exceeds the allowable temperature. Leave the ventilation covers attached for temperatures of 40 degrees C or below, and remove the cover for temperatures of between 40 and 50 degrees C. **If the cover needs to be removed, another type of enclosure may be required for safety purposes.**

CAUTION: If dynamic braking is desired on inverters rated 11 Kw (15 HP) and above, it is required that both a dynamic braking unit and dynamic braking resistor be added. Do not attempt to connect an external braking resistor to inverters rated 11 Kw (15 HP) and above without an external dynamic braking unit or damage will result.

CAUTION: When using an external braking resistor with inverters rated less than 7.5KW, first disconnect the inverter internal braking resistor wires from terminals P and DB, then connect the external DB resistor to the P and DB terminals. The internal braking resistor wires that have been disconnected must be protected with insulation.

CAUTION: Be sure to remove the desicant dryer packet(s) when unpacking inverter. (If not removed these packets may become lodged in the fan or air passages and cause the inverter to overheat.)

CAUTION: The mounting wall for the inverter must be of heat resistant material because during operation, the temperature of the inverter's cooling fins rises to approximately 90 degrees C (194 F).

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow all procedures step by step.*

NOTE: *The motor chassis should be grounded to earth through a separate ground lead from all other equipment ground leads to prevent noise coupling.*

NOTE: *Read and heed all* ***WARNING, CAUTION,*** *and* ***NOTE*** *labels posted on the equipment.*

NOTES:

NOTES:

2. DESCRIPTION, COMPONENT IDENTIFICATION, AND SPECIFICATIONS,

GENERAL DESCRIPTION

The AF-300B inverter is available in 0.5 to 30 HP versions (200-230 VAC) and 1.0 to 30 HP versions (380-460 VAC). All of the inverters use 32-bit "Digital Signal Processors" (DSP) for multifunction processing to provide high performance in all applications. The AF-300B inverters feature completely enclosed (except for the cooling fan) construction for improved performance in adverse environments. All the inverters are furnished with a detachable wiring lead-in plate for access so that wiring can be easily accomplished. (See Figure 2-3 for inverter external component identification.)

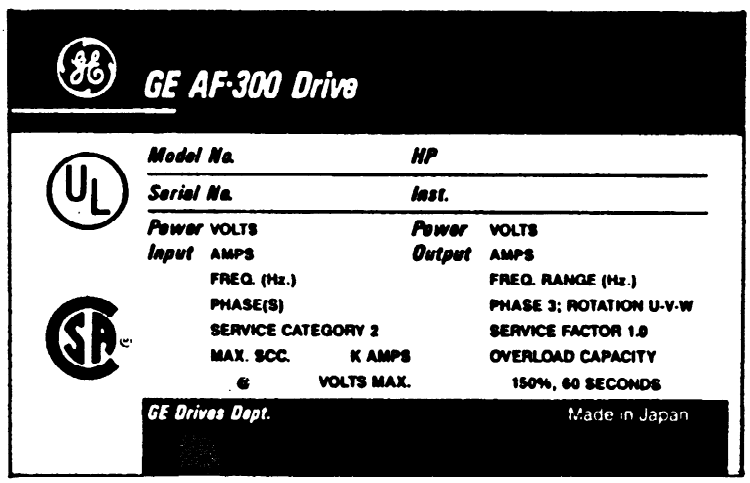
Inverter operation and code function setting is performed from the "Keypad Panel" that also features a Digital Monitor and Graphic Display. Five operation keys, STOP key, and the increment and decrement keys (for frequency/speed adjustment) control the inverter. (See Figure 2-2 for Keypad Panel component identification and function.)

General data and specifications for each individual inverter are listed on the nameplate attached to the inverter (see Figure 2-1 for nameplate example). See TABLE 1 for a complete listing of all AF-300B inverter specifications.

INSPECTION ITEMS UPON DELIVERY

Upon receipt of your inverter, inspect equipment for the following items:

1. Check the nameplate to insure that the specifications correspond to those ordered.
2. Inspect the unit for any damage that may have occurred during shipment.
   -- If there are any problems or questions regarding the above, contact the distributor from which the inverter was purchased.



*Not all models are UL Listed and CSA Approved; See Chart on Page 2-11.

Figure 2-1. NAMEPLATE DATA IDENTIFICATION

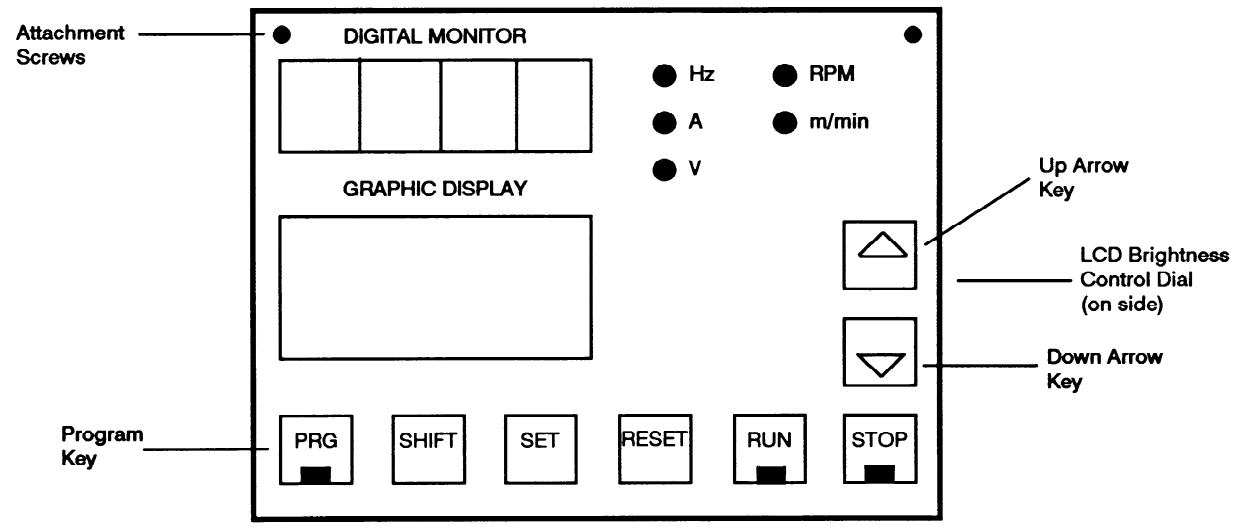


Figure 2-2. KEYPAD PANEL COMPONENT IDENTIFICATION

Attachment Screws -The Keypad Panel can be easily removed from the inverter unit by loosening the two attachment screws. With the optional extension cable, remote operation and display is possible.

Graphic Display - Frequency and output current are graphically displayed. The main input/output signal ON/OFF is also displayed. When in Program Setting Mode, the appropriate information is displayed graphically and alphabetically for easy Function Code selection setting.

Program (PGR) Key - Normal Mode or Program Setting Mode selection key. Key lights up when in Program Setting Mode.

SHIFT Key - When the Program key (**PRG**) is in Normal Mode, unit display can be changed while in either **RUN** or **STOP**. In the Program Setting Mode, this key allows Function Code selection by displaying each Function Code "column" in sequence. (See Figure 5-3 and TABLE 6.)

SET Key - Data display and settings for each Function Code are performed through this key. Also, when setting data on the graphic display, data accessed on the display can be changed.

RESET Key - Resets abnormal **STOP** conditions when Program key (**PRG**) is in the Normal Mode. Also changes from Data Setting Mode to Function Code Setting Mode when Program key (**PRG**) is in the Program Setting Mode.

RUN Key - Key is used for starting operation and lights up during operation. This key does not function when Data Code is set for external (control terminal) operation **[1501]** or serial link operation **[1502]**.

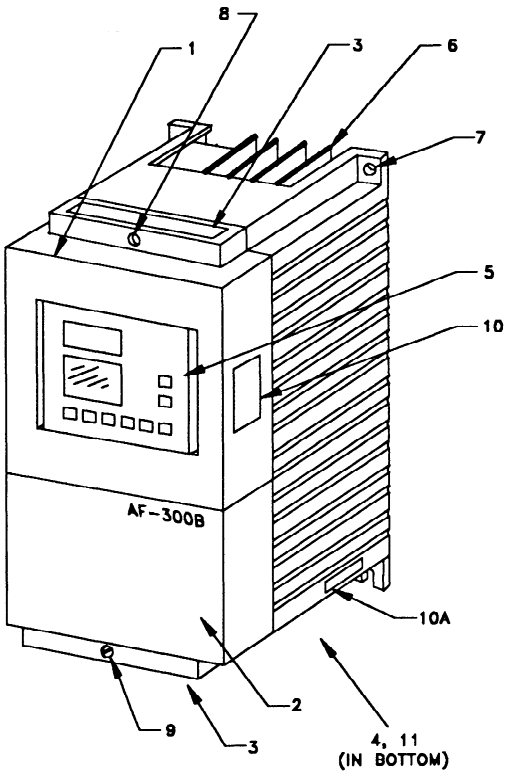
STOP Key - This key is used for stopping operation and will light up when in **STOP** Mode. If pressed when Data Code is set for external (control terminal) operation **[1501]** or serial link operation **[1502]**, Fault OH2 will be displayed and inverter will coast stop.

Up - Down Keys - These keys increase or decrease the frequency (or speed) of the inverter. When unit is in Program Setting Mode, they change the Function Code or Data Code values.

LCD Brightness - This control permits adjustment for easy to read brightness.

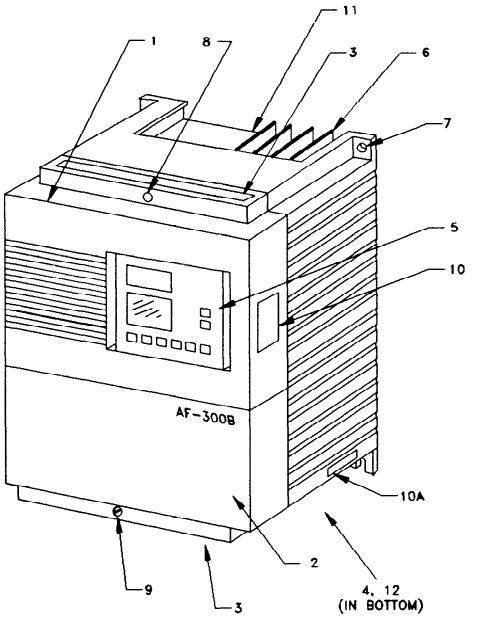
Unit Display - The unit information is displayed by LEDs.

Digital Monitor - When setting the program, the Function Code is indicated by the two digits on the left and the Data Code corresponding to the Function Code is indicated by the two digits on the right. During operation it displays the set frequency, current, voltage, etc. If a protective **STOP** occurs, the causes of the problem will be displayed as a fault code (see TABLE 11 and TABLE 12).



1/2 to 5 HP

NOTE: 1/2 and 1 HP models are not equipped with cooling fans



7-1/2 to 30 HP

1. Unit Cover
2. Terminal Cover
3. Ventilation Cover
4. Wiring Lead-In Plate (Rubber Bushings Included)
5. Keypad Panel
6. Cooling Fins
7. Mounting Screw Holes
8. Unit Cover Screws
9. Terminal Cover Screws
10. Rating Name Plate

10A. Model # and Serial # Tag

11. Cooling Fan(s) (if equipped)
12. Electroytic Capacitors

Figure 2-3. TYPICAL INVERTER COMPONENTS

TABLE 1: AF-300B INVERTER STANDARD SPECIFICATIONS

ITEM	SPECIFICATION										
Inverter (200-230V)	0.5	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
HP (380-460V)	–	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
Standard Applicable Motors-230V (KW)	0.4	0.75	1.5	2.2	3.7	5.5	7.5	11	15	18.5	22
Standard Applicable Motors-460V (KW)	--	0.75	1.5	2.2	3.7	5.5	7.5	11	15	18.5	22
OUTPUT:											
Rated Capacity-230V (KVA) (*1)	1.1	1.9	3.0	4.2	6.5	9.5	13	18	22	28	33
Rated Output Current-230V (Amps)	3	5	8	11	17	25	33	46	59	74	87
Rated Capacity-460V (KVA) (*1)	--	2.0	3.0	4.4	7.2	10	14	19	24	31	36
Rated Output Current-460V (Amps)	--	2.5	3.7	5.5	9.0	13	18	24	30	39	45
Rated Output Voltage (*2)	3-Phase, 3-Wire Type, 200-230 VAC or 380-460 VAC										
Rated Output Frequency	0.2 - 120 Hertz (Consult with distributor for operation above 120 Hz.)										
Overload Current Rating	150% for 1 minute duration (inverse time characteristic)										
POWER SUPPLY: **Rated Input AC Voltage (*3)**	3-Phase, 3-Wire, 200-230 VAC, 380-460 VAC-50 Hz, or 400-460 VAC-60 Hz										
Allowable Fluctuations (*3)	Voltage - +10%, -15%; Voltage Unbalance - Within 3%; Frequency - +/-5%										
Control System	Vector Dispersion PWM Control										
Output Frequency Range (*4)	0.2 - 120 Hz (0.2 to 60 Hz start frequency; 0.2 to 120 Hz base frequency)										
CONTROL: **Frequency Temperature Fluctuation**	Analog Setting - Maximum frequency +/-0.2% (at 25°C, +/-10°C) Digital Setting - Maximum frequency +/-0.01% (at -10°C to 50°C)										
Frequency Setting Resolution	Analog Setting - 0.02Hz (at maximum frequency of 60 Hertz) Digital Setting - 0.002Hz (at maximum frequency of 60 Hertz)										
Frequency Setting Resolution (*5)	0.002 Hertz (both analog and digital setting)										
Voltage/Frequency Characteristics (V/F) (*4)	Voltage - 200-230 VAC or 400-460 VAC (When in free selection, voltage or Frequency - 0.2 to 120 Hertz frequency can be selected.)										

(*1) Indicates rated capacity when rated output voltage is 230 VAC.
(*2) It is not possible for output voltage to exceed power supply voltage.
(*3) Units are Dual Rated. Inverter looks for a similar V/Hz ratio.
(*4) Consult distributor where equipment was purchased for operation above 120 Hertz.
(*5) Output frequency is changed at 0.002Hz intervals during acceleration/deceleration (when at max. frequency of 60Hz).

TABLE 1: AF-300B INVERTER STANDARD SPECIFICATIONS (continued)

ITEM	SPECIFICATION										
Inverter (200-230V)	0.5	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
HP (380-460V)	–	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
CONTROL: (continued) **Torque Boost**	32 patterns (squared decrease, including 10 step minute adjustment, automatic torque boost selection possible)										
Acceleration/Deceleration Chacteristics	0.01 to 3600 seconds (independent acceleration/deceleration) "S" Curve Characteristic (strong and weak) Automatic and Multi-step Accel./Decel. Selection (independent 4-step) (Both terminal and program selection is possible.)										
Internal Functions: **Operating Sound Selection**	The sound quality of the sound produced by the motor can be changed to reduce irrating noise.										
Frequency Meter Adjustment	Scale calibration of externally connected analog meter (6.5-10.5 VDC)										
Parameter Protection	Data lock is possible to ensure that the data codes are not changed.										
Terminal Link	Operation frequency setting for multiple inverters, multi-step operation, interlocking operation, and ratio operation are all possible at the same time.										
Brake Switch	Normal or strong brake selection is possible.										
Pattern Operation	Seven independent steps (freq. up to 120 Hz; 7 times up to 3600 sec. each)										
Program Operation	Based on the pattern operation; 1 cycle, repeat cycle, continuous operation at last step speed, etc. selections are possible.										
Momentary Power Failure Restart	After momentary power failure, automatic restart is possible.										
High/Low Limiter	Output frequency upper and lower range limit 0 to 100% (1% step) setting.										
Bias (See NOTE at bottom of page) (See WARNING in Section 1)	Magnitude of the bias that contains the frequency setting signal can be set from 0 to 100% (1% step).										
Gain (See NOTE at bottom of page)	Output frequency gain corresponding to the frequency setting signal can be set from 0 to 100% (1% step).										
Frequency Jump	A 3 point jump in width of sympathetic vibrations and resonance is possible. Width adjustable from 0 - 5 Hertz (1Hz step). (Resonant Frequency Bypass)										
Current Limit Control	Maximum output current can be limited. Limit is adjustable from 30% to 150% (1% step).										
8-Step Multiple Speed Switching	8-step multiple speed operation is possible.										
Momentary or Maintained Contact Operation	Selection between the maintained contact operation/stop command (2-wire operation) or the momentary contact (3-wire operation) is possible.										
Terminal Function Change	Multi-Use terminals can be changed via the Function Code setting (for 9 terminals).										

NOTE: *Bias and Gain do not affect operation when Frequency command comes from the keypad. (Function Code 16 = 00)*

TABLE 1: AF-300B INVERTER STANDARD SPECIFICATIONS (continued)

ITEM			SPECIFICATION									
Inverter (200-230V)			0.5	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
HP (380-460V)			–	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
CONTROL TORQUE:													
Std. Equip.: (230V)	Regen. Braking Tq (*5)		Up to 150%			Up to 100%		Up to 40%		Up to 20% (*6)			
	Duty Cycle		3%			3%		3%		----			
	DC Braking		Frequency 0.2 to 60 Hertz; Time 0.01 to 30 seconds; Voltage 0 to 15%										
Opt. Equip.: (230V)	Regen. Braking Tq (*5)		Braking Resistor Up to 150%			Braking Resistor Up to 100%				Braking Resistor and DB Unit Up to 100%			
	Duty Cycle		15%	12%	7%	7%	5%			5%			
Std. Equip.: (460V)	Regen. Braking Tq (*5)		--	150%	65%	45%	40%	Up to 35%		Up to 20% (*6)			
	Duty Cycle		--	3%	3%	3%	3%	3%		----			
	DC Braking		Frequency 0.2 to 60 Hertz; Time 0.01 to 30 seconds; Voltage 0 to 15%										
Opt. Equip.: (460V)	Regen. Braking Tq (*5)		--	Braking Resistor						Braking Resistor and DB Unit			
			--	Up to 150%		Up to 100%				Up to 100%			
	Duty Cycle		--	12%	7%	7%	5%			5%			
OPERATION:													
Frequency Setting Signal			Frequency setting device or voltage input: 0-10 VDC (0-5 VDC), 4-20mA DC.										
Input Signal (contact type)			Forward command, reverse command, self-holding selection (when operation is 3-wire), multi-step speed setting (8-step), multiple accel/decel time settings (4 settings), coast-to-stop command, external alarm, and reset. (Sampling time span is 1 millisecond or less.) **NOTE:** *By changing the terminal function, the input command or modes can be changed as follows: cycle operation command, frequency adder-subtractor setting, DC brake command, and link input.*										
External Output Signals:	Fault Contact		Batch alarm output (one "c" contact, contact point capacity - 250 VAC,0.3 A @ Cos *theta* $\geq$ 0.3)										
	Open Collector		During inverter operation, rated frequency attainment, overload prediction, and undervoltage. **NOTE:** *These can be changed to the following outputs through the terminal change function: finish signal of each stage/cycle in pattern operation, frequency level detection, inverter stop, current-limiting monitoring, undervoltage or restarting detection, and overvoltage detection.*										
PROTECTIVE FUNCTIONS:			Stall prevention, overcurrent, momentary power failure, undervoltage, over-voltage, inverter overheating, external problems (external thermal activation, etc.), motor overload (electronic thermal), setting error, communication error, DSP error, link error, and output terminal short-circuit. Short-to-ground protection is optional for 3.7KW (5 HP) or less.										
DISPLAY:													
Frequency Meter Output Signal			Analog - 0 to +10 VDC (adjustment range of 6.5 to 10.5 VDC)										

(*5) Short Time Rating
(*6) Condenser Regenerative Braking

TABLE 1: AF-300B STANDARD SPECIFICATIONS (continued)

ITEM	SPECIFICATION										
Inverter (200-230V)	0.5	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
HP (400-460V)	--	1.0	2.0	3.0	5.0	7.5	10	15	20	25	30
DISPLAY: (continued) **Keypad Panel:** **Frequency Meter Output Signal**	Digital Display - 4 digit LED Graphic Display - LCD, with brightness control										
Operation Display	Output frequency, output current, output voltage, synchronous rpm, and line speed. (Set Frequency when not in RUN or PROGRAM Mode)										
Setting Display	Function code and setting data displayed (see Operation Panel paragraph).										
Fault Display	OC1 - acceleration overcurrent; OC2 - deceleration overcurrent; OC3 - constant speed overcurrent; LU (LV) - undervoltage; OU (OV) - overvoltage; OH1 - inverter overheat; OH2 - external alarm (external thermal); OL1 - electronic thermal motor overload; Err1 - setting error; Err2 - communication error; Err3 - DSP error; Err4 - link error, operating conditions at time of fault (output frequency, etc.), (record of faults) (display of past three fault codes).										
Input Signal Display	Display of control signal at FWD, REV, HLD, X1, X2, X3, RT1, and RT2.										
Charge Lamp (LED)	Lights when DC Link capacitor voltage is present.										
CONDITION: **Installation Location**	Indoor, less than 1000 meters (3281 feet) elevation, not in contact with corrosive gas, oil mist or dust, and out of direct sunlight. Motor wiring should be less than 66 feet total length, otherwise wire separation of 4 inches may be required.										
Ambient Temperature	-10 to +50°C (+14 to +122°F) (remove ventilation covers if temperature is over +40°C [+104°F]).										
Humidity	Less than 90% relative humidity (without dew condensation).										
Vibration	0.6G or less.										
INSTALLATION:	Install on a panel (self-cooling); install with an external cooling system.										
PROTECTION/COOLING TYPES:	(*7)	Fully enclosed forced air cooling type (fan not enclosed).									
APPROXIMATE WEIGHT (230V): Kgs	3.0	3.0	4.3	4.4	4.5	10.7	10.9	14.7	15.0	19.0	19.5
Lbs	6.6	6.6	9.5	9.7	9.9	23.6	24.0	32.4	33.0	41.9	43.0
APPROXIMATE WEIGHT (460V): Kgs	--	3.6	4.4	4.5	4.6	9.3	9.5	12.9	13.0	16.6	16.9
Lbs	--	7.9	9.7	9.9	10.1	20.5	20.9	28.4	28.7	36.6	37.3

(*7) Fully enclosed self-cooling type.

TABLE 1: AF-300B STANDARD SPECIFICATIONS (continued)

ITEM	SPECIFICATION								
INVERTER HP (200-230V)	**0.5**	**1.0**	**2.0**	**3.0**	**5.0**	**7.5**	**10**		
Efficiency and Loss: Loss (W)	25	40	70	110	150	280	380		
Efficiency (%) Constant Torque Load: Base Speed Full Load	96.5	96.6	96.4	96.5	97.0	96.4	96.3		
0.75 Base Speed Full Load	95.5	95.4	95.6	95.5	96.0	95.5	95.5		
0.5 Base Speed Full Load	92.5	93.6	94.2	94.0	94.7	94.0	94.3		
0.25 Base Speed Full Load	88.0	90.1	91.1	90.8	91.5	91.3	91.5		
Efficiency (%) Pump or Fan 0.8 Base Speed 64% Load	94.2	95.4	96.2	96.2	95.7	95.8	95.8		
0.6 Base Speed 36% Load	89.5	90.0	92.0	91.0	94.3	91.0	93.2		
0.4 Base Speed 16% Load	80.0	80.5	82.2	83.8	90.2	84.3	88.8		
INVERTER HP (400-460V)	**3.0**	**5.0**	**7.5**	**10**	**15**	**20**	**25**	**30**	
Efficiency and Loss: Loss (W)	100	150	280	380	580	650	700	750	
Efficiency (%) Constant Torque Load: Base Speed Full Load	96.3	97.2	96.3	96.3	96.0	96.7	97.4	97.2	
0.75 Base Speed Full Load	95.4	96.1	95.2	95.3	95.2	95.9	96.6	96.3	
0.5 Base Speed Full Load	94.2	94.8	94.1	94.2	93.8	94.6	95.4	95.1	
0.25 Base Speed Full Load	90.6	91.4	91.5	91.5	91.3	92.1	93.2	92.7	
Efficiency (%) Pump or Fan 0.8 Base Speed 64% Load	96.3	95.9	95.9	95.8	96.0	96.0	97.0	95.8	
0.6 Base Speed 36% Load	91.2	94.2	91.1	93.1	94.0	94.6	95.4	95.3	
0.4 Base Speed 16% Load	83.6	90.1	84.2	88.5	90.2	92.4	92.4	92.4	

INVERTER OPTIONS

Various options are available for the AF-300B Inverter. These options can be ordered from the supplier of the inverter per the GE Catalog Number shown on the chart within this paragraph. The General Electric Company "6V" catalog numbers are coded as shown in the following Examples:

NOTE: *Option charts are divided into horsepower related and non-horsepower related items. Check the horsepower of the inverter before ordering options to be sure the correct option is ordered.*

NOTE: *Disregard steps on* ***Serial Link Option*** *usage unless inverter is so equipped.*

Example: (Non-HP Related Items)

6VKAAA3AANN*

6V - General Electric Drives Products Operation
K - Option Kit
AAA - Applicable Series (1, 2, or 3 characters)
3 - AF-300 Family of inverters
AA - Option Abreviation
NN - Option Size/Length (if applicable)

Example: (HP Related Items)

6VKA3NAANN*

6V - General Electric Drives Products Operation
K - Option Kit
AAA - Applicable Series
3 - AF-300 Family of inverters
N - Voltage
AA - Option Abreviation
NN - Horsepower Rating (2 or 3 characters)

**A* = alphabetical character, N = Numerical char.*

AF-300B NON-HORSEPOWER RELATED OPTIONS

Description	GE Catalog Number
RS-422/485 Card	6VKB3RS
BCD Card	6VKB3BC
Ground Fault Unit	6VKB3GF
Pulse Tach Feedback Card	6VKB3TF
Relay Unit	6VKB3RY
Keypad Cable - 2M*	6VKB3CB02
Keypad Cable - 5M*	6VKB3CB05
Keypad Cable - 10M*	6VKB3CB10
Speed Potentiometer	6VKAB3SP
RFI Reactor	6VKABC3RF
Terminal Link Adaptor	6VKB3JM
RS485 Interface	6VKB3RS
Parameter Copy Unit	6VKB3PC

See* ***Figure 2-7* *for drill pattern for remote mounting of Keypad.*

AF-300B HORSEPOWER RELATED OPTIONS

Description	GE Catalog Number
Brake Switching Unit	6VKB32BU025
Brake Switching Unit	6VKB32BU030
Brake Switching Unit	6VKB34BU030
Brake Resistors	6VKB32BR01
Brake Resistors	6VKB32BR05
Brake Resistors	6VKB32BR07
Brake Resistors	6VKB32BR10
Brake Resistors	6VKB32BR20
Brake Resistors	6VKB32BR25
Brake Resistors	6VKB32BR30
Brake Resistors	6VKB34BR01
Brake Resistors	6VKB34BR05
Brake Resistors	6VKB34BR07
Brake Resistors	6VKB34BR10
Brake Resistors	6VKB34BR20
Brake Resistors	6VKB34BR25
Brake Resistors	6VKB34BR30

CAUTION: The AC Supply Power Transformer KVA rating should be greater than 1.5 times the Horsepower rating of the inverter, but not greater than 500 KVA. If an AC Supply Power Transformer greater than 500 KVA is to be used, install a line reactor (optional; must be purchased seperately) on the line side of the inverter. A Line Reactor is also required if an SCR type controller and/or Power Factor Correction Capacitors are used on the same power lines.

Refer to the following Charts for the recommended line reactor for the applicable inverter.

AC LINE REACTORS FOR AF-300B INVERTERS (230 VAC)

HP	Volts	KVA	Input Amps*	Part # 104X220	Inductance (in *u*H)
0.5	230	1.2	3	FB004	4000
1.0	230	2.0	5	FB005	2500
2.0	230	3.2	8	FA001	950
3.0	230	4.38	11	FA001	950
5.0	230	6.77	17	FA001	950
7.5	230	9.96	25	FA002	485
10	230	13.2	33	FA002	485
15	230	18.3	46	FA003	240
20	230	23.5	59	FA003	240
25	230	29.5	74	FA003	240
30	230	34.7	87	FA004	120

**Amps shown are at 200 VAC input voltage.*

NOTE: *If reactors are supplied by others, be sure that the peak (non-saturating) current rating of the reactor is greater than 2.8 times the input current rating of the inverter.*

AC LINE REACTORS FOR AF-300B INVERTERS (460 VAC)

HP	Volts	KVA	Input Amps*	Part # 104X220	Inductance (In *u*H)
1.0	460	1.43	2.6	FB002	10000
2.0	460	2.95	3.7	FB004	4000
3.0	460	4.38	5.8	FB004	4000
5.0	460	7.17	9.5	FB005	2500
7.5	460	10.4	13	FA001	950
10	460	14.3	18.9	FA001	950
15	460	19.1	24.5	FA002	485
20	460	23.9	31.5	FA002	485
25	460	31.1	39	FA002	485
30	460	35.9	45	FA003	240

**Amps shown are at 400 VAC input voltage.*

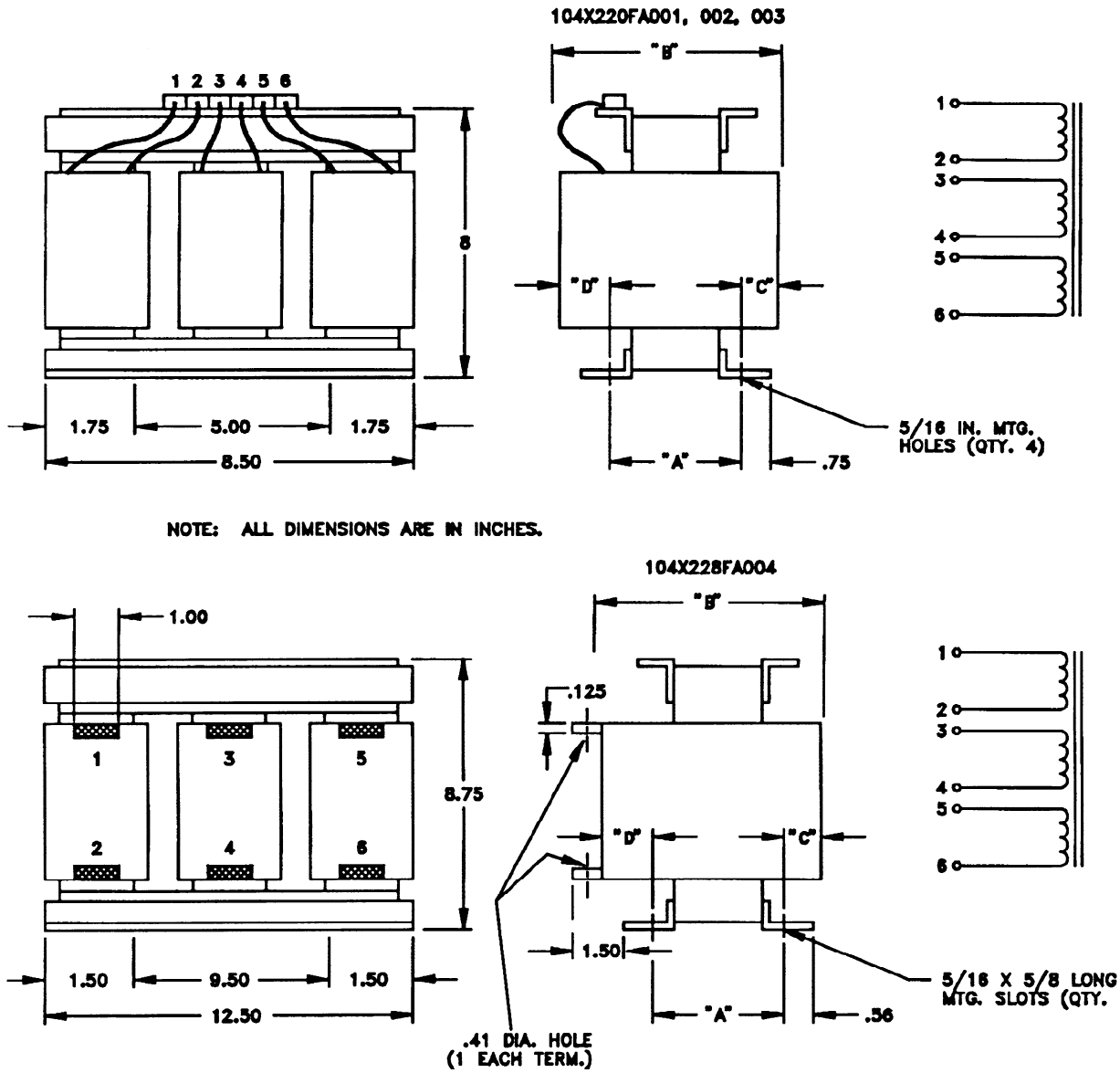
NOTE: *If reactors are supplied by others, be sure that the peak (non-saturating) current rating of the reactor is greater than 2.8 times the input current rating of the inverter.*

AF-300B Catalog Numbers (230 VAC*)	
HP Rating	Catalog No.
0.5	6VAF323F50B -__(1)
1.0	6VAF323001B -__(1)
2.0	6VAF323002B -__(1)
3.0	6VAF323003B -__(1)
5.0	6VAF323005B -__(1)
7.5	6VAF323007B -__(1)
10	6VAF323010B -__(1)
15	6VAF323015B (2)
20	6VAF323020B (2)
25	6VAF323025B (2)
30	6VAF323030B (2)
**200-230 VAC, 3-Phase, 50/60 Hertz*	

AF-300B Catalog Numbers (460 VAC*)	
HP Rating	Catalog No.
1.0	6VAF343001B -__(1)
2.0	6VAF343002B -__(1)
3.0	6VAF343003B -__(1)
5.0	6VAF343005B -__(1)
7.5	6VAF343007B -__(1)
10	6VAF343010B -__(1)
15	6VAF343015B -__(1)
20	6VAF343020B -__(1)
25	6VAF343025B -__(1)
30	6VAF343030B -__(1)
**400-460 VAC, 3-Phase, 60 Hertz or 380-460 VAC, 3-Phase, 50 Hertz*	

(1) *The two character positions following a dash in the catalog number designate the following:*
No dash or characters *- inverter is not UL listed or CSA approved.*
A1 *- inverter is UL listed only.*
A2 *- inverter is UL listed and CSA approved.*

(2) *15 - 30 HP, 230 VAC inverters are not available as UL listed or CSA approved.*

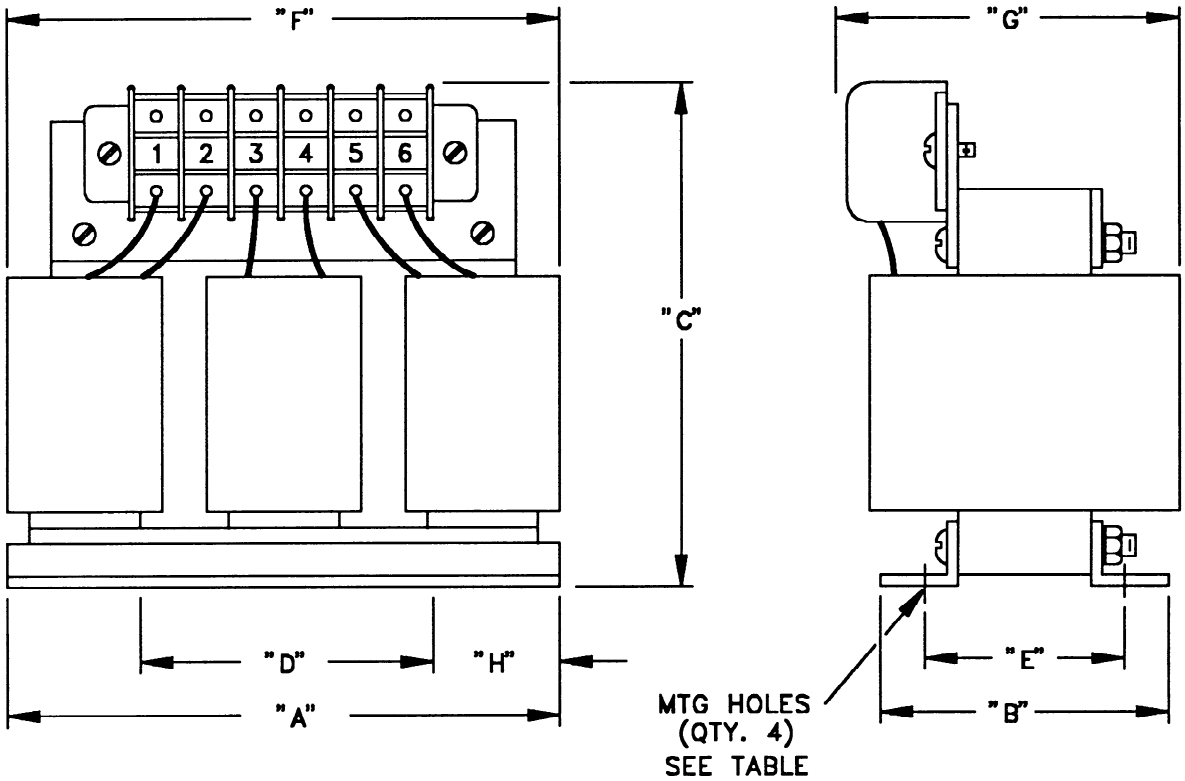


104X220FA	Dimensions (Inches)				Terminal Board	Approximate Weight (lbs)
	A	B	C	D		
001	2.0	4.375	.875	1.625	14 - 10 AWG	18.5
002	2.375	4.938	.875	1.875	14 - 4 AWG	25.0
003	3.125	5.563	.875	1.750	16 - 6 AWG	37.0
004	3.625	7.250	.625	1.50	----	67.0

**Tolerance of +/-1/8 inch.*
***If 4 AWG wire is required, use #4 solid or two #10 stranded; #4 stranded will not fit in connector.*

Figure 2-4. 104X220FA001, 002, 003, and 004 LINE REACTOR DIMENSIONS

NOTE: ALL DIMENSIONS ARE IN INCHES.



104X220FB	Dimensions (Inches)								
	A	B	C	D	E	F	G	H	Mtg. Holes
002	3.437	3.312	5.50	1.437	2.563	4.062	3.00	1.00	.25 x .50
004	5.25	4.125	6.25	2.00	3.375	6.125	3.25	1.625	.25 x .37
005	5.25	4.562	6.25	2.00	3.812	6.125	3.75	1.625	.25 x .37

Figure 2-5. 104X220FB002, 004, and 005 LINE REACTOR DIMENSIONS

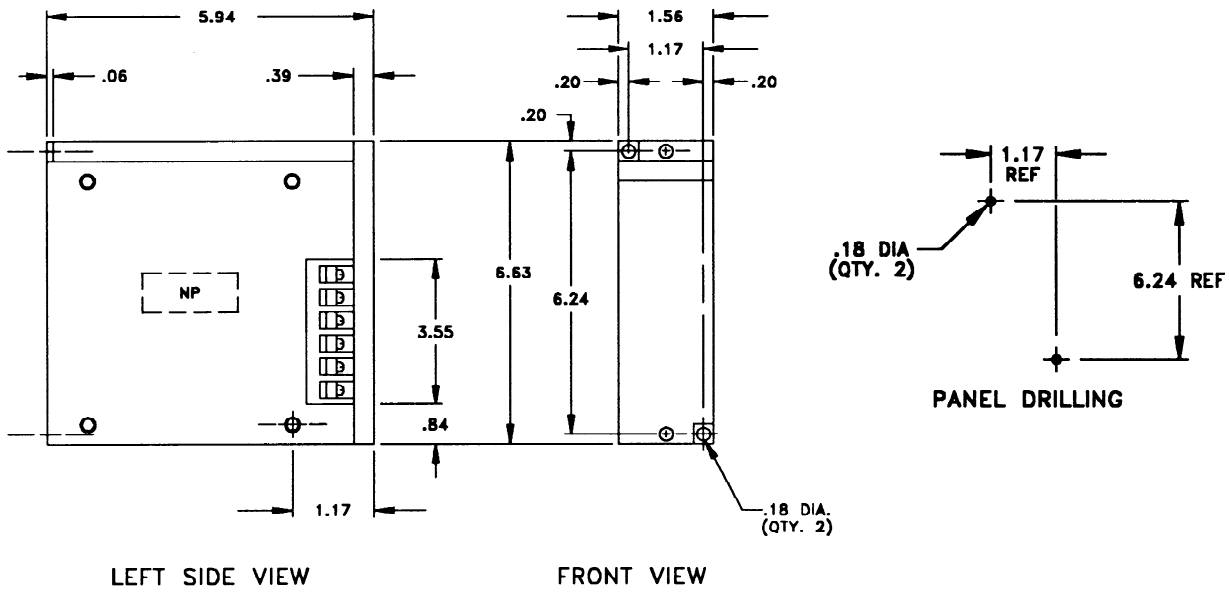


Figure 2-6. DYNAMIC BRAKING UNIT DIMENSIONS
(6VKB32BU25, 6VKB32BU30, and 6VKB34BU30 Braking Units)

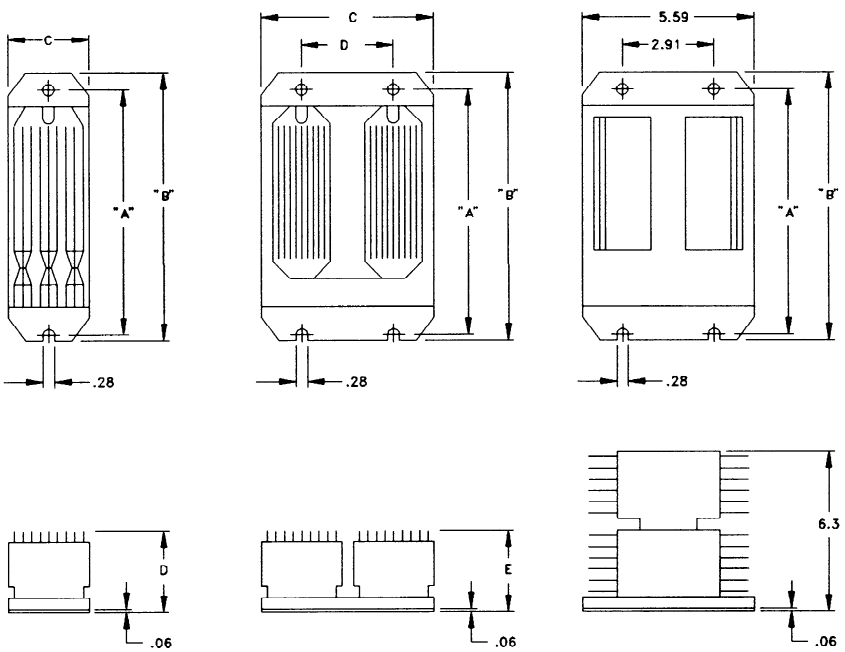


FIG. A FIG. B FIG. C

NOTE:
All dimensions (drawing and chart) are in inches.

Figure C		
Catalog #	**"A"**	**"B"**
6VKB32BR20 6VKB34BR20	16.34	16.93
6VKB32BR25 6VKB34BR25	19.49	20.08
6VKB32BR30 6VKB34BR30	19.49	20.08

Figure A				
Catalog #	**"A"**	**"B"**	**"C"**	**"D"**
6VKB32BR01 6VKB34BR01	11.61	12.20	2.52	2.64
6VKB32BR05	13.07	13.58	2.99	3.70
6VKB34BR05	17.91	18.50	2.99	2.64

Figure B					
Catalog #	**"A"**	**"B"**	**"C"**	**"D"**	**"E"**
6VKB32BR07	16.93	17.72	5.59	3.54	2.66
6VKB34BR07	17.91	18.50	5.59	2.91	2.64
6VKB32BR10	14.57	15.35	6.14	3.54	3.54
6VKB34BR10	19.49	20.08	5.59	2.91	2.64

Figure 2-7. DYNAMIC BRAKING RESISTOR DIMENSIONS

This relay unit permits the inverter open collector output signal to be transformed into a relay output signal. The inverter running signal (RUN), frequency equivalence detection signal (FAR), overload early warning signal (OL), and undervoltage detection signal (LV) are output from a relay contact.

Power Supply Voltage: 85 to 265 VAC, 50/60 Hz

Relay Rating: 250 VAC, 0.3 A inductive @ Cos *theta* $\geq$ 0.3 SPDT Contact

NOTE: *When the power supply voltage is greater than 265 VAC, provide an auxiliary transformer.*

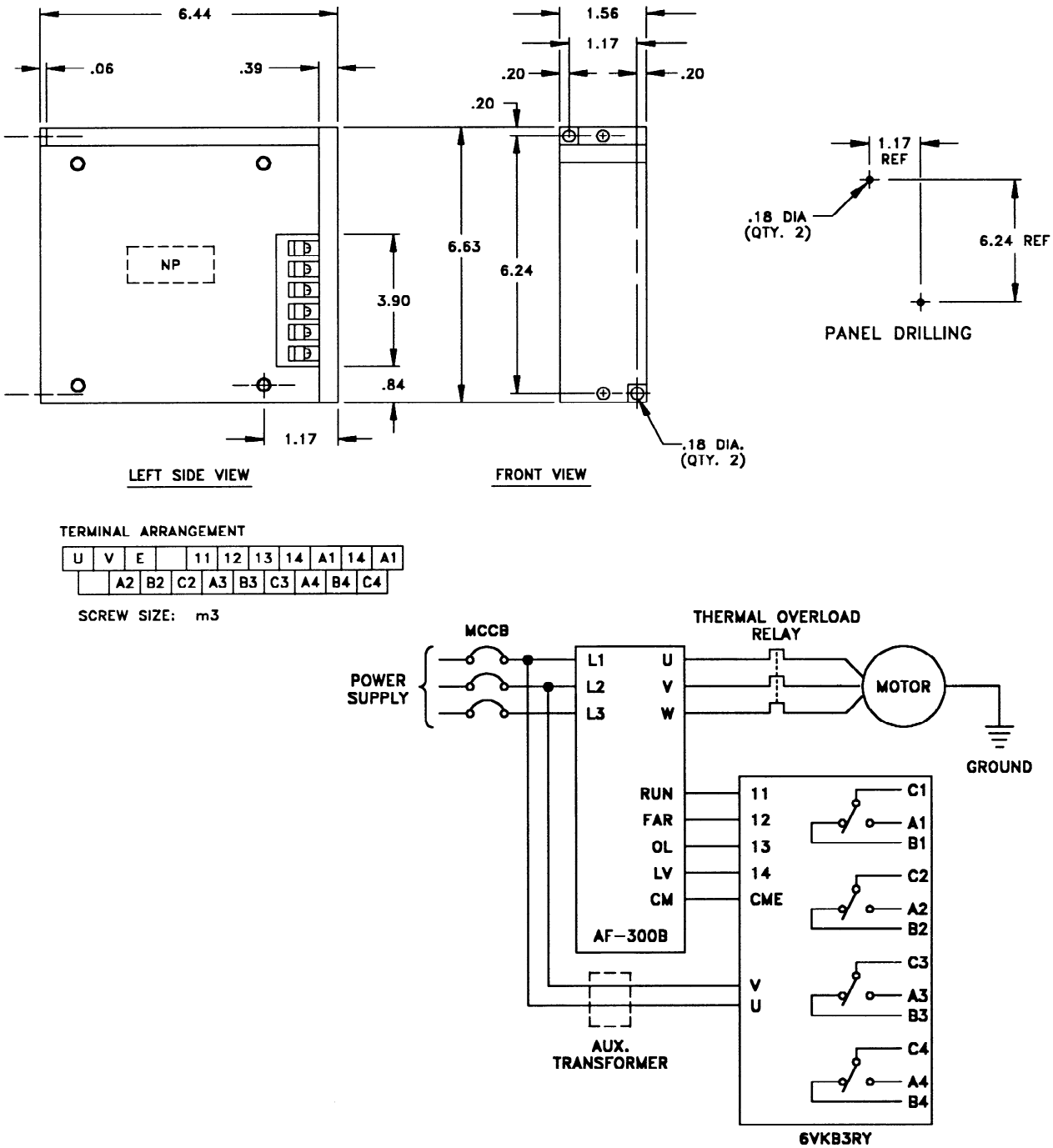


Figure 2-8. 6VKB3RY RELAY UNIT DIMENSIONS, SPECIFICATIONS, and WIRING

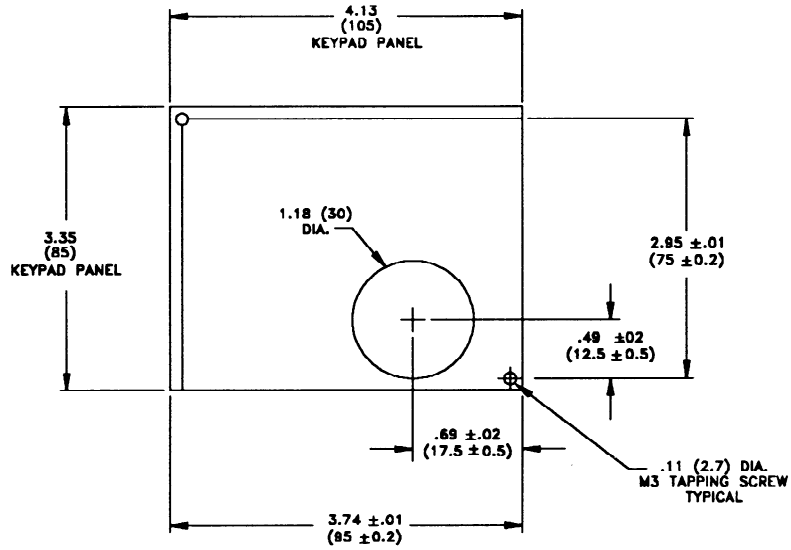


Figure 2-9. HOLE DRILLING FOR KEYPAD PANEL INSTALLATION WHEN USING KEYPAD EXTENSION CABLE OPTION

NOTES:

3. INSTALLATION GUIDELINES

INSTALLATION ENVIRONMENT

Install the inverter in a location that meets the following requirements:

- -- The ambient temperature is between -10°C and +50°C (+14°F to +122°F). (Remove the ventilation cover when the temperature exceeds +40°C [+104°F].)
- -- The relative humidity is between 20% and 90%. Avoid any location subject to condensation, freezing, or where the inverter would come in contact with water.
- -- Do not install in any location subject to direct sunlight, dust, corrosive gas, inflammable gas, or oil mist.
- -- The inverter should be installed at an elevation below 1000 meters (3281 feet) and vibration should be less than 0.6G.
- -- Wiring from inverter to motor should be less than 66 feet, otherwise separation of 4 inches may be required.

INSTALLATION MOUNTING CLEARANCE

CAUTION: Because the ambient temperature greatly affects inverter life and reliability, do not install the inverter in any location that exceeds the allowable temperatures.

- -- Install the inverter perpendicular to the ground and with the lettering right side up. (If the inverter is installed up-side-down or horizontally, heat build-up will occur.)
- -- Install at a sufficient distance from other equipment, walls, or wiring ducts as shown in Figure 3-1 (these clearances are required to allow the heat generated by the inverter to escape).

CAUTION: The mounting wall for the inverter must be of heat resistant material because during operation, the temperature of the inverter's cooling fins rises to approximately 90 degrees C (194 F).

NOTE: *When installing **two or more inverters** in close proximity, allow sufficient space as shown in Figure 3-1 and install them in a horizontal row. If they must be installed in a vertical row, at least 19.7 inches (50cm) internal space must be provided between each one or a ventilation system should be provided to prevent the ambient temperature from rising.*

- -- Mounting screws or bolts should be of appropriate size for weight of inverter .
- -- See the appropriate view in Figure 3-2 for the location of mounting holes.
- -- Install the rubber bushings supplied with the inverter into the cable openings in the wiring lead-in plate to prevent cable damage and to minimize dust entry.

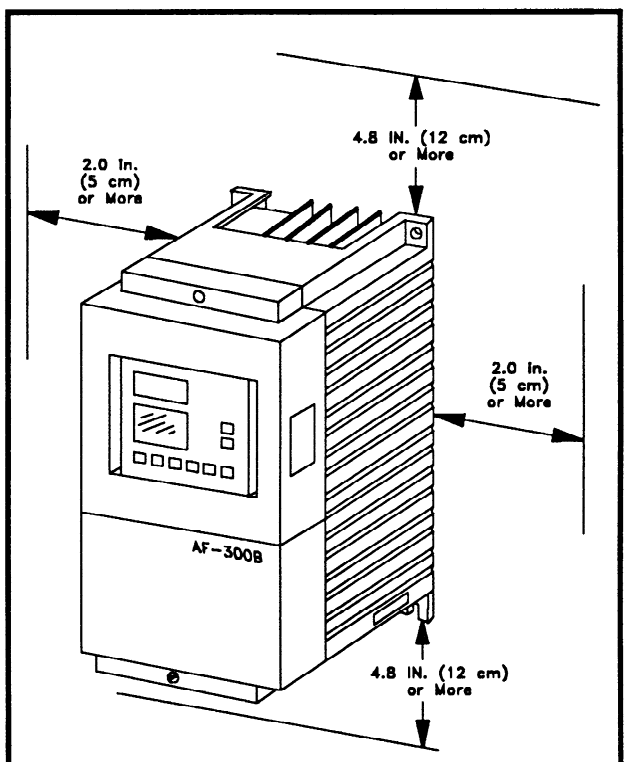
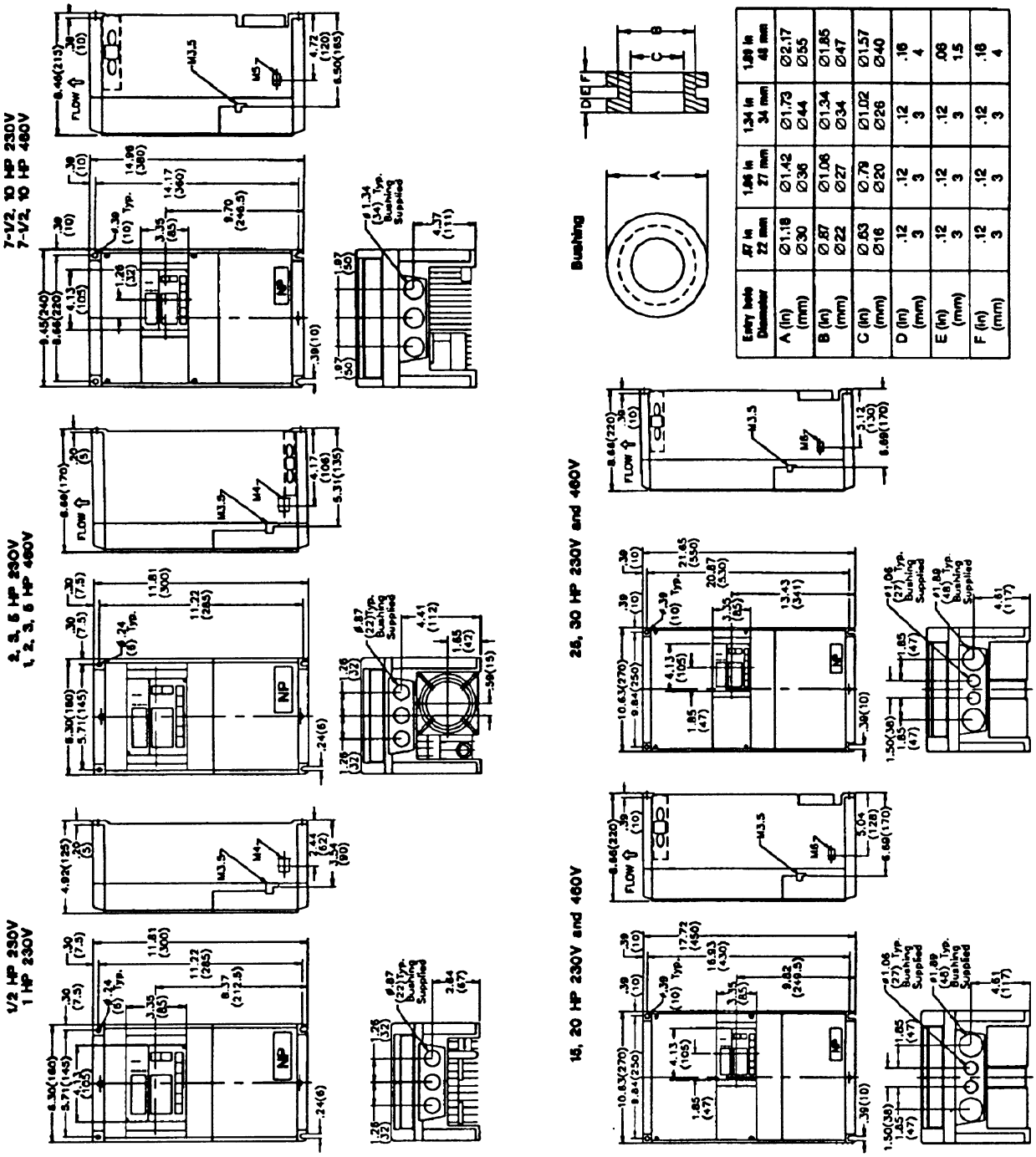


Figure 3-1. INVERTER MOUNTING CLEARANCE



Entry hole Diameter	.87 in 22 mm	1.06 in 27 mm	1.34 in 34 mm	1.69 in 43 mm
A (in) (mm)	Ø1.18 Ø30	Ø1.42 Ø36	Ø1.73 Ø44	Ø2.17 Ø55
B (in) (mm)	Ø.87 Ø22	Ø1.06 Ø27	Ø1.34 Ø34	Ø1.85 Ø47
C (in) (mm)	Ø.63 Ø16	Ø.79 Ø20	Ø1.02 Ø26	Ø1.57 Ø40
D (in) (mm)	.12 3	.12 3	.12 3	.16 4
E (in) (mm)	.12 3	.12 3	.12 3	.08 1.5
F (in) (mm)	.12 3	.12 3	.12 3	.16 4

Figure 3-2. INVERTER EXTERNAL DIMENSIONS

4. WIRING PROCEDURES

REMOVE TERMINAL BLOCK COVER

Remove the terminal block cover as follows (see Figure 4-1):

1. Remove the screw located at the bottom of the cover.
2. Press upward on bottom of cover and lift off.
3. See Figure 4-1 for the location of the **Main Circuit Terminal Block** and the **Control Circuit Terminal Block.**

MAIN CIRCUIT WIRING

CAUTION: Be sure that the power supply is never connected to the U, V, W terminals or the N, P, P, DB terminals.

1. Connect the power supply wires to the L1, L2, and L3 terminals of the Main Circuit Terminal Block as shown in the appropriate view of Figure 4-2. (See TABLE 2 for description of all terminals and TABLE 3 for application wiring list.)

2. Connect the 3-phase motor wires to the U, V, and W terminals of the Main Circuit Terminal Block as shown in the appropriate view of Figure 4-2. (See TABLE 2 for description of all terminals and TABLE 3 for application wiring list.)

NOTE: *Motor will rotate counterclockwise when viewed from the load side when connected normally. If the motor rotates in reverse direction, interchange any two of the U, V, or W terminal connections.*

3. Connect the ground terminal as shown in the appropriate view of Figure 4-2. (Do not operate without the unit being grounded.)
   -- The ground wire must be as large and short as possible (see TABLE 3 for application wiring list).

CONTROL CIRCUIT WIRING

Inverter is wired at shipment for operation and frequency setting through the **keypad panel** (frequency is set at 60 Hertz).

-- See Figure 4-3 for wiring connections.
-- See TABLE 2 for description of all terminals.

Make wire connections as shown in Figures 4-4 thru 4-26 for **desired mode of external operation** through control circuit terminals.

-- See TABLE 2 for description of all terminals.

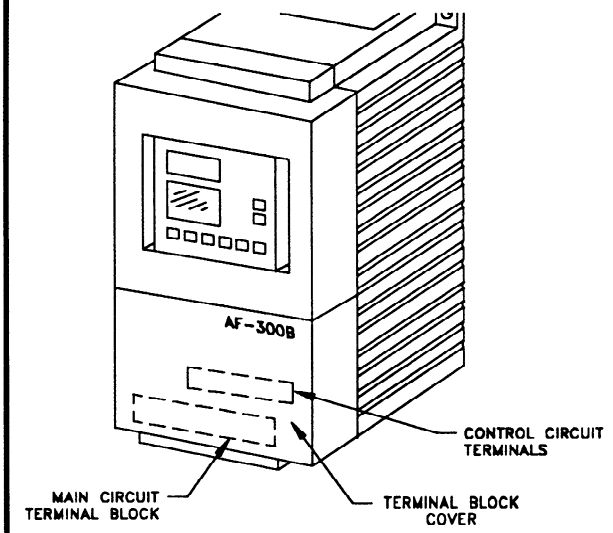
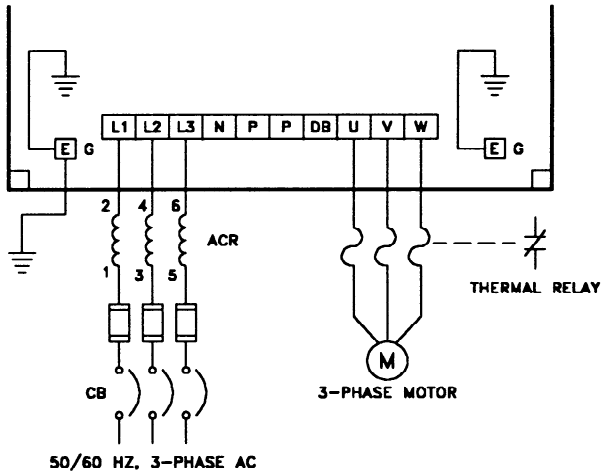
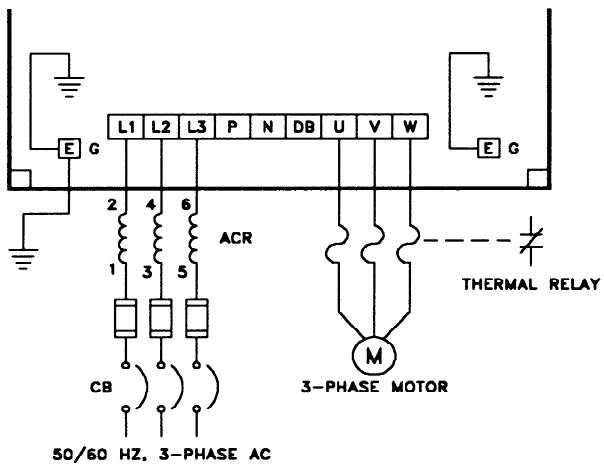


Figure 4-1. TERMINAL BLOCK COVER

UNITS UP TO 5 HP



UNITS FROM 7.5 TO 10 HP



UNITS FROM 15 TO 30 HP

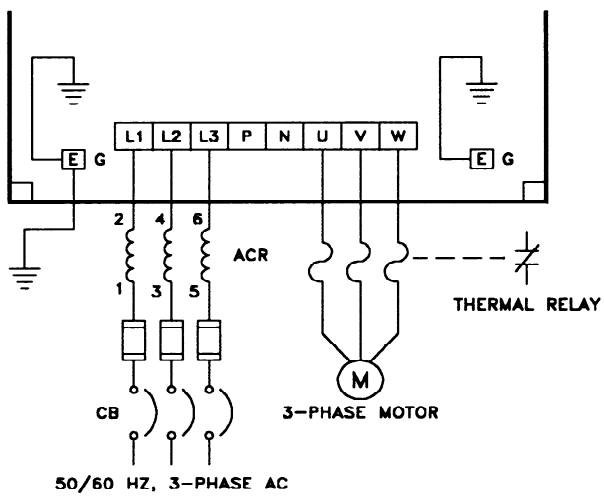
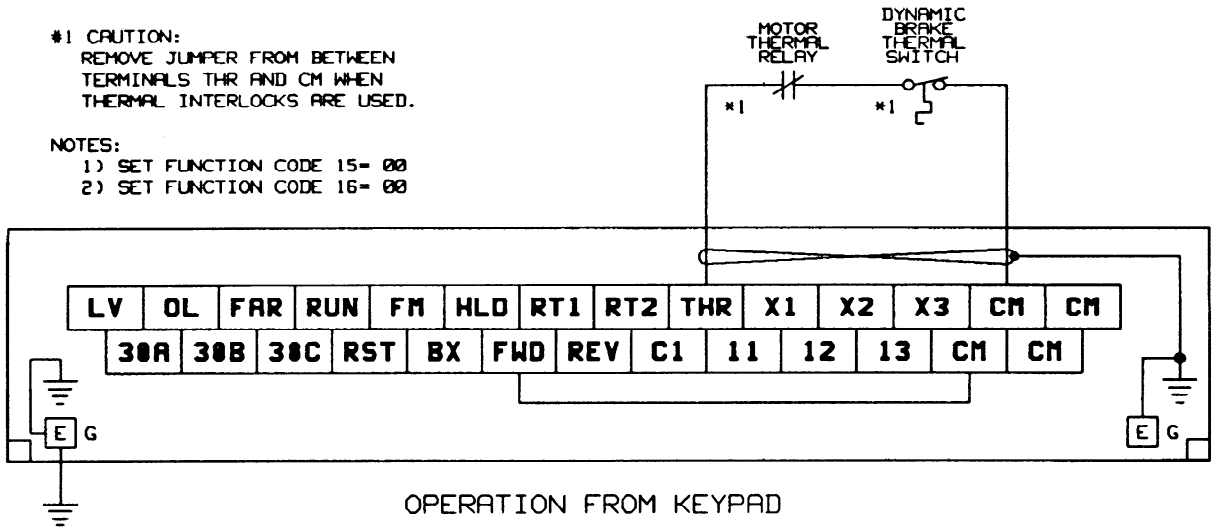
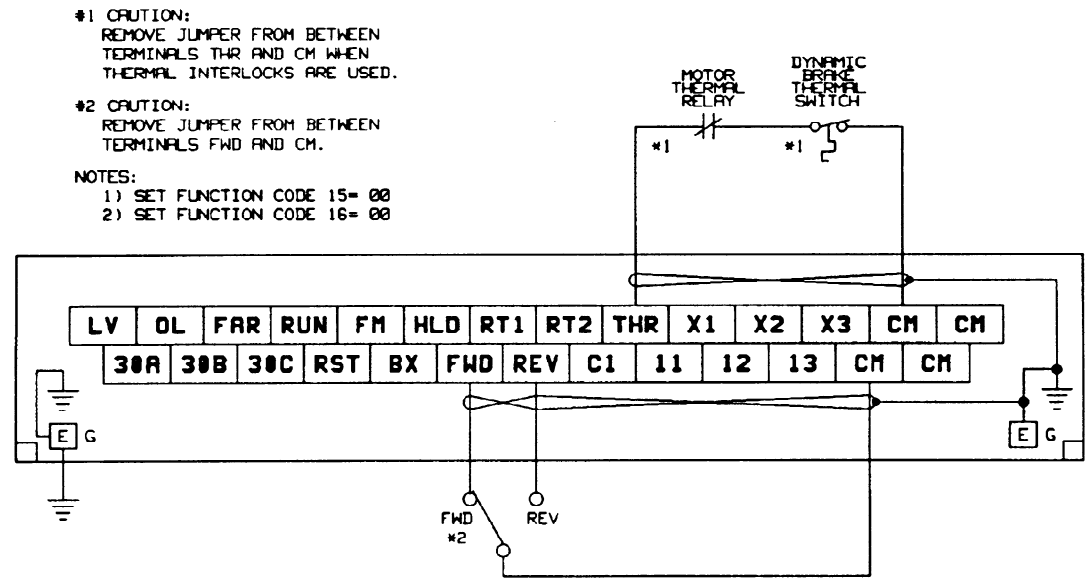


Figure 4-2. POWER CIRCUIT WIRING CONNECTIONS



OPERATION FROM KEYPAD



OPERATION FROM KEYPAD
WITH REVERSING CAPABILITY
(EXTERNAL SWITCH IS REQUIRED)

Figure 4-3. WIRING CONNECTIONS FOR OPERATION THROUGH KEYPAD PANEL

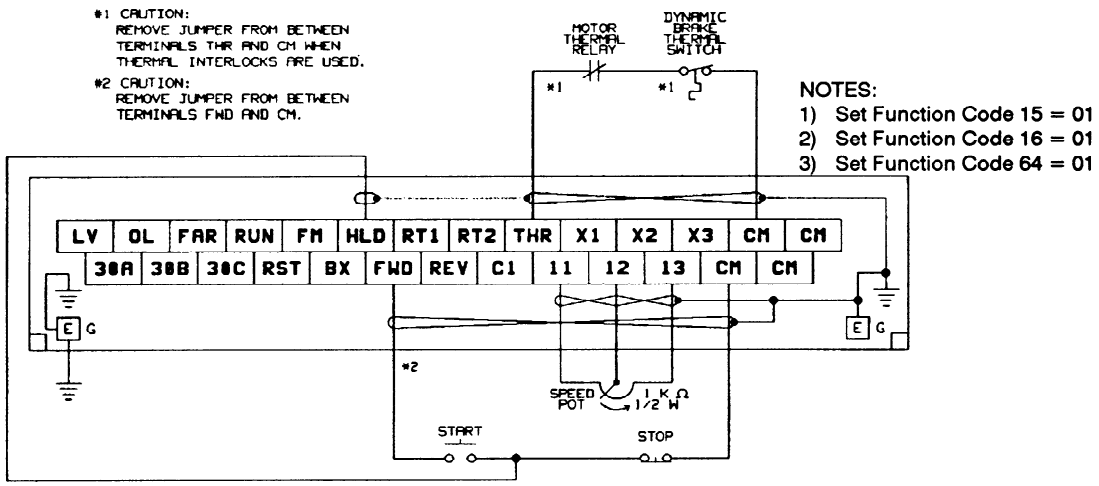


Figure 4-4. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND SPEED REFERENCE FROM SPEED POTENTIOMETER

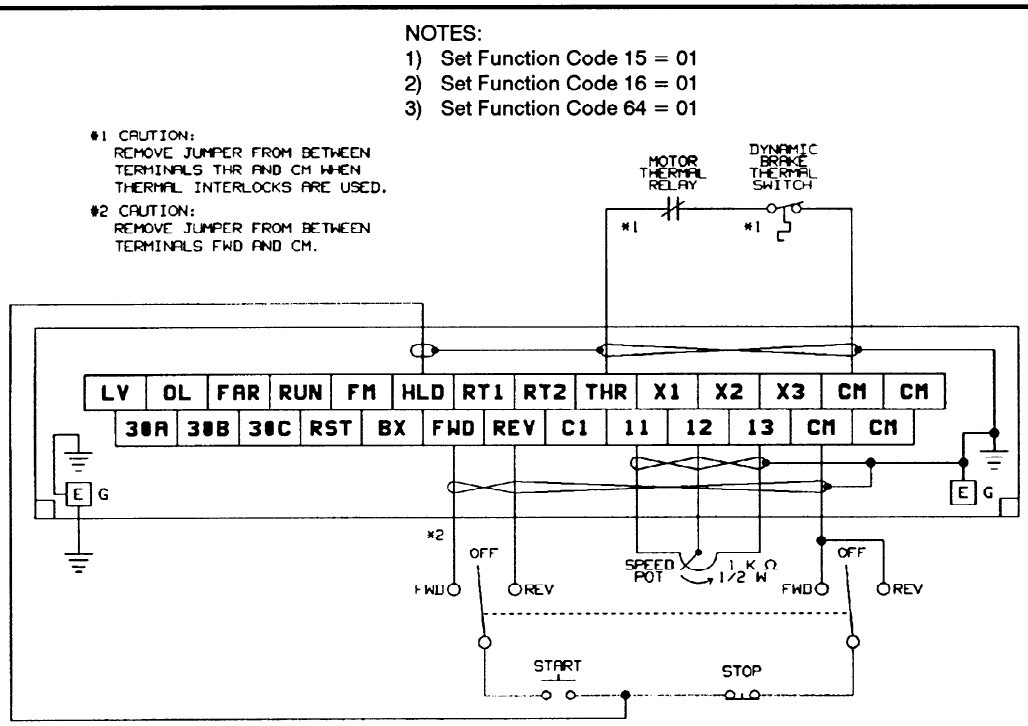


Figure 4-5. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS, REVERSING CAPABILITY, AND SPEED REFERENCE FROM SPEED POTENTIOMETER

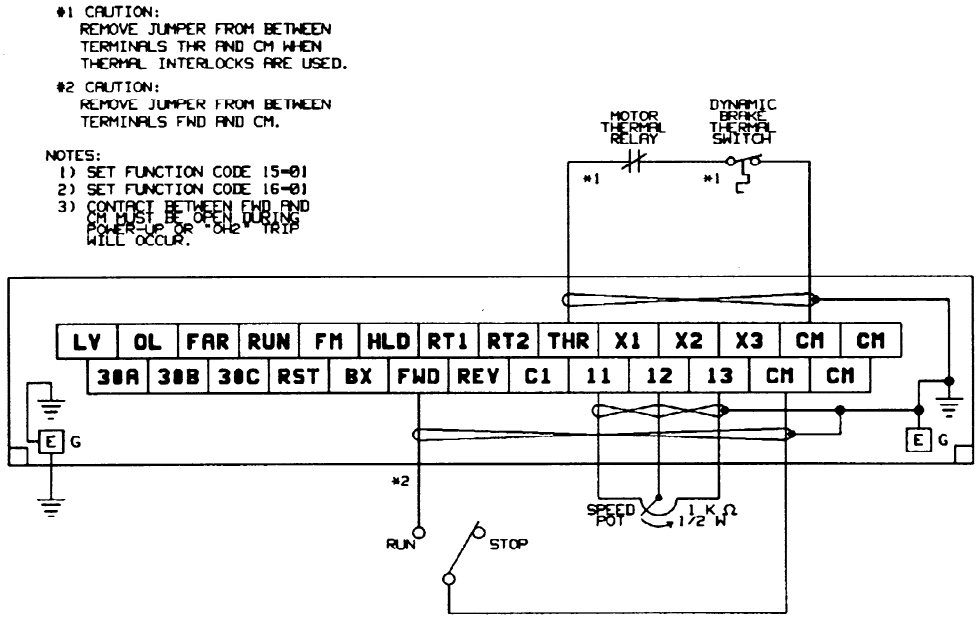


Figure 4-6. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS AND SPEED REFERENCE FROM SPEED POTENTIOMETER

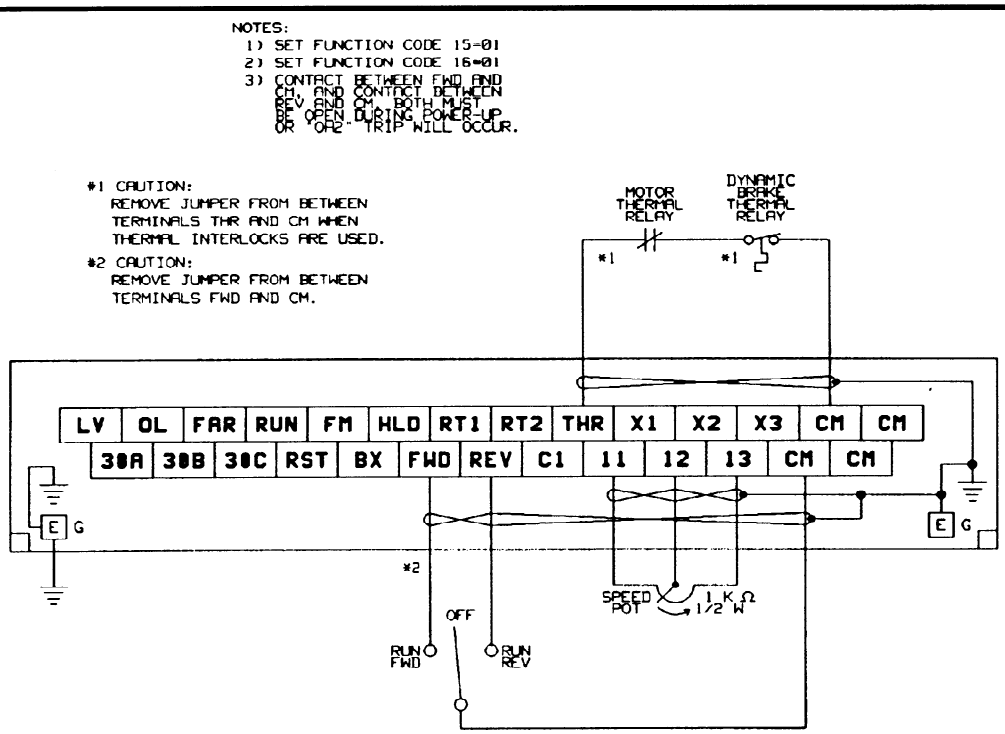


Figure 4-7. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS, REVERSING CAPABILITY, AND SPEED REFERENCE FROM SPEED POTENTIOMETER

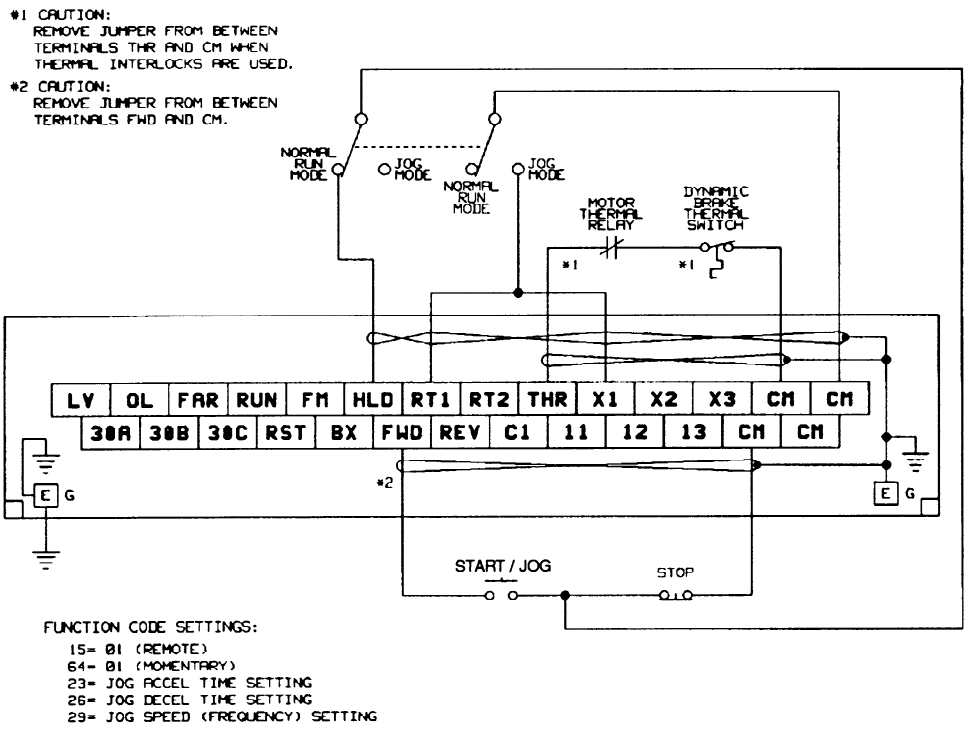


Figure 4-8. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND JOG FUNCTION

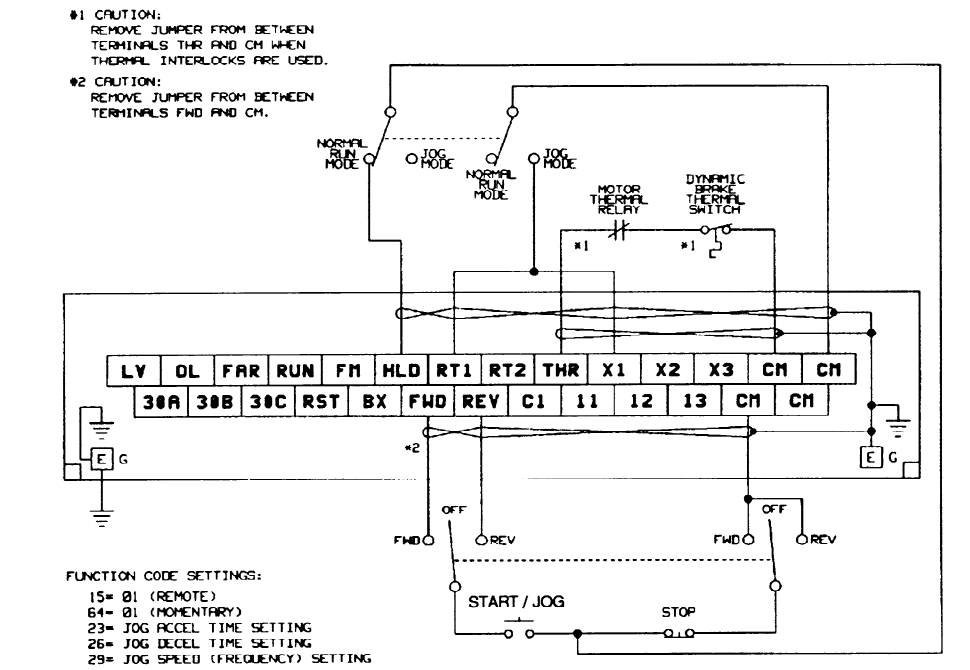


Figure 4-9. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS, JOG FUNCTION , AND REVERSING CAPABILITY

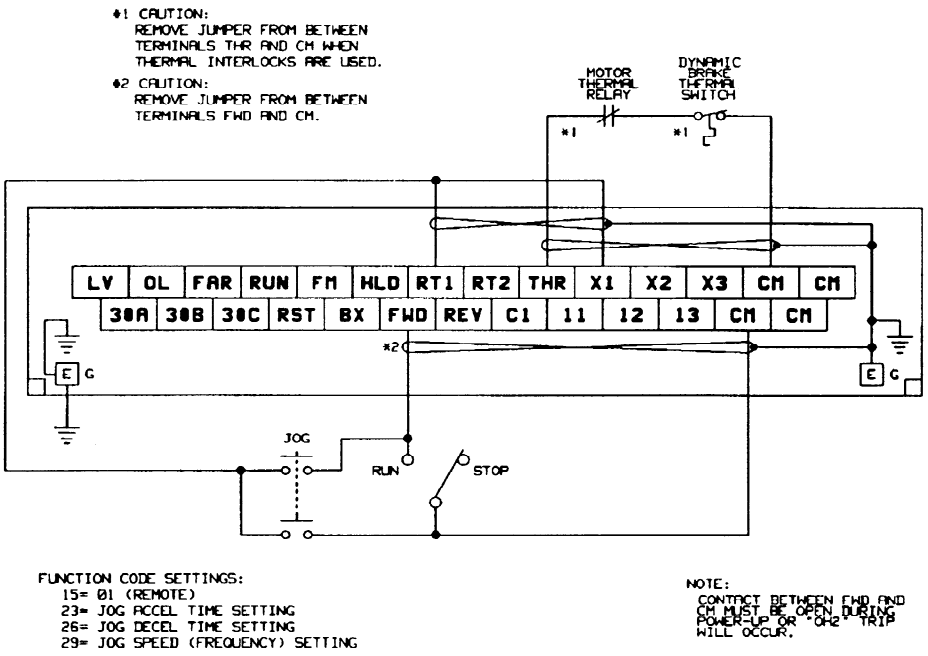


Figure 4-10. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS AND JOG FUNCTION

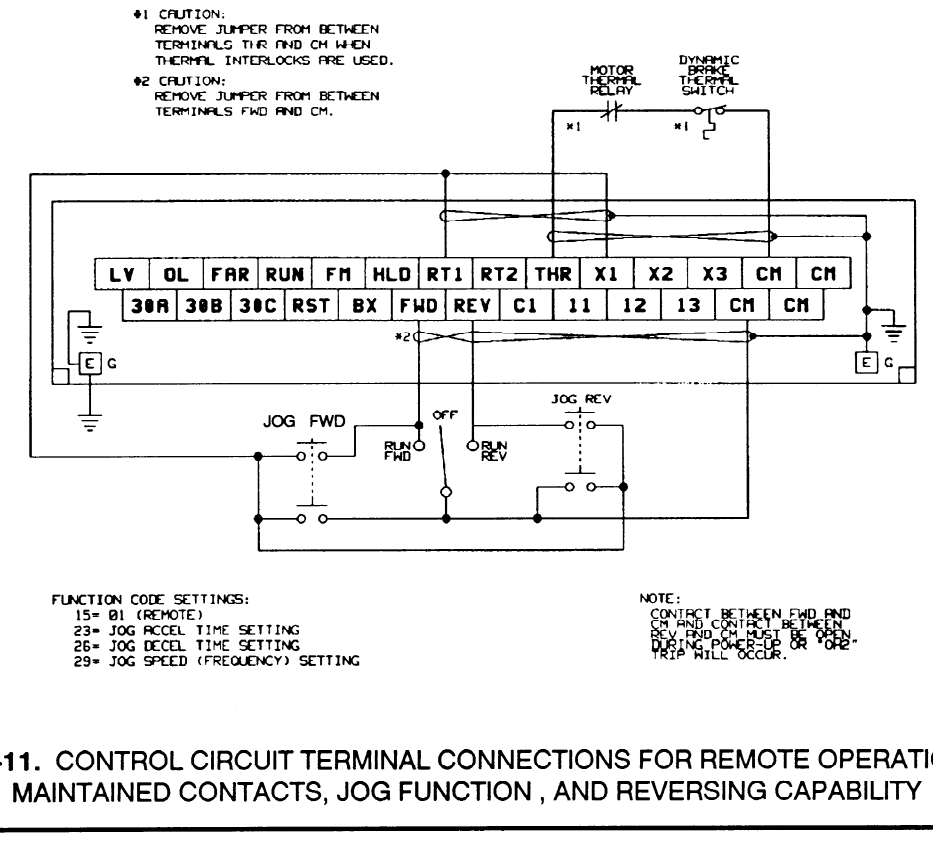


Figure 4-11. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS, JOG FUNCTION , AND REVERSING CAPABILITY

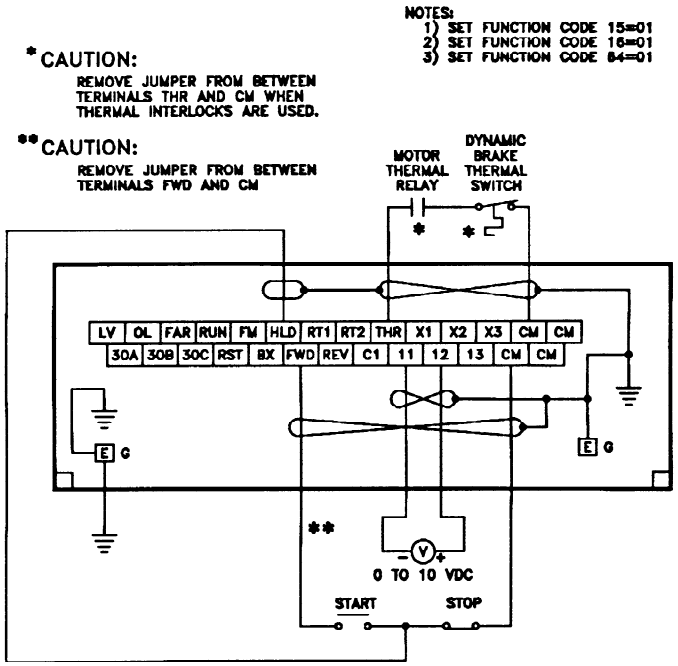


Figure 4-12. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND SPEED REFERENCE FROM 0 TO 10 VDC SIGNAL

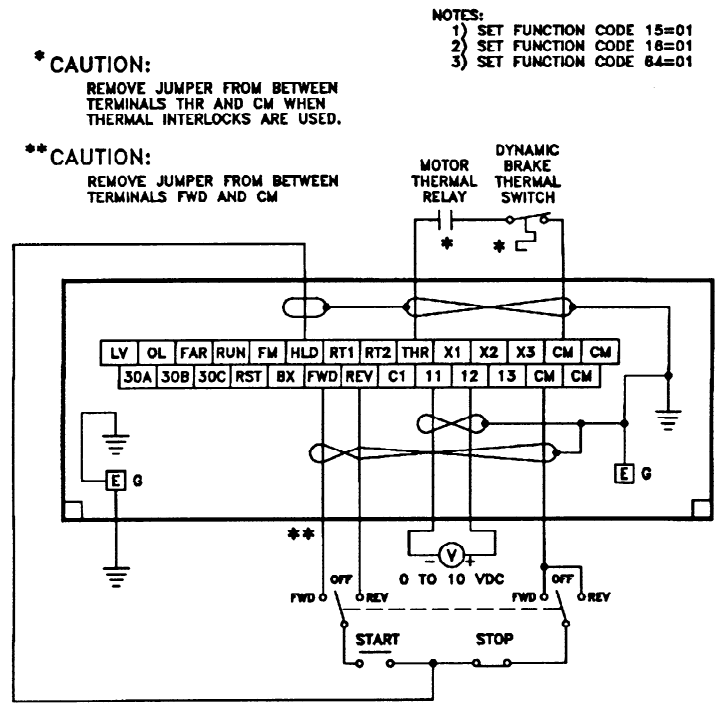


Figure 4-13. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS, REVERSING CAPABILITY, AND SPEED REFERENCE FROM 0 TO 10 VDC SIGNAL

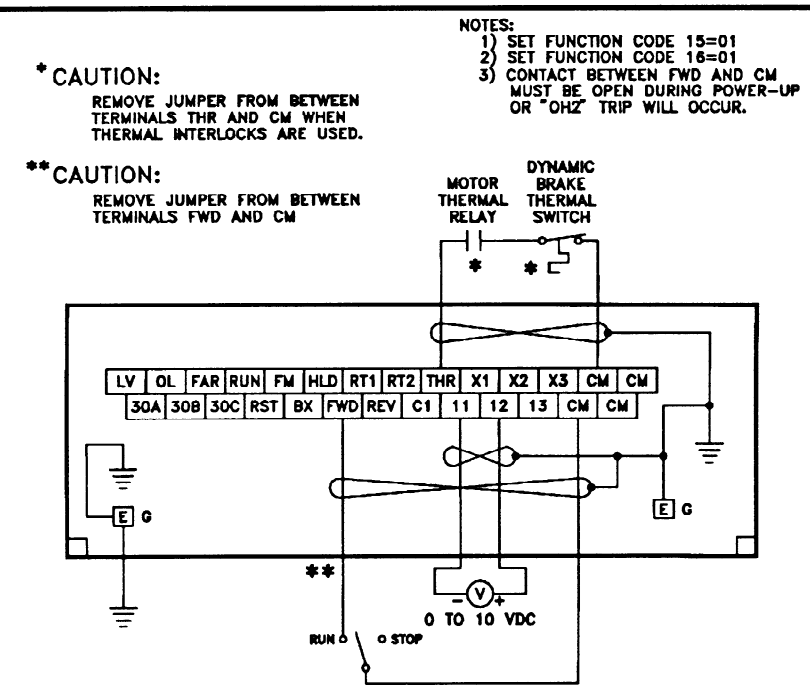


Figure 4-14. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS AND SPEED REFERENCE FROM 0 TO 10 VDC SIGNAL

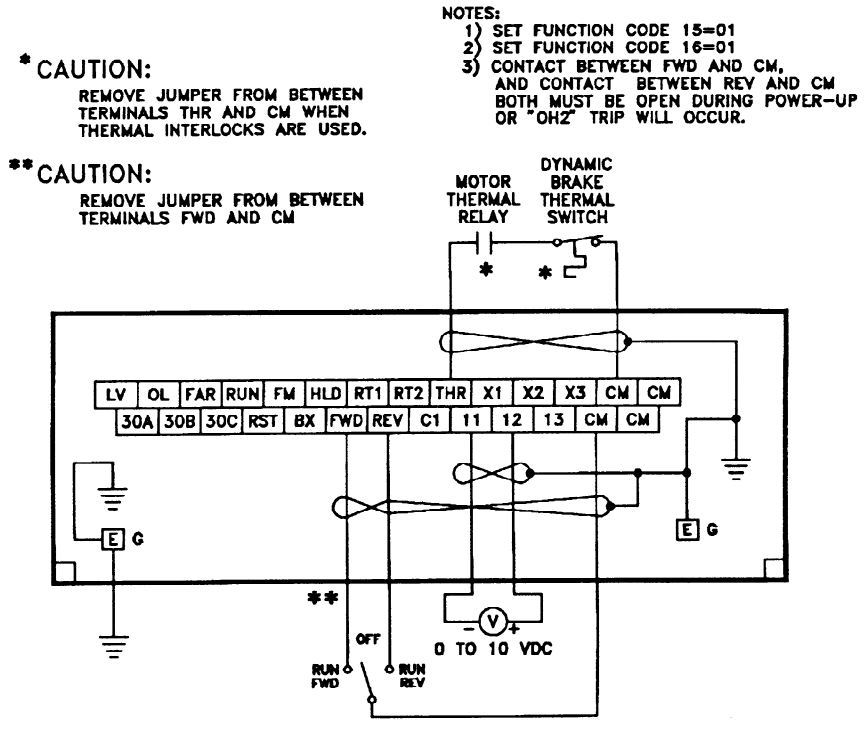


Figure 4-15. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS, REVERSING CAPABILITY, AND SPEED REFERENCE FROM 0 TO 10 VDC SIGNAL

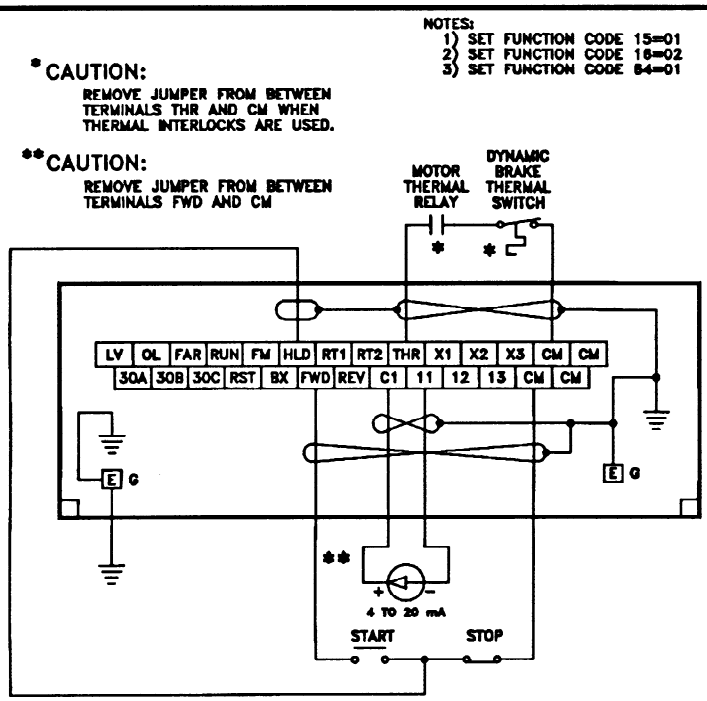


Figure 4-16. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND SPEED REFERENCE FROM 4 TO 20 mA SIGNAL

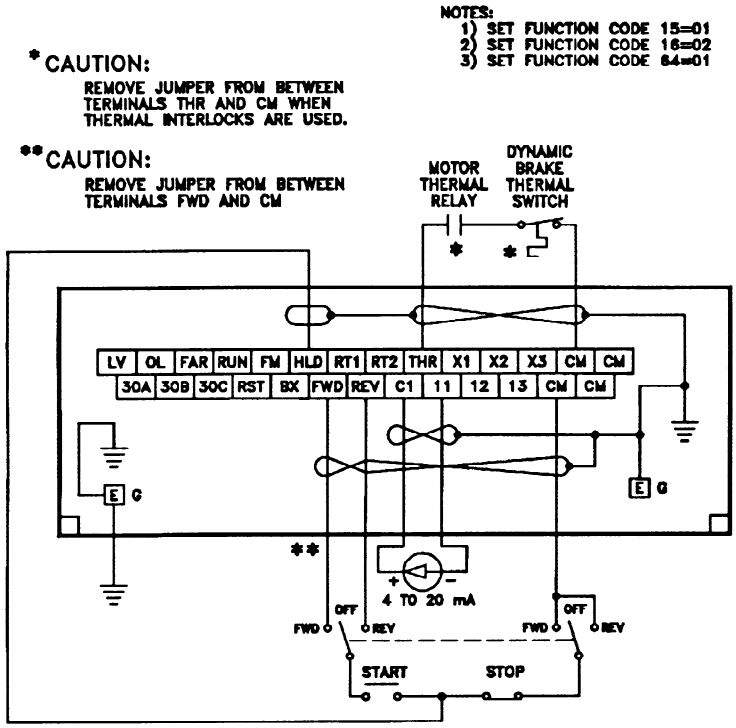


Figure 4-17. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS, REVERSING CAPABILITY, AND SPEED REFERENCE FROM 4 TO 20 mA SIGNAL

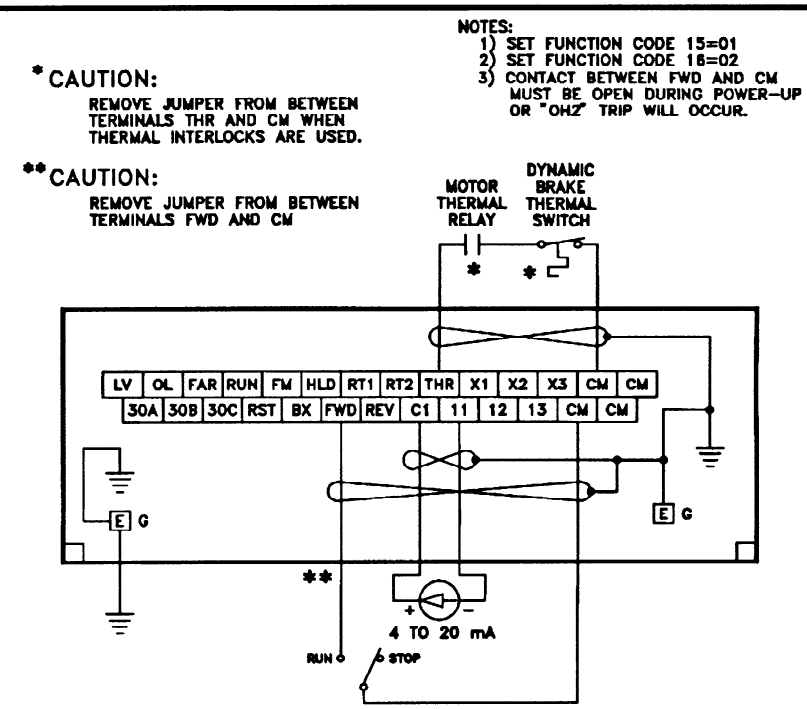


Figure 4-18. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS AND SPEED REFERENCE FROM 4 TO 20 mA SIGNAL

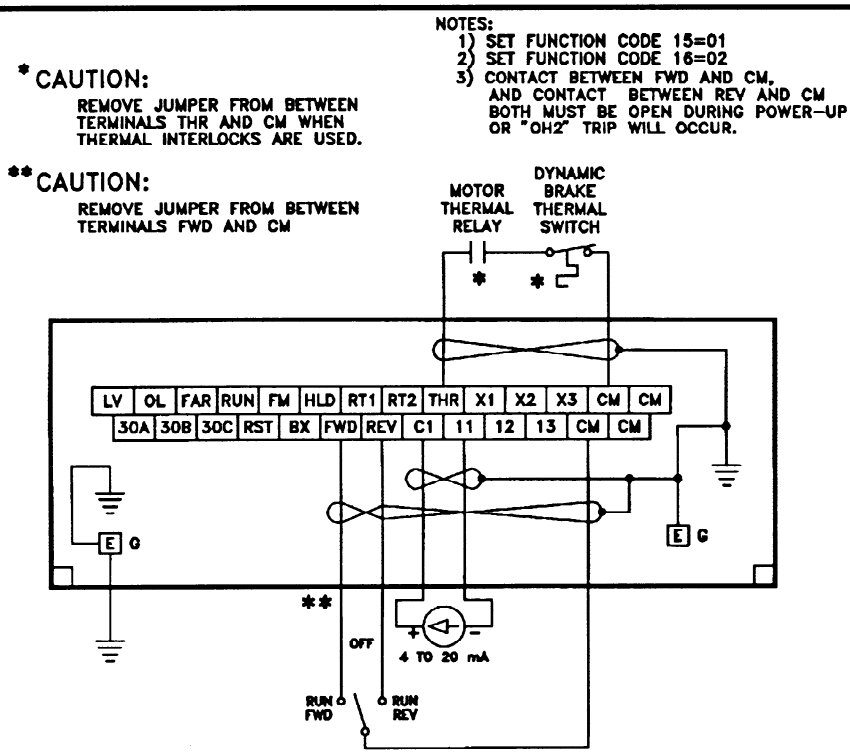


Figure 4-19. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS, REVERSING CAPABILITY, AND SPEED REFERENCE FROM 4 TO 20 mA SIGNAL

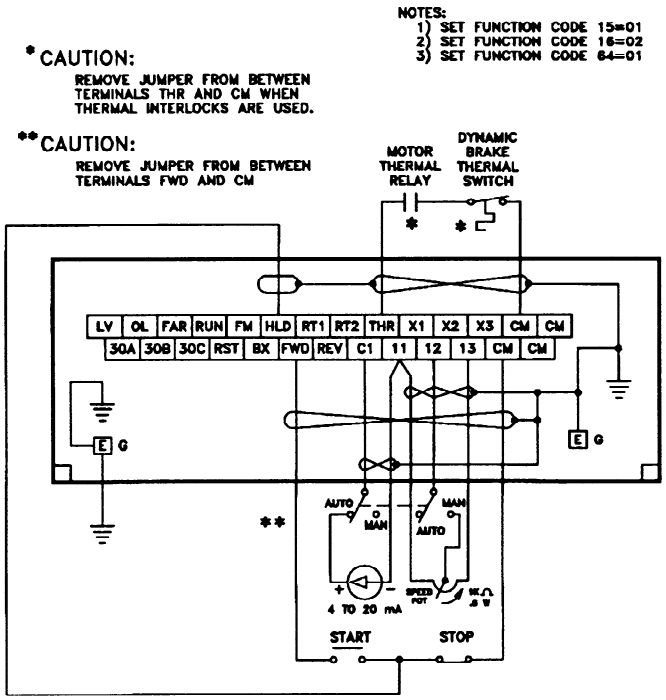


Figure 4-20. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND AUTO/MANUAL SELECTION

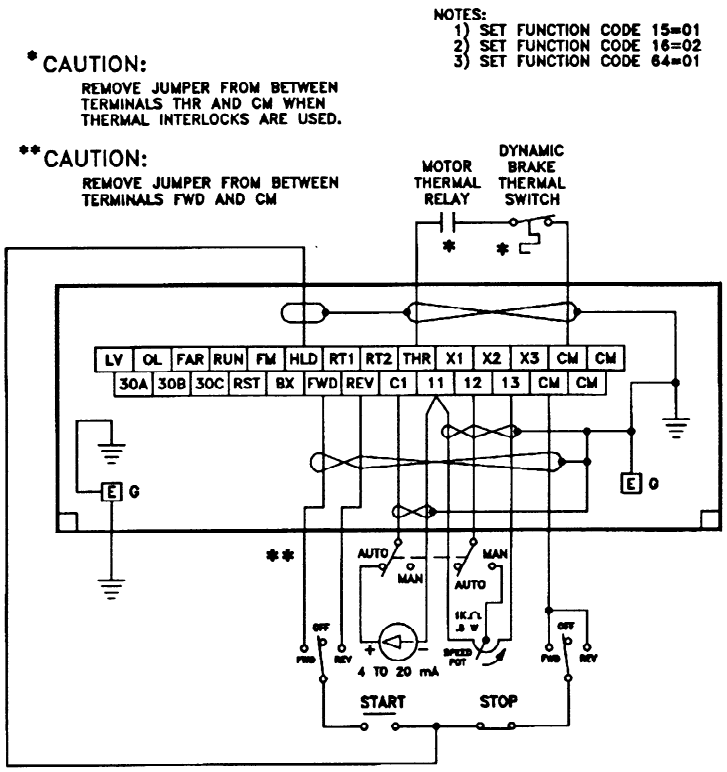


Figure 4-21. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS, REVERSING CAPABILITY, AND AUTO/MANUAL SELECTION

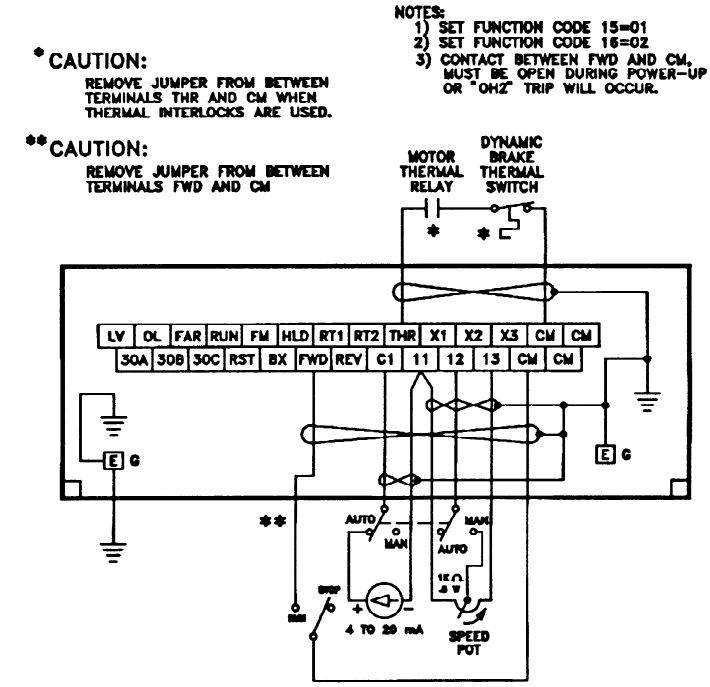


Figure 4-22. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS AND AUTO/MANUAL SELECTION

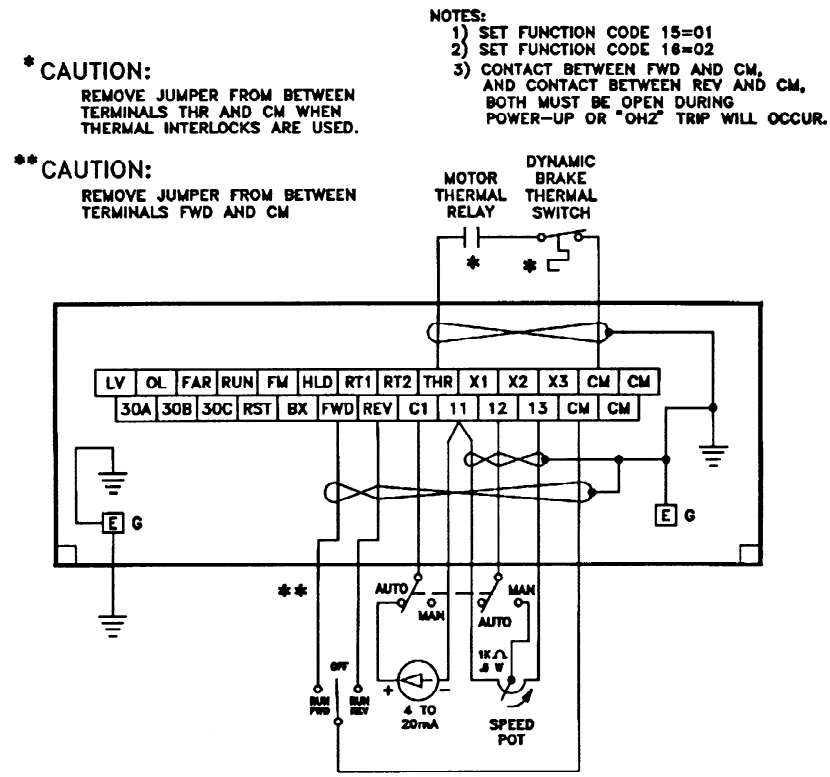


Figure 4-23. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MAINTAINED CONTACTS, REVERSING CAPABILITY, AND AUTO/MANUAL SELECTION

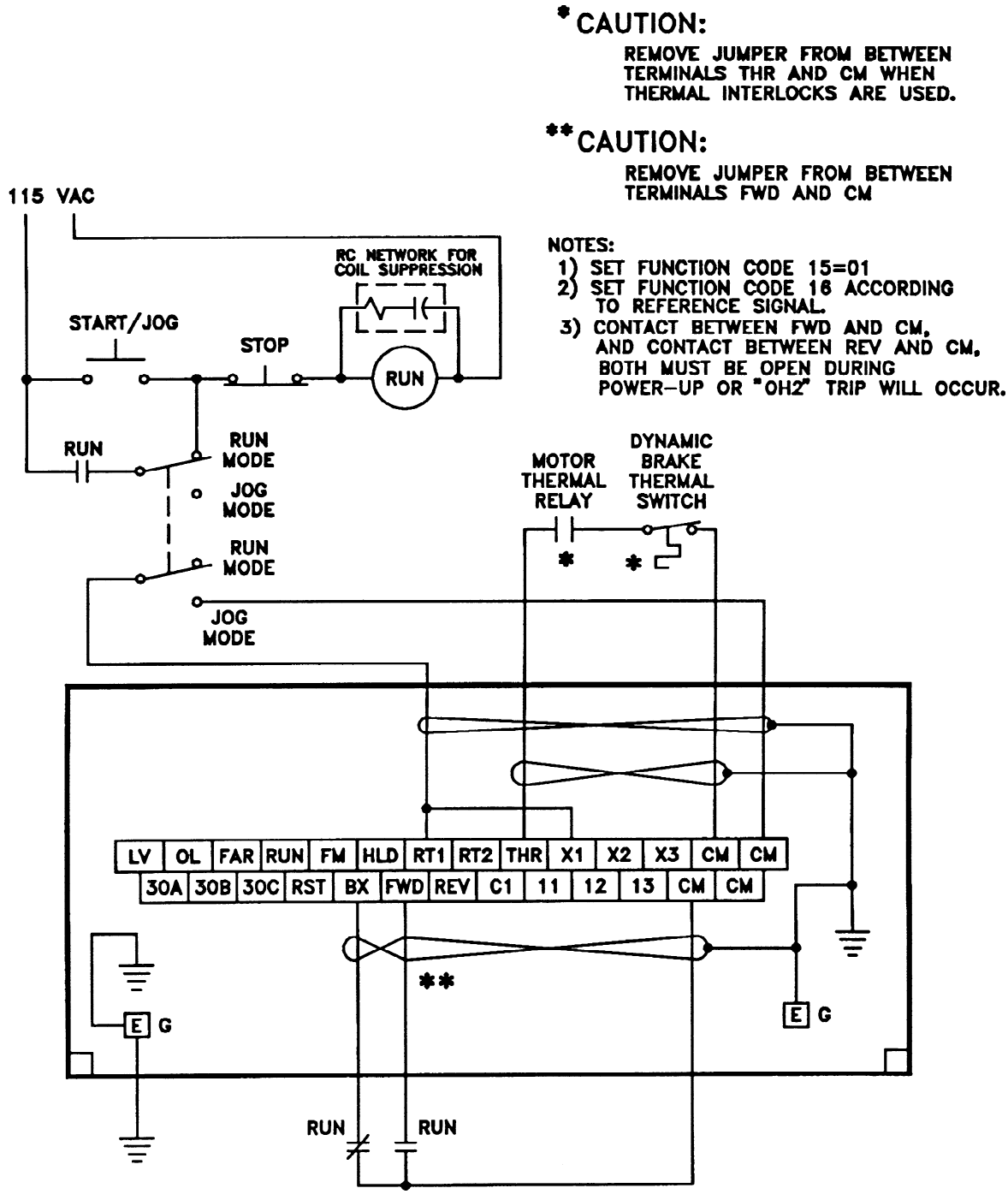


Figure 4-24. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND COAST STOP AFTER RUN OR JOG OPERATION

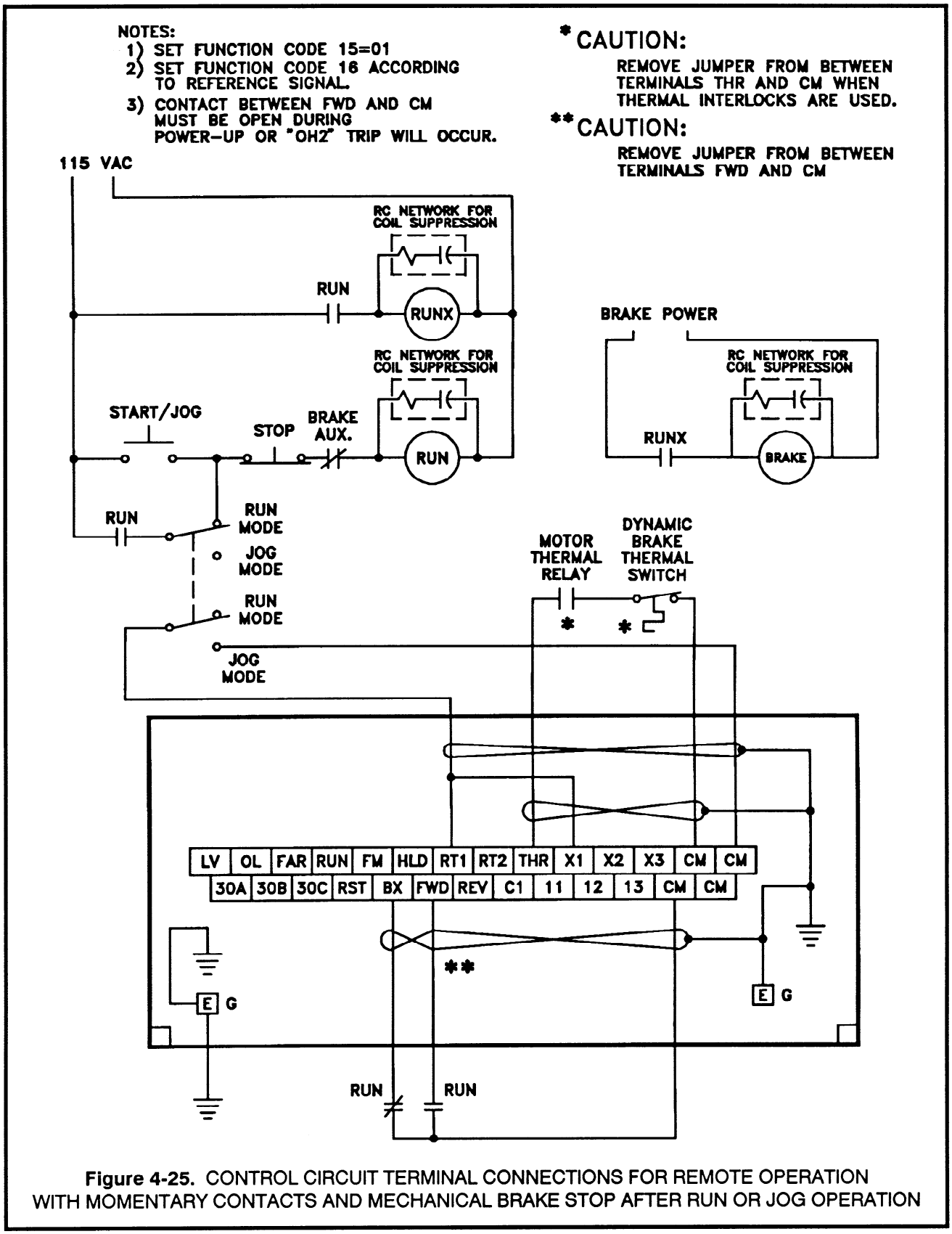


Figure 4-25. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR REMOTE OPERATION WITH MOMENTARY CONTACTS AND MECHANICAL BRAKE STOP AFTER RUN OR JOG OPERATION

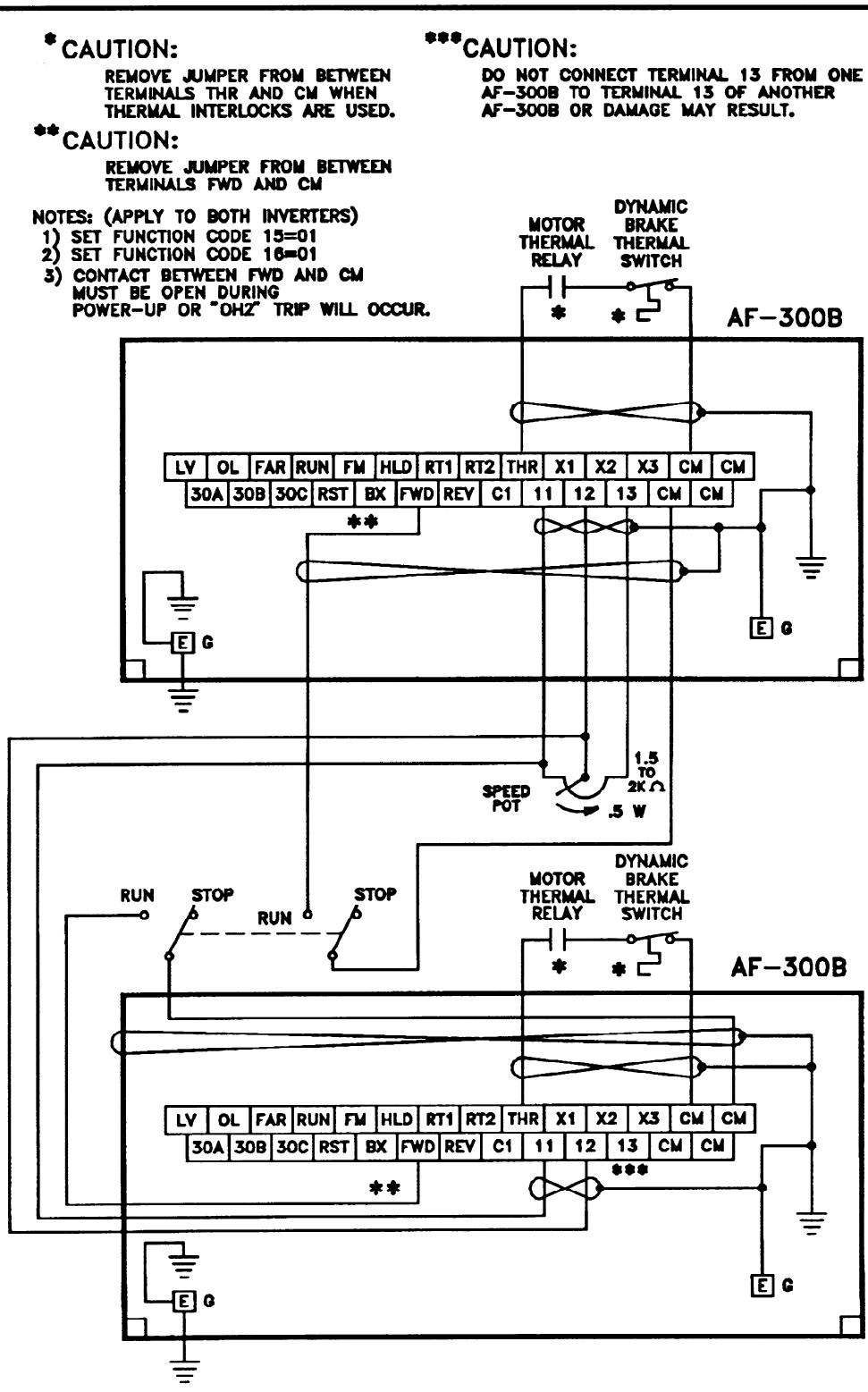


Figure 4-26. CONTROL CIRCUIT TERMINAL CONNECTIONS FOR RUNNING TWO AF-300B INVERTERS FROM ONE SPEED POTENTIOMETER

CAUTION: The control circuit terminal wiring should be kept as far away as possible from the main power wiring to prevent operational error due to noise interference. Never install both types of wiring in the same duct or conduit. A separation distance of 10 centimeters (4 inches) or more is recommended. If the control circuit wiring must cross the main power wiring, it should cross at a right angle.

CAUTION: Use shielded and twisted wire for the control circuit wiring. Wiring should be as short as possible, i.e. 20 meters (65 feet) or less. Connect shields to the inverter ground terminal and leave the other end disconnected and insulated from ground.

CAUTION: Install a suppressor in parallel with any relays, solenoid type coils, brake coils, etc. that are close to the inverter or wired to the inverter.

CAUTION: If the P-DB terminals or the P-N terminals are inadvertently short-circuited, damage to the inverter will result.

CAUTION: If dynamic braking is desired on inverters rated 11Kw (15 HP) and above, it is required that both a dynamic braking unit and a dynamic braking resistor be added. Do not attempt to connect an external braking resistor to inverters rated 11Kw (15 HP) and above without an external dynamic braking unit or damage will result.

CAUTION: When using an external braking resistor with inverters rated less than 7.5 KW (10 HP), first disconnect the inverter internal braking resistor wires from P and DB, then connect the external DB resistor to the P and DB terminals. The internal braking resistor wires that have been disconnected must be protected with insulation.

CAUTION: Do not apply an external voltage to THR or CM. Use dry contacts only for the external overload circuit.

WARNING - FIRE HAZARD: Use only dynamic braking resistors that are provided by GE Drive Systems. These are special resistors that have a thermal fuse built in that opens if the resistor gets too hot. This fuse should only open if the DB transistor in the inveter or DB unit fails short circuited. The resistor will need to be replaced if this happens.

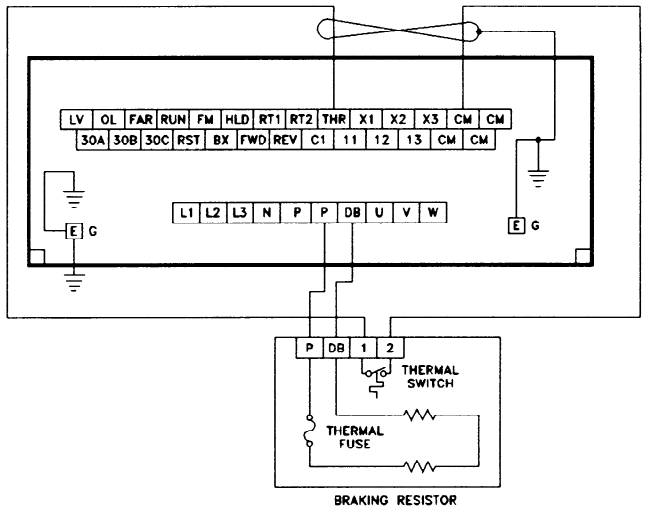
OPTIONAL EXTERNAL DYNAMIC BRAKING RESISTOR UNIT WIRING

NOTE: *Remove the factory installed jumper from the CM - THR terminals. If the jumper is not removed, the OH2 alarm will not function during operation.*

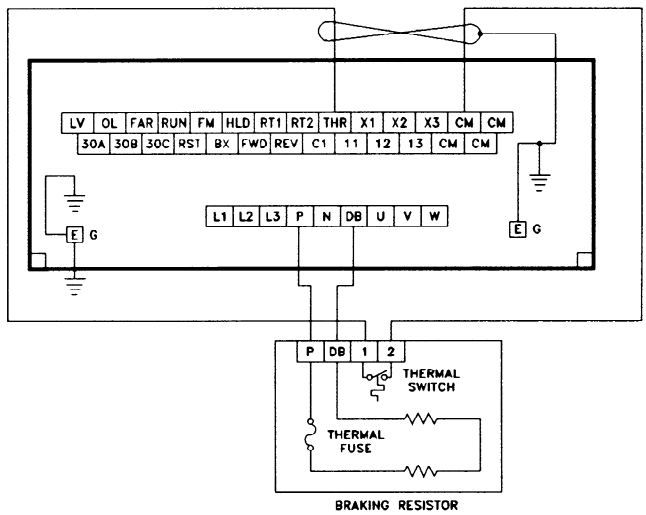
When frequent braking or high torque braking is required, connect the optional braking resistor as shown in the appropriate view of Figure 4-27.

- Option 1: Dynamic Braking (DB) resistor (0.4 to 7.5 KW; 0.5 to 10 HP).
- Option 2: Braking Unit and Dynamic Braking (DB) resistor (11 to 22 KW; 15 to 30 HP).

UNITS UP TO 5 HP



UNITS FROM 7.5 TO 10 HP



UNITS FROM 15 TO 30 HP

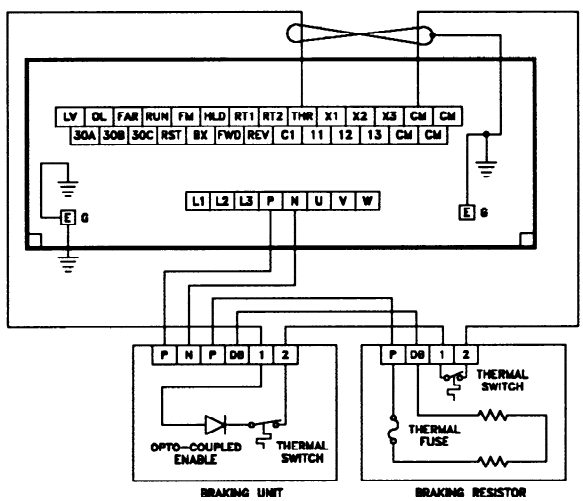


Figure 4-27. OPTIONAL EXTERNAL DYNAMIC BRAKING RESISTOR UNIT WIRING CONNECTIONS

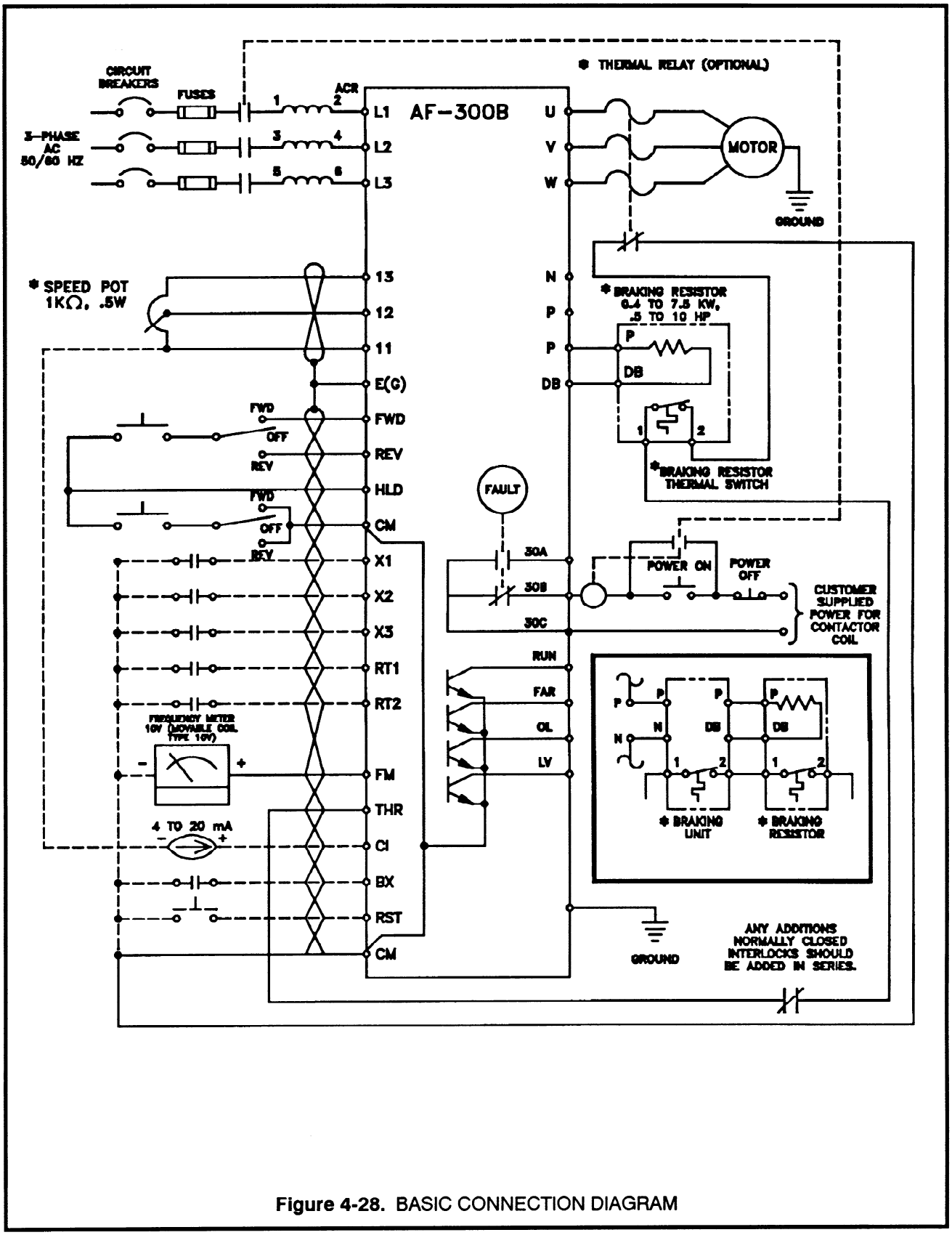


Figure 4-28. BASIC CONNECTION DIAGRAM

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION

Terminal Label	Terminal Name	Function
MAIN CIRCUIT:		
L1,L2, L3	AC Supply Line Input Terminals	Connection for commercial power (200-230 VAC or 380-460 VAC).
U,V,W	Inverter Output Terminals	Connection for 3-phase induction motor.
P,DB	External Braking Resistor Terminals	Connection for external braking resistor. (Only on 7.5 Kw [10 HP] or less.)
N,P	External Braking Unit Terminals	Connection for external braking resistor via external braking unit. (Braking Unit only required on 11 Kw [15 HP] and greater.)
E (G)	Ground Terminal	Connection for ground.
CONTROL INPUT:		
11	Frequency Setting Common Terminal	Voltage setting and current setting common terminal. (Do not connect to CM terminal if possible or some electrical noise immunity may be lost.)
12	Frequency Setting Voltage Input	When 0 to +10 VDC (0 to 5V) is input, the maximum frequency is reached at +10 VDC (5V) and is proportional to output frequency down to 0 VDC. Input impedance is 22K ohm.
13	Frequency Setting Voltage Output Term.	+10 VDC power supply, 10mA or less (to terminal 11). (For use with frequency (speed) setting potentiometer.)
C1	Frequency Setting Current Input	When 4 to 20mA DC is input, the maximum frequency is reached at 20mA and is proportional down to 4mA. Input Impedence = 250 ohms between terminals C1 and 11.
CM	Control Circuit Common Terminal	Common terminal for control input/output commands. (Do not connect to terminal 11 if possible or some electrical noise immunity may be lost.)
FWD **REV**	Forward Command Input Terminal Reverse Command Input Terminal	Forward command via FWD-CM (closed). Reverse command via REV-CM (closed). When FWD-CM is closed and REV-CM is closed at the same time, the inverter will decelerate to stop.
BX	Motor Coast-To-Stop Input Terminal	Motor will coast-to-stop with BX-CM (closed). (For use when applying machanical brake with inverter in operation.)
HLD	3-wire Operation - Stop Command (Operation From Momentary Contacts)	When 3-wire operation is selected and HLD-CM is closed, the pulse signal input from FWD, REV terminals is held internally. Operation from momentary pushbuttons is enabled (also set Function Code 64 to 01). See Figure 4-28 for Connection Diagram.

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION (continued)

Terminal Label	Terminal Name	Function
CONTROL INPUT: (cont.) **X1** **X2** **X3**	Multi-step speed operation command input terminals *or* Frequency Increase/Decrease Control *or* DC Injection Brake Control	X1, X2, and X3 can be used to select one of seven preset speeds (frequencies). These frequencies are adjustable via Function Codes 29, 31, 33, 35, 37, 39, and 41. (See Table 10.) **NOTE:** *When there is no input to X1, X2, or X3, frequency is set by Function Code 16. (Also see Function Code 63 in Table 10.)* X1 increases frequency, X2 decreases frequency. See Function Code 63 in Table 10. X1 turns DC Injection Braking ON. See Function Code 63 in Table 10.
RT1 **RT2**	Multi-step accel/ decel time command input terminals	RT1 and RT2 can be used to select one of four ACCEL/DECEL groups. ACCEL and DECEL rates are independantly adjustable via Function Codes 08, 09, and 23 through 28. See Table 10. **NOTE:** *When there is no input to RT1 and RT2, acceleration/ deceleration time is set by Function Codes 08 and 09.*
THR	External thermal relay, external braking resistor thermostat terminal	With THR-CM (open), inverter will trip on OH2 and motor will coast-to-stop. **NOTE:** *With no external thermal relay or external braking resistor thermostat, the THR-CM terminals must be closed or the inverter will not operate.* **NOTE:** *When using the external DB Unit, it is important to keep the polarity of terminals 1 and 2 on the DB Unit consistant with the drawings in Section 4 of this instruction.*
RST	Reset signal input terminal	The fault is reset when the RST-CM (closed) terminals are short-circuited for more than 0.1 seconds. **NOTE:** *RUN command must be removed in order to reset the inverter.*
FM	Frequency meter connection terminal	0 to 10 VDC output - Voltage is proportional to output frequency, 10 VDC = maximum frequency. DC Voltage Meter (0-10 Volt) - Internal impedance over 10K ohm. DC Current Meter (1mA full scale) - 10K ohm, 1/2 watt resistor should be wired in series. **NOTE:** *See Table 10, Function Code 70.*

TABLE 2: TERMINAL IDENTIFICATION/FUNCTION (continued)

Terminal Label	Terminal Name	Function
CONTROL OUTPUT:		
RUN (See Notes 1 and 2)	Inverter running signal output terminal	RUN-CM turns ON when output frequency is greater than the frequency set in Function Code 61. The function of this terminal can be changed via Function Code 69. **Open Collector Output:** Maximum externally applied voltage = +27 VDC Maximum current sinking capability = 50 mA DC Maximum leakage in OFF state = 10 uA
FAR (See Notes 1 and 2)	Frequency equivalence detection signal output terminal. (Up-To-Speed or Frequency at Reference)	When the set frequency is reached, FAR-CM will turn ON. **Open Collector Output:** Maximum externally applied voltage = +27 VDC Maximum current sinking capability = 50 mA DC Maximum leakage in OFF state = 10 uA **NOTE:** *Also see Function Codes 65 and 68.*
OL (See Notes 1 and 2)	Inverter overload early warning signal output terminal	**CAUTION:** Function Codes 62, 65, and 67 must be set properly to ensure proper overload early warning signal operation. OL-CM turn ON when the inverter output current excedes the current level set in Function Code 62 for longer than 10 seconds. The function of this terminal can be changed via Function Codes 65 and 67. **Open Collector Output:** Maximum externally applied voltage = +27 VDC Maximum current sinking capability = 50 mA DC Maximum leakage in OFF state = 10 uA
LV (See Notes 1 and 2)	Undervoltage signal output terminal	LV-CM turns ON when the DC link voltage drops below the undervoltage detection level. The function of this terminal can be changed via Function Codes 65 and 66. **Open Collector Output:** Maximum externally applied voltage = +27 VDC Maximum current sinking capability = 50 mA DC Maximum leakage in OFF state = 10 uA
30A **30B** **30C**	Fault Relay Output Terminals	During normal operation, the relay is not energized and contact is made between 30B and 30C. When a fault is detected, the relay is energized and contact is made between 30A and 30C. (Contact rating: 250 VAC, 0.3 Amps @ cos *theta* $\geq$ 0.3)

NOTE 1:
RUN, FAR, OL, and LV are open collector outputs. The relay option or an external DC power supply and relays are required if a contact output is desired.

NOTE 2:
The RUN, FAR, OL, and LV terminals are open collector type outputs. This means that external power is required to pick up a relay or power a light. See applicable drawings in Section 4 for recommended wiring.

TABLE 3: APPLICATION OF WIRING AND EQUIPMENT

ITEM	SPECIFICATION										
Inverter (200-230V) HP (400-460V)	0.5 --	1.0 1.0	2.0 2.0	3.0 3.0	5.0 5.0	7.5 7.5	10 10	15 15	20 20	25 25	30 30
Standard Applicable Motor Output-230V (KW)	0.4	0.75	1.5	2.2	3.7	5.5	7.5	11	15	18.5	22
Standard Applicable Motor Output-460V (KW)	--	0.75	1.5	2.2	3.7	5.5	7.5	11	15	18.5	22
Rated Capacity-230V (KVA) (*1)	1.1	1.9	3.0	4.2	6.5	9.5	13	18	22	28	33
Rated Capacity-460V (KVA) (*1)	--	2.0	3.0	4.4	7.2	10	14	19	24	31	36
Applicable Wire Sizes (mm^2) (*2)	3.5	3.5	5.5	5.5	5.5	8	14	14	22	30	38
Main Circuit Wiring (230 V)	14	14	14	14	12	8	6	4	2	2	2
Main Circuit Wiring (460 V)	--	14	14	14	14	14	12	10	8	8	4
Control Circuit Wiring	20 (twisted, shielded)										

(*1) Indicates rated capacity when rated output voltage is 230 VAC.
(*2) Standard wire is 600V vinyl wire.

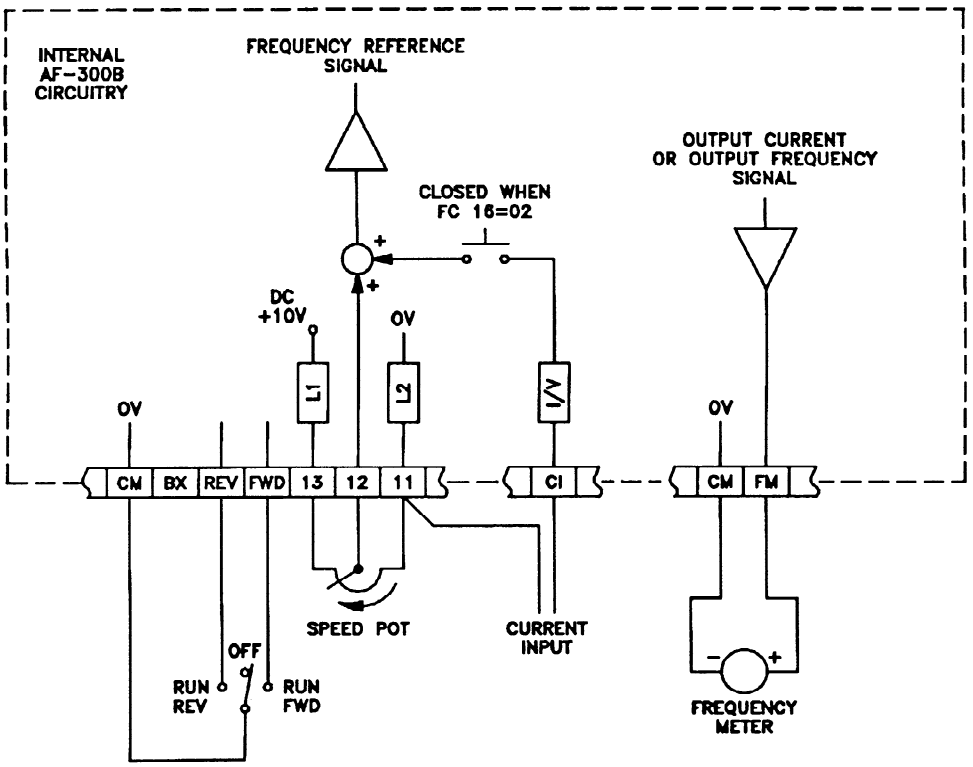


Figure 4-29. FREQUENCY SETTING/MONITORING TERMINAL CONNECTIONS

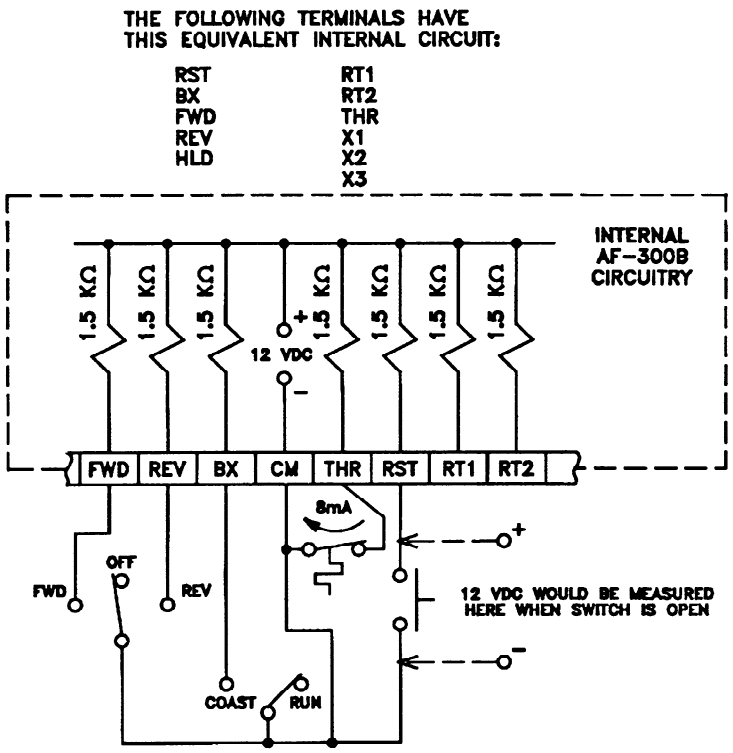
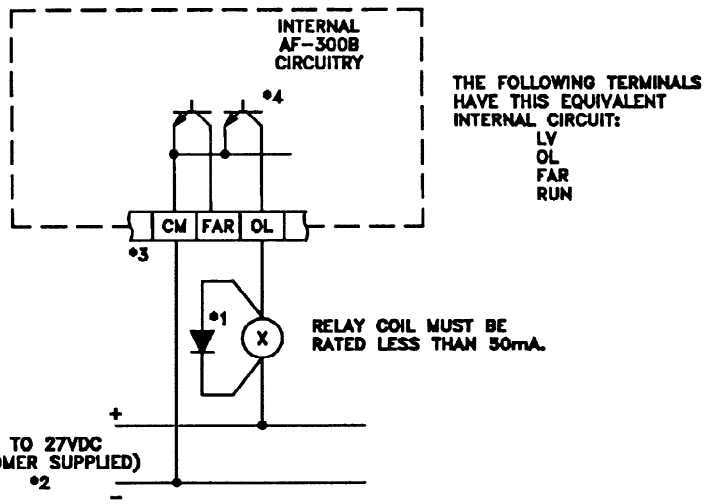


Figure 4-30. EQUIVALENT CONTROL CIRCUITS FOR AF-300B



CAUTION:

•1) REVERSE BIASED DIODE MUST BE INSTALLED FOR COIL SUPPRESSION OR OPEN COLLECTOR OUTPUT TRANSISTOR MAY BE DAMAGED.

NOTE:

•2) CUSTOMER SUPPLIED POWER SUPPLY FOR RELAY SHOULD BE ISOLATED FROM GROUND IF POSSIBLE. (THIS RECOMMENDATION WILL GIVE THE INVERTER A HIGHER LEVEL OF NOISE IMMUNITY IN ELECTRICALLY NOISY ENVIRONMENTS.)

•3) USE TWISTED/SHIELDED WIRE.

•4) THESE TRANSISTORS ARE RATED AT 27VDC MAXIMUM APPLIED VOLTAGE AND 50mA MAXIMUM CURRENT SINKING

Figure 4-31. OPEN COLLECTOR OUTPUT CIRCUIT FOR AF-300B

5. INVERTER OPERATION

PRE-OPERATION INSPECTION

After mounting and wiring has been completed, check the inverter for the following items before applying AC power:

- -- Check for wiring errors (especially main circuit wiring).
- -- Verify that there are no wiring chips, screws, etc. remaining in the inverter.
- -- Check that all screw and terminal connections are tight.
- -- Verify that no exposed wire ends are touching other terminals.

If a test run is desired, run the inverter at a low frequency of around 5 Hertz. Conduct the test run in a safe manner and check for the following:

- -- Smooth rotation of motor.
- -- Correct direction of rotation.
- -- Any abnormal vibration or noise in the motor.
- -- Smooth speed increase and speed reduction.

CAUTION: Do not conduct megger tests between the inverter terminals or control circuit terminals. For megger testing method, refer to Section 7, Maintenance and Inspection.

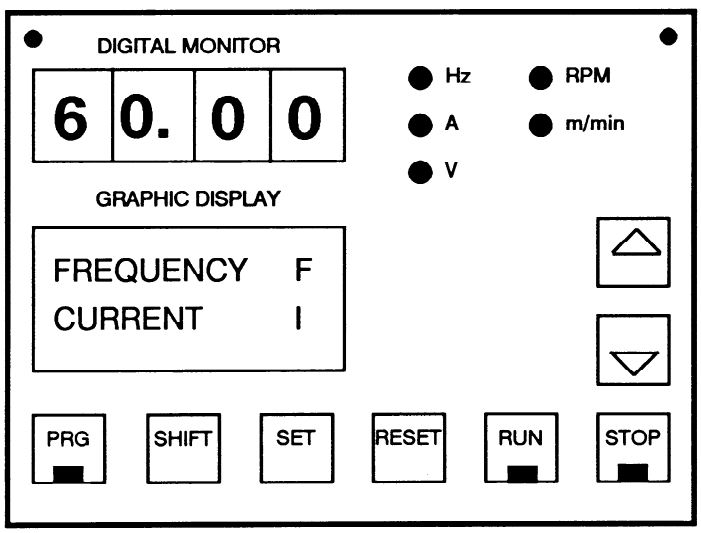


Figure 5-1. PANEL DISPLAY WHEN AC POWER IS APPLIED

KEYPAD PANEL IDENTIFICATION/ OPERATION

See TABLE 4 for Display and Keypad Operation description when in **Normal Operation**, **Program Setting Mode**, and **Trip Mode**. Refer to Figure 5-2 for identification of the components of the Keypad Panel.

FUNCTION CODE AND DATA CODE DESCRIPTION/SELECTION

When AC power is applied to the inverter, the operation panel display will be as shown in Figure 5-1 and will be flashing on and off. If the **RUN** key is pressed at this point, operation will be at 60 Hertz according to the Function Code set at the factory. (Use the **STOP** key to halt operation.)

- -- A Flashing display indicates the set frequency when a run command is not present.
- -- A Solid display indicates the actual output frequency when the inverter is running.

NOTE: *Function Codes are divided into three different "Blocks" according to their degree of complexity. Set Data Code of Function Code* **[22]** *as follows in order to access the three Function Code "Blocks":*

- -- **[2200]** - *Basic Function Codes only* **[00]** *to* **[22]**.
- -- **[2201]** - *Standard Function Codes* **[23]** *to* **[62]** *(in addition to Basic Function Codes).*
- -- **[2202]** - *High Level Function Codes* **[63]** *to* **[82]** *(in addition to Basic and Standard Function Codes.*

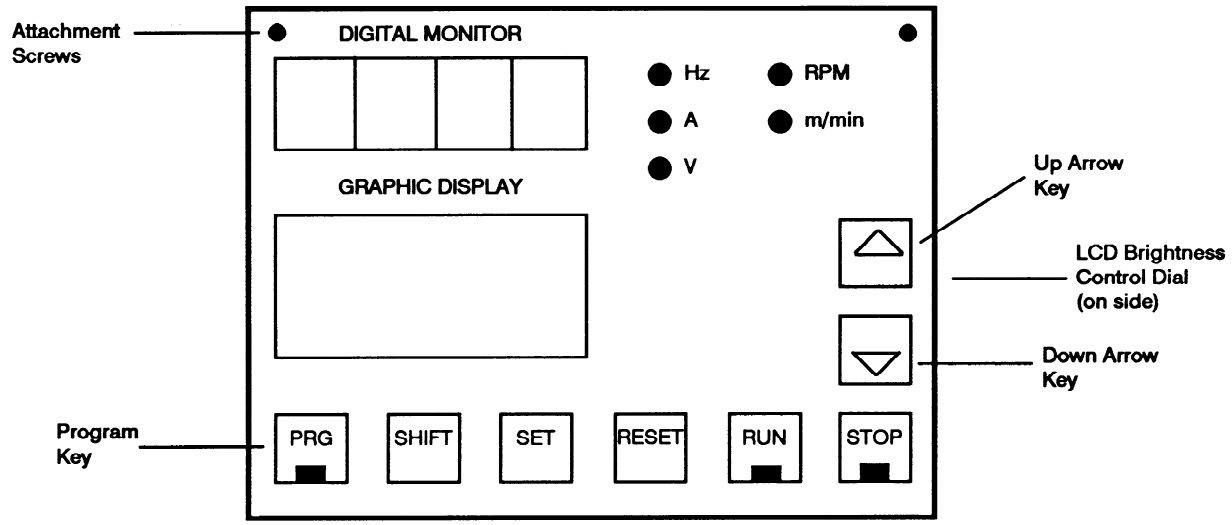


Figure 5-2. KEYPAD PANEL COMPONENT IDENTIFICATION

Attachment Screws -The Keypad Panel can be easily removed from the inverter unit by loosening the two attachment screws. With the optional extension cable, remote operation and display is possible.

Graphic Display - Frequency and output current are graphically displayed. The main input/output signal ON/OFF is also displayed. When in Program Setting Mode, the appropriate information is displayed graphically and alphabetically for easy Function Code selection setting.

Program (PGR) Key - Normal Mode or Program Setting Mode selection key. Key lights up when in Program Setting Mode.

SHIFT Key - When the Program key (**PRG**) is in Normal Mode, unit display can be changed while in either **RUN** or **STOP**. In the Program Setting Mode, this key allows Function Code selection by displaying each Function Code "column" in sequence. (See Figure 5-3 and TABLE 6.)

SET Key - Data display and settings for each Function Code are performed through this key. Also, when setting data on the graphic display, data accessed on the display can be changed.

RESET Key - Resets abnormal **STOP** conditions when Program key (**PRG**) is in the Normal Mode. Also changes from Data Setting Mode to Function Code Setting Mode when Program key (**PRG**) is in the Program Setting Mode.

RUN Key - Key is used for starting operation and lights up during operation. This key does not function when Data Code is set for external (control terminal) operation **[1501]** or serial link operation **[1502]**.

STOP Key - This key is used for stopping operation and will light up when in **STOP** Mode. If pressed when Data Code is set for external (control terminal) operation **[1501]** or serial link operation **[1502]** Fault OH2 will be displayed and inverter will coast stop.

Up - Down Keys - These keys increase or decrease the frequency (or speed) of the inverter. When unit is in Program Setting Mode, they change the Function Code or Data Code values.

LCD Brightness - This control permits adjustment for easy to read brightness.

Unit Display - The unit information is displayed by LEDs.

Digital Monitor - When setting the program, the Function Code is indicated by the two digits on the left and the Data Code corresponding to the Function Code is indicated by the two digits on the right. During operation it displays the set frequency, current, voltage, etc. If a protective **STOP** occurs, the causes of the problem will be displayed as a fault code (see TABLE 11 and TABLE 12).

TABLE 4: KEYPAD AND DISPLAY OPERATION

NORMAL OPERATION

0 **RUN** Frequency Setting Method (digital setting)

0 Function Setting
- RUN Mode (15)
- 00 or 01
- Frequency Command (16)
- 00

UP/DOWN Keys change operation frequency setting

	One Digit Shift to Left Resolution	Normal Resolution
Max. Frequency: To 60 Hz	0.002 Hz	0-10.0 Hz - 60.0 Hz 0.01 Hz 0.1Hz
Max. Frequency: 60 to 300 Hz	0.01 Hz	0-10.0 Hz - 100.0 Hz 0.01 Hz 0.1 Hz 1 Hz
Max. Frequency: 300 to 400 Hz	0.02 Hz	0-10.0 Hz - 100.0 Hz 0.02 Hz 0.1 Hz 1 Hz

SET Key for writing operation frequency (STOP Mode only). (Must be in frequency display mode.)

0 DIGITAL MONITOR
0 Function Setting
- Digital Monitor (00)
- 00: Frequency
- 01: Output Current
- 02: Output Voltage
- 03: Synchronous RPM
- 04: Line Speed

SHIFT Key: Frequency Display (flashes during STANDARD/STOP Mode)
↓
SHIFT Key: Frequency Display (flashes during one digit shift and STOP Mode)
↓
SHIFT Key: Output Current Display
↓
SHIFT Key: Output Voltage Display
↓
SHIFT Key: Synchronous RPM Display
↓
SHIFT Key: Line Speed Display

0 GRAPHIC MONITOR
0 Function Setting
Graphic Monitor (01)

00	Output frequency and output current level display (10 levels) FREQUENCY F ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ CURRENT C ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
01	Control Terminal Block Display 1 ■ FWD ■ RT1 ■ REV ■ RT2 ■ HLD
02	Control Terminal Block Display 2 ■ LV ■ X1 ■ OL ■ X2 ■ FAR ■ X3 ■ RUN

PROGRAM MODE OPERATION

0 DIGITAL MONITOR

Function Code Display
- **SHIFT** Key: Advances Function Code to the code block containing the desired Function Code.
- **UP/DOWN** Keys: Selects desired Function Code.
  - **UP** Key: Function Code plus one.
  - **DOWN** Key: Function Code minus one.

TABLE 4: KEYPAD AND DISPLAY OPERATION (continued)

PROGRAM MODE **(continued)** 0 DIGITAL MONITOR (continued)	Data Display Mode **SET** Key: Data Readout **UP/DOWN** Keys: Data Change (update) **UP** Key: Data plus one. **DOWN** Key: Data minus one. **SET** Key: Data write (After data write, Do Not turn power supply OFF for 5 seconds.) **RESET** Key: End data change.
0 GRAPHIC MONITOR	Function Code Display Mode **SHIFT** Key: Advances Function Code to the code block containing the desired Function Code. **UP/DOWN** Key: Selects desired Function Code. **UP** Key: Moves cursor up. **DOWN** Key: Moves cursor down.
	Data Display Mode **SET** Key: Data Readout **UP/DOWN** Keys: Data Change (update) **UP** Key: Data plus one. **DOWN** Key: Data minus one. **SET** Key: Data write (After data write, Do Not turn power supply OFF for 5 seconds.) **RESET** Key: End data change.
TRIP MODE 0 DIGITAL MONITOR	Trip Data Display OC1, 0C2, 0C3, OU, LU, OH1, OH2, OL1, Err1, Err2, Err3, Err4 **RESET** Key: Trip reset.
0 GRAPHIC MONITOR	Trip Condition Display F = XXX.X Hz I = X.XA V = XXXV Temperature* = XX°C Trip Record Display 0 = LV (current) -1 = OV (1 time before) -2 = OC1 (2 times before) -3 = OH2 (3 times before)

**Temperature is displayed only during Thermal Trips (OH1, OH2). No temperature is displayed on OL1, OC1, OU, LU, ERR3.*

Function Code Selection - Press the **PRG** key to select Program Setting Mode (see Figure 5-3).

-- Each time the **SHIFT** key is pressed, it changes the Function Code in the direction of the arrows shown in Figure 5-3 (**Example: [00] > [04] > [08]**).

-- Each time the **UP/DOWN** keys are pressed, it changes the Function Code in the vertical direction as shown in Figure 5-3 (**Example: [00] <> [01] <> [02]**).

Data Code Selection - After selecting the Function Code, press the **SET** key and the **UP/DOWN** keys will change the Data Code.

-- Some data is not shown and is selected on the Graphic Display.

-- Selected data is written by the **SET** key.

-- The **RESET** key terminates Data Code Setting Mode and returns to the Function Code Setting Mode.

FUNCTION CODE AND DATA CODE SETTING PROCEDURES

In order that the inverter (including the motor) may operate under optimum conditions, in addition to the codes mentioned previously (**[15]** and **[16]**), other setting changes may be required. This paragraph is a general explanation of the code setting method.

NOTE: *Complete details for Function Code settings are shown in TABLE 9: Function Codes, and TABLE 10: Function Code Descriptions. Be sure to fully understand all the basic codes.*

Changing Code Settings In STOP Mode:

1. Verify operation signal (**RUN** key, FWD and REV terminals) is completely in **OFF** mode.
   -- The frequency that has been set will flash on the display - **[60.00]**.
2. Press **PRG** key to change to Function Code Setting Mode.
   -- Display will show **[00]**.
3. Press **SHIFT** key to advance to a different Function Code "column" (advances each time key is pressed) if desired (see Figure 5-3).
4. Press **UP/DOWN** keys to select Function Code to be changed within the same "column".
5. Press **SET** key to access existing Data Code for the selected Function Code.
   -- Display will show existing data.
6. Press **UP/DOWN** keys to adjust code or adjust actual number to the desired data.
   -- Display will show any adjustments as they are made.
7. Press **SET** key to store the data adjusted in step 6 (if **SET** key not pressed, changes will be ignored).
   -- Display will show last adjustment made.
   -- Do not turn power OFF for **5 seconds** after a data adjustment is made.
8. Press **RESET** key to return to Function Code Changing Mode.
9. **Within same "column"** - Press **UP/DOWN** keys to select a different Function Code and repeat steps 5 thru 8.

   In different "column" - Press **SHIFT** key to advance to a different Function Code column (advances each time key is pressed) and repeat steps 4 thru 8.
10. Press **PRG** key to end Function Setting Mode and return to **STOP** mode.

Changing Code Settings In RUN Mode:

Data Code setting of Function Codes **[00]** to **[10]**, **[73]** to **[75]**, and **[80]** to **[82]** is possible in the **RUN** Mode for inverters with serial numbers beginning with 29 and earlier. For inverters with serial numbers beginning with 2X and later, it is not possible to adjust Function Codes 0, 4, 6, or 7 while running.

-- Use same procedure as when in **STOP** Mode (see previous paragraph).

Verification of all Function Codes and Data Codes is also possible when in the **RUN** mode.

-- Use same procedure, except skip steps 6 and 7, as when in **STOP** Mode. (Inverter will not perform steps 6 and 7 if inadvertently called for.)

OPERATION METHOD SELECTION

The methods shown in TABLE 5 select the **RUN/STOP** Signal Transmission Method and the Frequency Setting Signal Transmission Method.

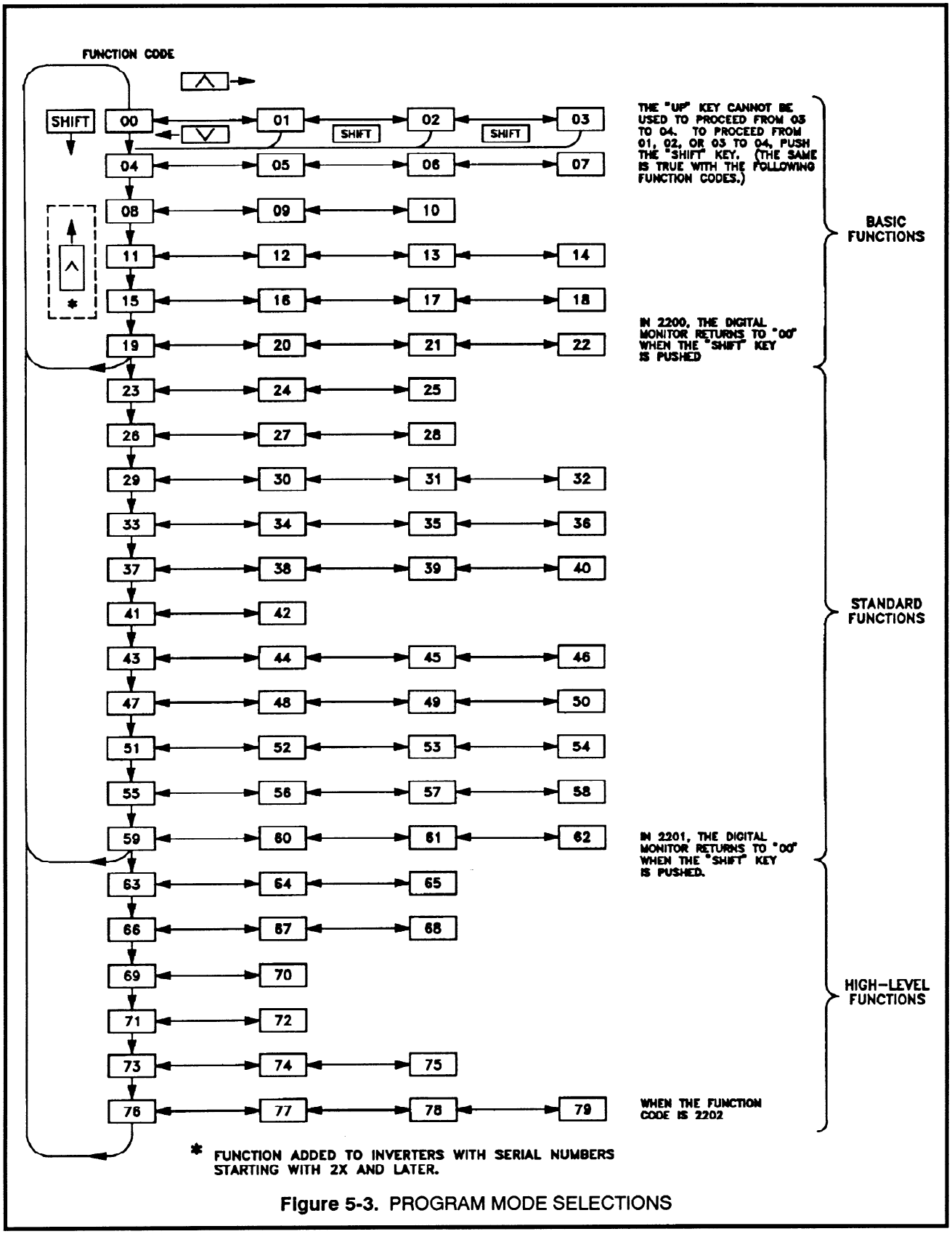


Figure 5-3. PROGRAM MODE SELECTIONS

TABLE 5: OPERATION METHODS

	RUN/STOP	Operation Method Code Setting	Frequency Setting	Operation Method Code Setting
1	Operation Panel **RUN** and **STOP** Keys	**1500**	**UP/DOWN** Keys	**1600**
2			Speed Pot or Analog Voltage Signal (0 - 10 VDC). Terminals C1, 11, 12, and 13.	**1601**
3			Analog Signal (4 - 20 mA DC) and (0 - 10 VDC). Terminals C1, 11, 12, and 13.	**1602**
4	External Signal (FWD, REV Terminals)	**1500**	**UP/DOWN** Keys	**1600**
5			Speed Pot or Analog Voltage Signal (0 - 10 VDC). Terminals C1, 11, 12, and 13.	**1601**
6			Analog Signal (4 - 20 mA DC) and (0 - 10 VDC). Terminals C1, 11, 12, and 13.	**1602**

Multi-Step Frequency Operation:.

-- *For RUN/STOP and manual speed frequency setting, the Function Codes are* **15** *and* **16** *for setting as above.*

-- *For Step 1 - 7 frequencies, the Function Codes are* **29, 31, 33, 35, 37, 39,** *and* **41.** *(Then select the external terminal X1, X2, X3.)*

CONTROL CIRCUIT CONNECTION AND OPERATION

Basic connections that are required for Function Codes **[1500]** and **[1600]** are shown in Figure 5-4 (these connections are done at factory).

Common terminal connections that are optional for either Keypad Panel Operation or External Operation (through control circuit terminals) are shown in Figure 5-5.

Connection description is as follows:

- **30A, 30B, 30C** - Fault Relay Output used for alarm circuits, etc.
- **FM** - Used for external analog meter when frequency indication is needed.
- **THR** - Connected to **CM** terminal at factory. Connects to a normally closed contact. When inverter STOP is desired through external alarm, open this contact. The inverter will trip on "OH2" and the motor will coast stop.
- **RST** - Connects to a normally open momentary contact for resetting a fault externally.
- **BX** - Connects to a normally open contact. When motor coast stop is desired during normal operation, close this contact and remove the RUN command.

Keypad Panel Operation [1500] terminal connections that are required are shown in Figure 5-6.

Connection description/operation is as follows:

- Common terminal connections are as described in same titled subparagraph.
- Rotation Direction - Make contact between **FWD-CM** for forward rotation or between **REV-CM** for reverse rotation.
- Frequency Setting - Select from the following three forms of Function Code **[16]**:

  [1600] UP/DOWN keys used for frequency setting.

  [1601] Connection of 1K ohm, 1/2 watt potentiometer to terminals **11, 12, 13** or input of 0 to +10 VDC signal to terminals **11, 12** (**12** is positive).

  [1602] Added setting of 4 to 20mA DC current signals to terminals **C1, 11** (C1 is positive) and 0 to +10 VDC signal to terminals **11, 12** (**12** is positive).
- See Figure 5-7 for **RUN** Operation example (Function Code **[1600]**).

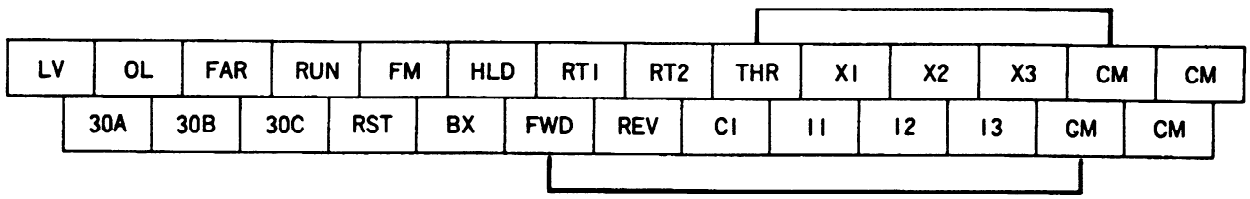


Figure 5-4. FACTORY CONNECTIONS

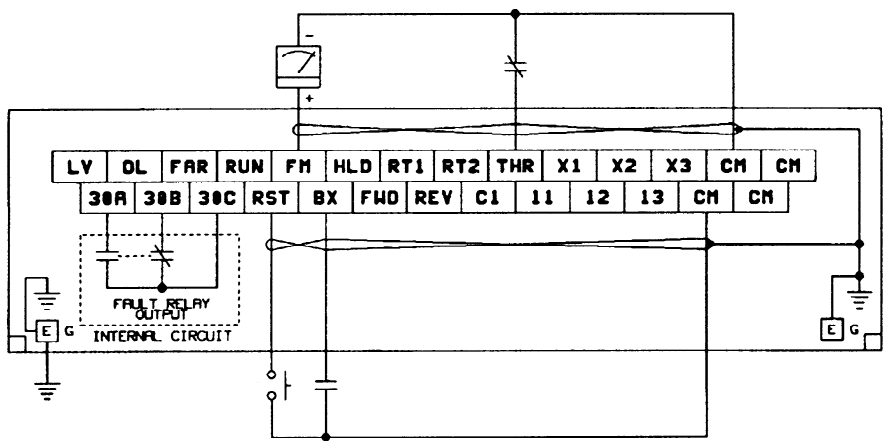


Figure 5-5. COMMON TERMINAL CONNECTION EXAMPLE

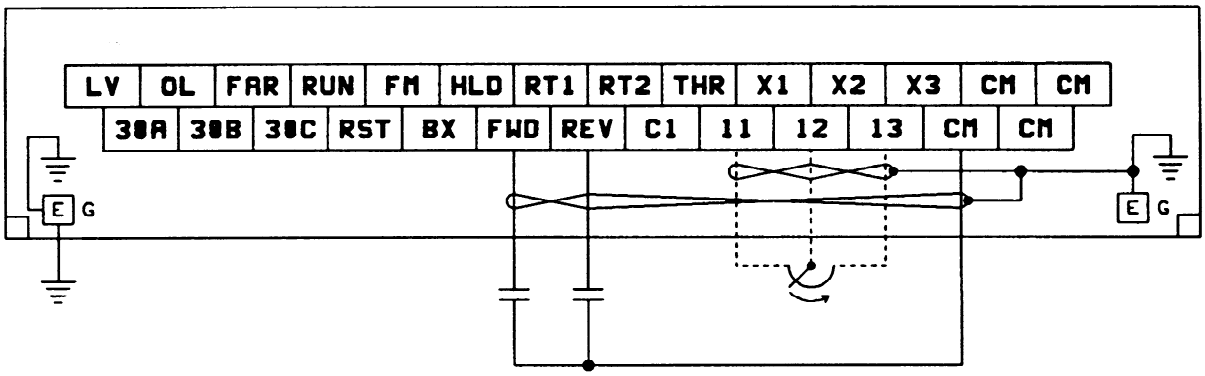


Figure 5-6. KEYPAD PANEL OPERATION [1500] CONNECTIONS

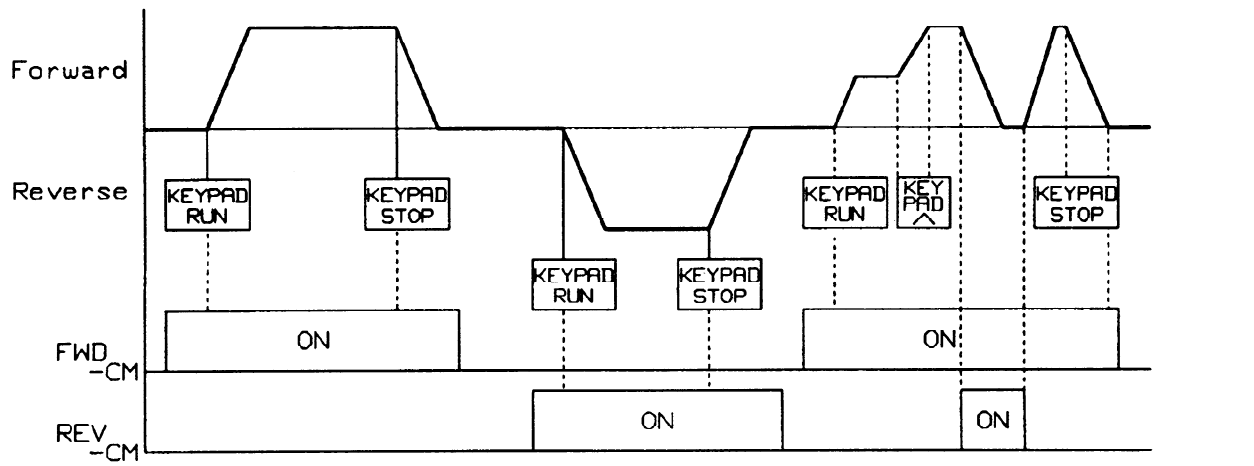


Figure 5-7. KEYPAD PANEL "RUN" OPERATION [1500] and [1600] EXAMPLE

External Signal Operation [1501] terminal connections that are required are shown in Figure 5-8.

Connection description/operation is as follows:

-- Common terminal connections are as described in same titled subparagraph.

-- Rotation Direction - Make contact (momentary) between **FWD-CM** for forward rotation or between **REV-CM** for reverse rotation.

-- Frequency Setting - Select from the following three forms of Function Code **[16]**:

[1600] **UP/DOWN** keys used for digital setting.

[1601] Connection of 1K ohm, 1/2 watt potentiometer to terminals **11, 12, 13** or input of 0 to +10 VDC signals to terminals **11, 12** (**12** is positive).

[1602] Added setting of 4 to 20mA DC current signals to terminals **C1, 11** (**C1** is positive) and 0 to +10 VDC signal to terminals **11, 12** (**12** is positive).

-- See Figure 5-9 for **RUN** Operation example.

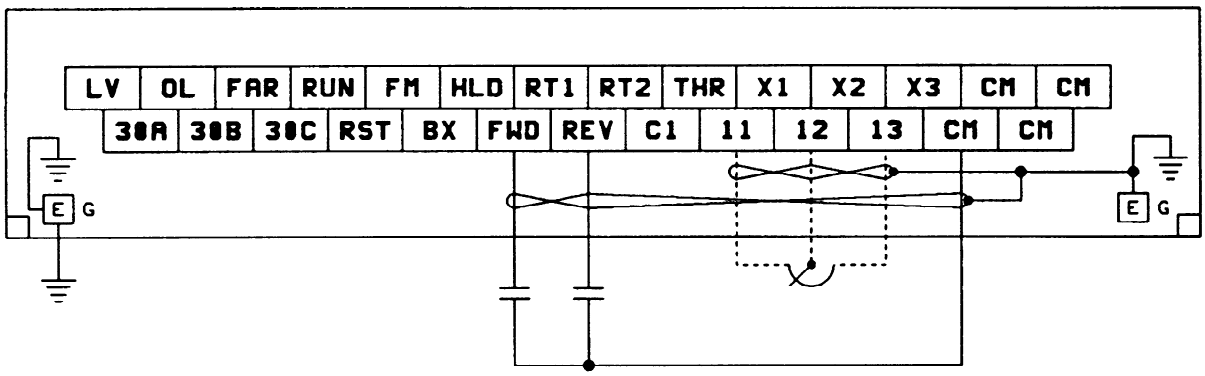


Figure 5-8. EXTERNAL SIGNAL OPERATION [1501] CONNECTIONS

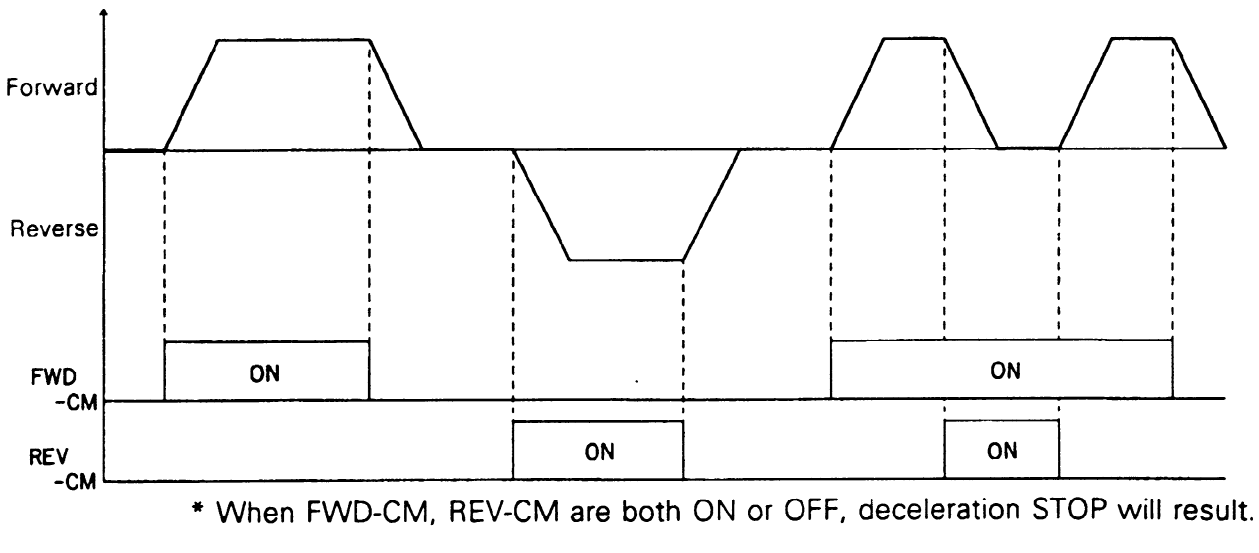


Figure 5-9. EXTERNAL SIGNAL "RUN" OPERATION [1501] and [1601] EXAMPLE

ACTUAL OPERATION

Terminal connections and Function Code selections for Multi-Step Frequency Operation, Multi-Step Acceleration and Deceleration Operation, and Pattern Operation are as described in the following subparagraphs.

Multi-Step Frequency Operation terminal connections (**X1, X2, X3** terminals) that are required are shown in Figure 5-10. Multi-step Frequency Operation can be used to select one of eight different speed settings (one manual setting [Function Code 16] and seven preset settings).

Operation is as follows:

1. Function Code setting is **[15]** for Operation Method and **[16]** for Manual Frequency Setting Method.
2. The multi-step frequency Function Code settings are:
   - **[29]** for multi-step frequency 1.
   - **[31]** for multi-step frequency 2.
   - **[33]** for multi-step frequency 3.
   - **[35]** for multi-step frequency 4.
   - **[37]** for multi-step frequency 5.
   - **[39]** for multi-step frequency 6.
   - **[41]** for multi-step frequency 7.
3. See Figure 5-11 for **RUN** Operation example.

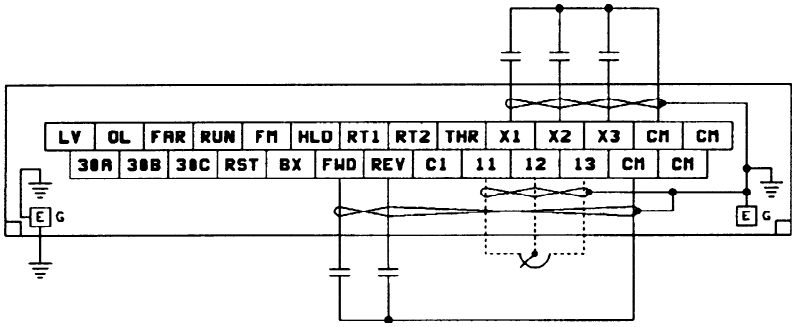


Figure 5-10. MULTI-STEP FREQUENCY OPERATION CONNECTIONS

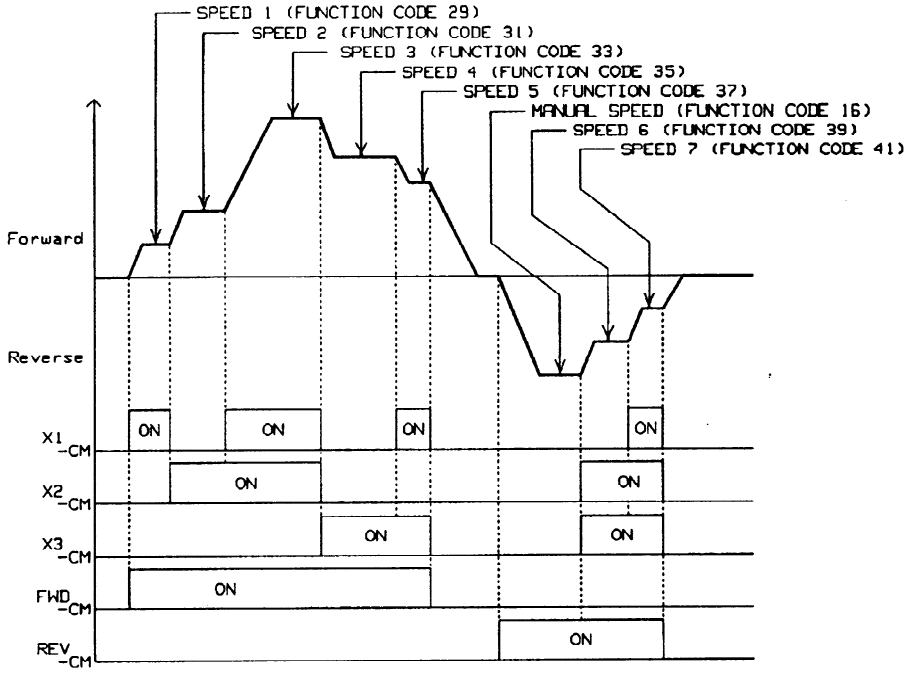


Figure 5-11. MULTI-STEP FREQUENCY "RUN" OPERATION EXAMPLE

Multi-Step Acceleration and Deceleration Operation terminal connections (RT1, RT2, terminals) are shown in Figure 5-12. Four different acceleration and deceleration times can be externally switched.

Operation is as follows:

1. Function Code setting is **[15]** for Operation Method and **[16]** for Manual Frequency Setting Method.
2. The multi-step acceleration and deceleration Function Code settings are:

[08] for acceleration time 1.
[23] for acceleration time 2.
[24] for acceleration time 3.
[25] for acceleration time 4.
[09] for deceleration time 1.
[26] for deceleration time 2.
[27] for deceleration time 3.
[28] for deceleration time 4.

3. See Figure 5-13 for **RUN** Operation example.

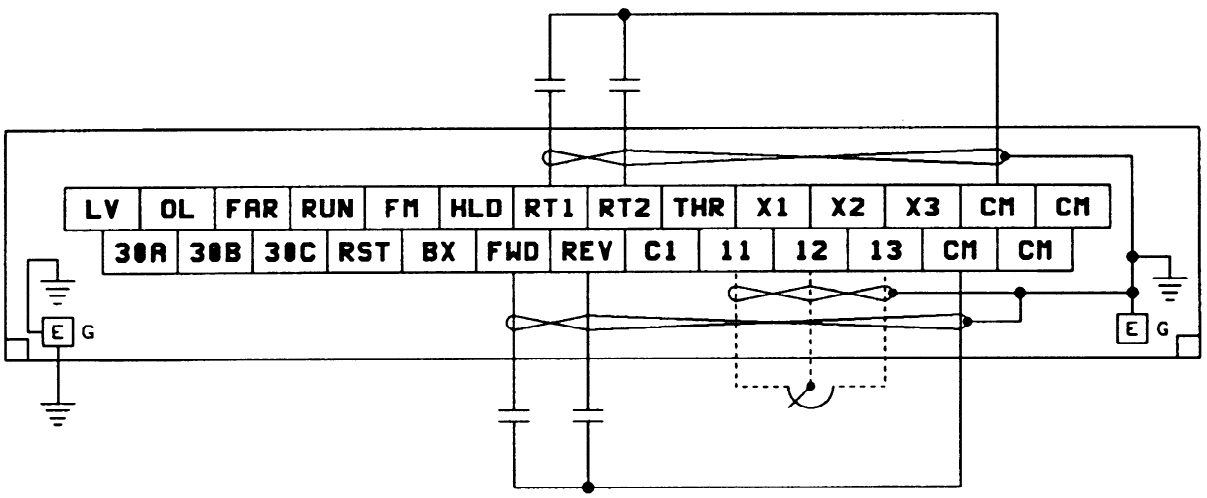


Figure 5-12. MULTI-STEP ACCELERATION AND DECELERATION OPERATION CONNECTIONS

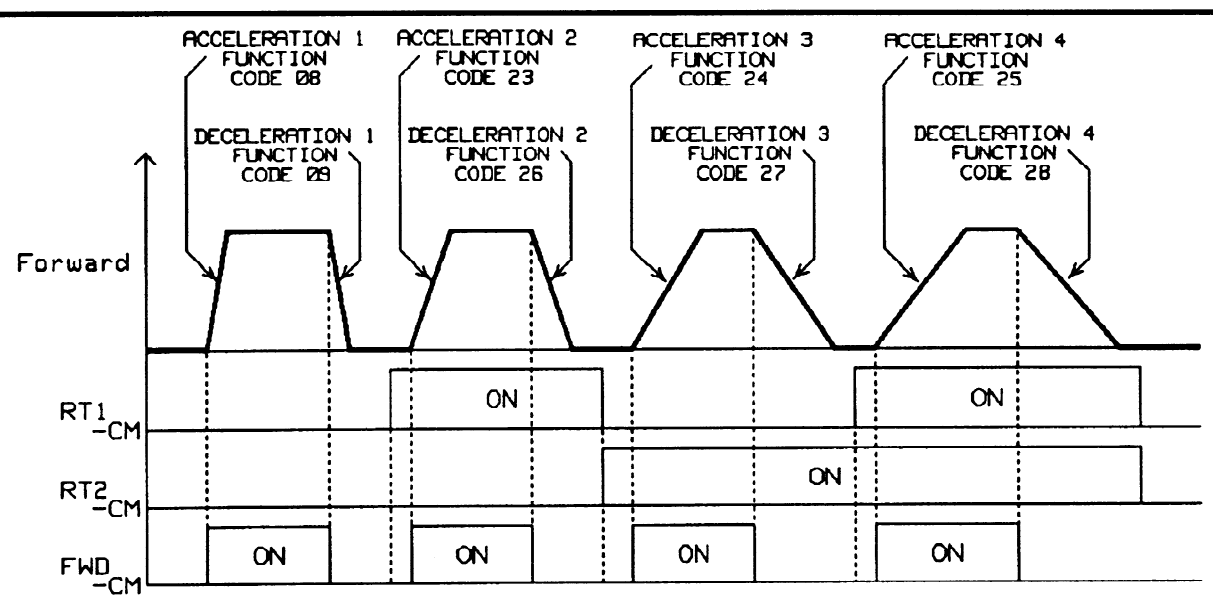


Figure 5-13. MULTI-STEP ACCELERATION AND DECELERATION "RUN" OPERATION EXAMPLE

Pattern Operation terminal connections that are required are shown in Figure 5-14. Pattern Operation is accomplished by aligning the Multi-Step Frequency Settings and Timer Settings.

Operation is as follows:

1. Function Code setting must be **[1901]** for Pattern Operation selection (if **[1900]** selected, Function Codes **[30]**, **[32]**, **[34]**, **[36]**, **[38]**, **[40]**, and **[42]** cannot be accessed).
2. The frequency, rotation direction, timer, and acceleration/deceleration time Function Code settings are:
   - **[29]** sets the frequency for Multi-step Frequency 1.
   - **[30]** sets the time for the Multi-Step Frequency 1 timer, rotation direction, and acceleration/deceleration time.
   - **[31]** sets the frequency for Multi-step Frequency 2.
   - **[32]** sets the time for the Multi-Step Frequency 2 timer, rotation direction, and acceleration/deceleration time.
   - **[33]** sets the frequency for Multi-step Frequency 3.
   - **[34]** sets the time for the Multi-Step Frequency 3 timer, rotation direction, and acceleration/deceleration time.
   - **[35]** sets the frequency for Multi-step Frequency 4.
   - **[36]** sets the time for the Multi-Step Frequency 4 timer, rotation direction, and acceleration/deceleration time.
   - **[37]** sets the frequency for Multi-step Frequency 5.
   - **[38]** sets the time for the Multi-Step Frequency 5 timer, rotation direction, and acceleration/deceleration time.
   - **[39]** sets the frequency for Multi-step Frequency 6.
   - **[40]** sets the time for the Multi-Step Frequency 6 timer, rotation direction, and acceleration/deceleration time.
   - **[41]** sets the frequency for Multi-step Frequency 7.
   - **[42]** sets the time for the Multi-Step Frequency 7 timer, rotation direction, and acceleration/deceleration time.

NOTE: *When in timer display mode, use the **SET** key for changing rotation direction or acceleration/deceleration time.*

3. For Keypad Panel Operation Mode (**[1500]**):
   - **RUN** key: Starts pattern operation.
   - **STOP** key: Stops motor, but pattern timing continues.
   - **RESET** key: Resets timer (must press **STOP** key first).
4. For External Control Terminal Operation Mode (**[1501]**):
   - **FWD-CM** ON: Starts pattern operation.
   - **FWD-CM** OFF: Stops pattern operation and resets timer.
   - **REV-CM** ON: Stops motor, but pattern timing continues.
5. See Figure 5-15 for **RUN** operation example.

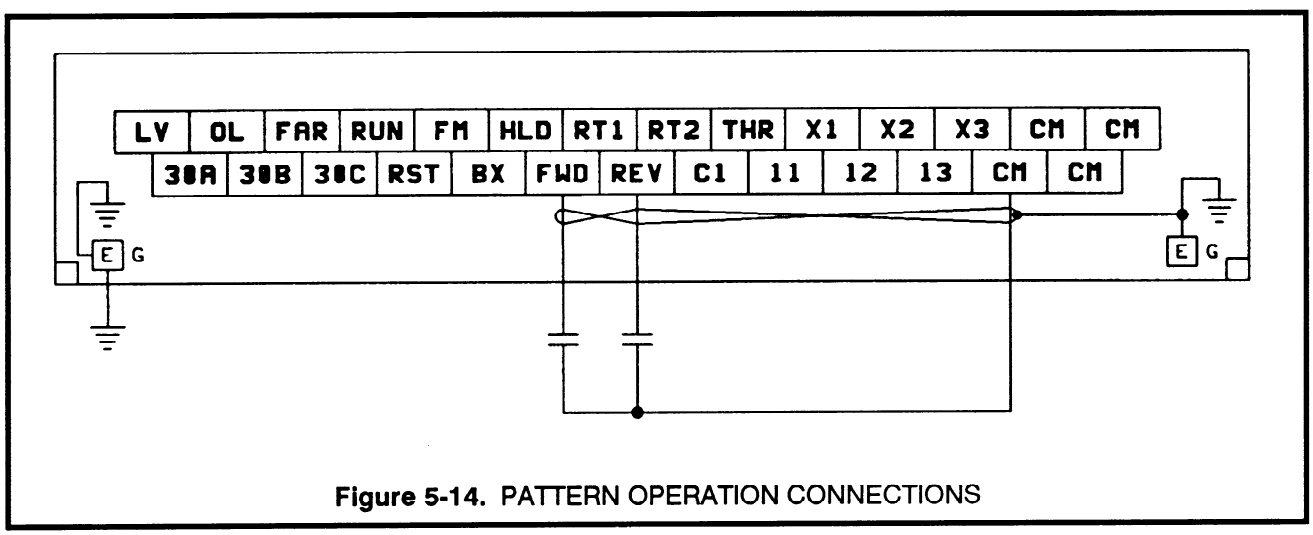


Figure 5-14. PATTERN OPERATION CONNECTIONS

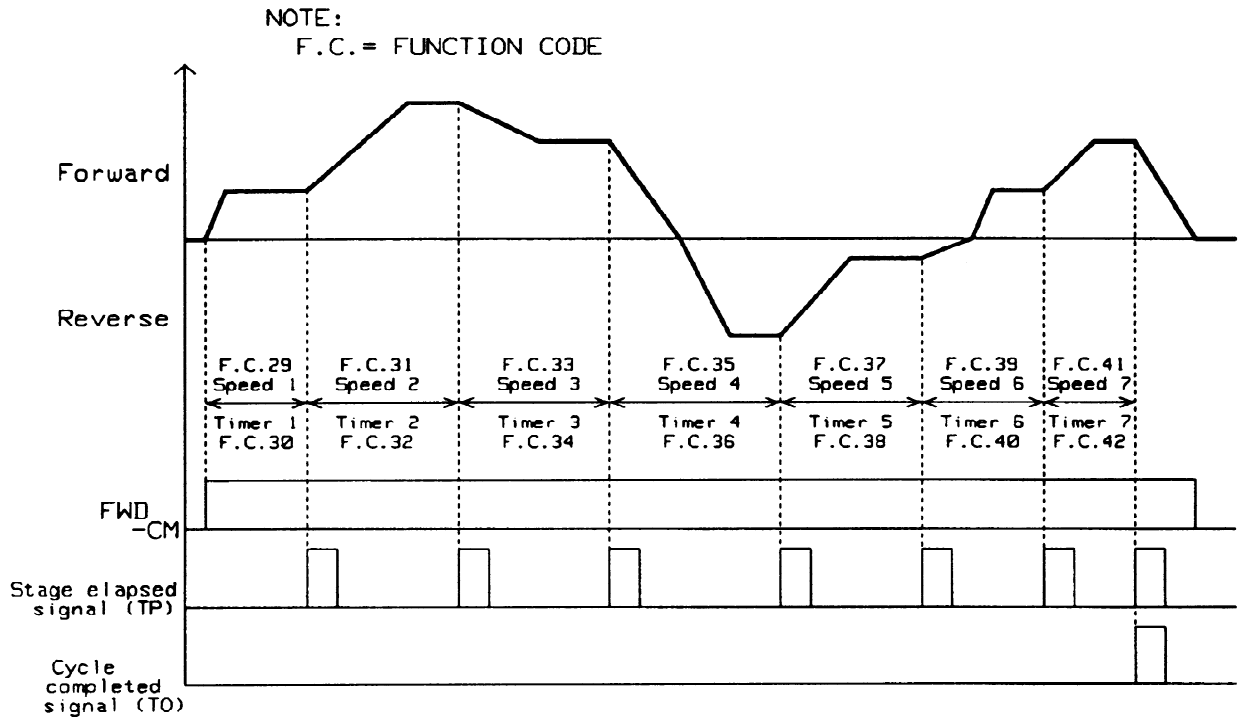


Figure 5-15. PATTERN OPERATION "RUN" OPERATION EXAMPLE

ACCESSING FAULT HISTORY

The inverter maintains an internal record of the previous three inverter trips. This internal record can be accessed during operation or **STOP** as follows:

1. Press **PROGRAM** key to change to program mode.
2. Press and hold **SHIFT** key.
3. Press **RESET** key to advance Digital Monitor and Graphic Display to the following display (example):
   Digital Monitor: **[OH2]**
   Graphic Display*: F = 60.00 Hz
   I = 1.0 A
   V = 230 V
4. Release the **RESET** and **SHIFT** keys.
5. Press **SHIFT** key to change display to show inverter trip record as in the following (example):
   Digital Monitor: **[OH2]**
   Graphic Display*: 0 = OH2 (present fault)
   -1 = OC3 (one time before)
   -2 = Err4 (two times before)
   -3 = OV (three times before)
6. Press **RESET** key to return to Function Mode.
7. Press **PRG** key to return to **RUN/STOP** Mode.

* *For display, the values shown for F, I, V are present operation values. When a trip occurs, the values shown are the values at the time of the trip.*

For display, the value for V indicates the operation output voltage. For trip, the value indicates the DC intermediate voltage.

For display, when trip record is displayed during operation, present values are displayed with the ***SHIFT*** *key. Also, when trip occurs and the values at the time of the trip are displayed, the trip history is displayed with the* ***SHIFT*** *key.*

LINK OPERATION FUNCTION CODE SETTINGS

When using Link Operation, Function Codes should be set as indicated in TABLES 6, 7, or 8 for the operation desired.

TABLE 6: FUNCTION CODE SETTING FOR LINK OPERATION

Function Code	Setting Data Master(*3)	Slave	Description
15	02	02	Link Operation selection
16	00, 01, 02	--	Link Set Frequency Monitor
47	0 - 200%	0 - 200%	Percentage Setting
71	00	01 - 15(*1)	Setting for Inverter Number
72	00 - 15	--	Setting for Number of Inverters Connected
73(*2)	03	03	Link Operation Select
74	00 or 01	--	Panel Operation/Terminal Operation Select
75	16	--	All Inverter Selection
82	00	00	Operation Command Selection in Link Oper.

(*1) *Set the slave inverter numbers in sequence from 01 without skipping any.*

(*2) *Set Function code 73 after setting other Function Codes.*

(*3) *After setting the slave and auxiliary (Aux) inverters, set the master inverter.*

TABLE 7: FUNCTION CODE SETTING FOR FREQUENCY SETTING IN LINK OPERATION

Function Code	Setting Data					Description
	Master(*3)	Aux. 1	Aux. 2	Aux. 3	Slave	
15	00 or 01	00 or 01	00 or 01	00 or 01	00 or 01	Panel/Terminal Operation Sel.
16	01 or 02	01 or 02	01 or 02	01 or 02	01 or 02	Link Set Frequency Monitor
71	00	01(*1)	02(*1)	03(*1)	04-15(*1)	Inverter Number Setting
73(*2)	01	01	01	01	01	Frequency Setting Selection
80	00-02	00-02	00-02	00-02	00-02	Bit Length Setting

TABLE 8: FUNCTION CODE SETTING FOR MONITORING SIGNAL IN LINK OPERATION

Function Code	Setting Data		Description
	Master(*3)	Slave	
15	00 or 01	00 or 01	Link Operation selection
71	00	01 - 15(*1)	Setting for Inverter Number
72	00 - 15	--	Setting for No. of Slave Inverters Connected
73(*2)	02	02	Monitoring Signal Selection

(*1) *Set the slave inverter numbers in sequence from 01 without skipping any.*

(*2) *Set Function Code 73 after setting other Function Codes.*

(*3) *After setting the slave and auxiliary (Aux) inverters, set the master inverter.*

TABLE 9: FUNCTION CODE SETTINGS

Function Code	Function	Display, Setting, Range	Factory Setting	Min. Unit
00	LED Digital Monitor Selection	**00:** Output Frequency (Hz) **01:** Output Current (Amps) **02:** Output Voltage (Volts) **03:** Synchronous rpm (rpm) **04:** Line Speed (m/min.)	00	---
01	Graphics Monitor Selection	**00:** Hz Amp Monitor **01:** Terminal Signal 1 **02:** Terminal Signal 2	00	---
02	Motor Noise Reduction	00 to 05 (code)	03	---
03	FM Terminal Output Level Calibration	00 to 99 (code) (6.5 to 10.5 VDC)	85	---
04	Automatic Torque Boost Control	**00:** Non-Operational **01:** Operational	00	---
05	Torque Boost	00 to 31 (code)	02*	---
06	Torque Boost Fine Adjustment	00 to 09 (code)	00	---
07	Automatic Accel/ Decel Control	**00:** Non-Operational **01:** Operational	00	---
08	Acceleration Time 1	(LCD) 0.01 to 3600 seconds	6/12**	0.01
09	Deceleration Time 1	(LCD) 0.01 to 3600 seconds	6/12**	0.01
10	Data Protection	**00:** Not Protected (changes) **01:** Protected (no changes)	00***	---
11	Maximum Frequency	**00:** 50 Hertz **01:** 60 Hertz **02:** 100 Hertz **03:** 120 Hertz **04:** Adjustable (0.2 - 400 Hertz)	01	1 Hz
12	Base Frequency	**00:** 50 Hertz **01:** 60 Hertz **02:** Adjustable (0.2 - 400 Hertz)	01	1 Hz

**Factory setting is 13 for inverters of 3.7 KW (5 HP) or less; setting is 8 for inverters of 5.5 KW (7.5 HP) or greater; it is recommended to change settings to 02 for all AF-300B inverters (see Section 6 for further explanation).*

***Factory setting for Accel Time 1 and Decel Time 1 is 6.0 for inverters of 7.5 KW (10 HP) or less; setting is 12.0 for inverters of 11 KW (15 HP) or greater.*

****Press and hold **STOP**, then use **UP/DOWN** arrows to change setting; press **SET** to store change and release **STOP**.*

Function Code	Function	Display, Setting, Range	Factory Setting	Min. Unit
13	Rated Output Voltage	**00:** 200 (400) VAC **01:** 220 (440) VAC **02:** 230 (460) VAC **03:** Adjustable (0 - 230 VAC) (0 - 460 VAC)*	02	1 V (2V)
14	Number of Motor Poles	02:, 04:, 06:, 08:, 10:, 12:	04	---
15	Operation Command	**00:** Keypad Panel Operation **01:** Terminal Block Operation **02:** Link Adaptor Operation	00	---
16	Frequency Command	**00:** Digital (Keypad UP/DOWN key) **01:** Analog (voltage, 0 - 10 VDC) **02:** Analog (voltage & current, 0 - 10 VDC plus 4 - 20 mA)	00	---
17	Accel/Decel Pattern	**00:** Linear **01:** Weak "S" Curve **02:** Strong "S" Curve	00	---
18	Normal/High Torque Dynamic Brakes	**00:** Normal Brake **01:** High Torque Brake	00	---
19	Pattern Operation	**00:** Normal Operation **01:** Pattern Operation	00	---
20	Restart After Instantaneous Power Failure	**00:** Non-Operational **01:** Operational	00	---
21	Coefficient for Line Speed	(LCD) 0.00 - 200	0.01	0.0
22	Function Code Blocks Accessible	**00:** Basic Functions (00 - 22) **01:** Basic & Standard Functions (00 - 62) **02:** Basic, Standard, & High Level Functions (00 - 79) See Fig. 5-3	00	---
23	Acceleration Time 2	(LCD) 0.01 - 3600 seconds	10.0	0.01
24	Acceleration Time 3	(LCD) 0.01 - 3600 seconds	15.0	0.01
25	Acceleration Time 4	(LCD) 0.01 - 3600 seconds	3.0	0.01
26	Deceleration Time 2	(LCD) 0.01 - 3600 seconds	10.0	0.01

**0 VAC setting means the output voltage is relative to the AC supply line voltage.*

Function Code	Function	Display, Setting, Range	Factory Setting	Min. Unit
27	Deceleration Time 3	(LCD) 0.01 - 3600 seconds	15.0	0.01
28	Deceleration Time 4	(LCD) 0.01 - 3600 seconds	3.0	0.01
29	Multistep Speed Setting 1	(LCD) 0.00 - 400 Hertz	0.0	0.002
30	Timer 1	(LCD) 0.01 - 3600 seconds	5.0	0.01
31	Multistep Speed Setting 2	(LCD) 0.01 - 400 Hertz	0.0	0.002
32	Timer 2	(LCD) 0.01 - 3600 seconds	5.0	0.01
33	Multistep Speed Setting 3	(LCD) 0.01 - 400 Hertz	0.0	0.002
34	Timer 3	(LCD) 0.01 - 3600 seconds	5.0	0.01
35	Multistep Speed Setting 4	(LCD) 0.01 - 400 Hertz	0.0	0.002
36	Timer 4	(LCD) 0.01 - 3600 seconds	5.0	0.01
37	Multistep Speed Setting 5	(LCD) 0.01 - 400 Hertz	0.0	0.002
38	Timer 5	(LCD) 0.01 - 3600 seconds	5.0	0.01
39	Multistep Speed Setting 6	(LCD) 0.01 - 400 Hertz	0.0	0.002
40	Timer 6	(LCD) 0.01 - 3600 seconds	5.0	0.01
41	Multistep Speed Setting 7	(LCD) 0.01 - 400 Hertz	0.0	0.002
42	Timer 7	(LCD) 0.01 - 3600 seconds	5.0	0.01
43	Electronic Thermal Overload Relay	00: Non-Operational 01: Operational (%)	01 (LCD) 90%	1%
44	High Limiter	(LCD) 0 - 100 %	100	1%
45	Low Limiter	(LCD) 0 - 100 %	0	1%
46	Bias Frequency*	(LCD) 0 - 100 %	0	1%

Frequency bias and frequency gain are operational only when the frequency setting method is analog (Function Code 16 = 01 or 02). See **WARNING in Section 1.*

Function Code	Function	Display, Setting, Range	Factory Setting	Min. Unit
47	Frequency Setting Signal Gain*	(LCD) 0 - 200 %	100	1%
48	Jump Frequency 1** (Frequency Rejection)	(LCD) 0 - 400 Hertz	0	1 Hz
49	Jump Frequency 2** (Frequency Rejection)	(LCD) 0 - 400 Hertz	0	1 Hz
50	Jump Frequency 3** (Frequency Rejection)	(LCD) 0 - 400 Hertz	0	1 Hz
51	Jump Frequency Width	(LCD) 0 - 5 Hertz	0	1 Hz
52	DC Brake	**00:** Non-Operational **01:** Operational	00	---
53	DC Brake Starting Frequency	(LCD) 0 - 60 Hertz	0	1 Hz
54	DC Brake Voltage	0 to 15 (code)	00	1%
55	DC Braking Time	(LCD) 0.01 - 30 seconds	0.10	0.01
56	Starting Frequency	(LCD) 0 (0.2) - 60 Hertz	1	1 Hz
57	Current Limiter	**00:** Non-Operational **01:** Operational (30 - 150%)	01 (LCD) 150%	1%
58***	Slip Compensation Control	**00:** Non-Operational **01:** Operational	00	---
59	Frequency Level Detection	(LCD) 0 - 400 Hertz	60	1 Hz
60	FDT and FAR Signal Hysterisis	(LCD) 0 - 30 Hertz	10	1 Hz
61	Run Signal Finishing Frequency	(LCD) 0 - 400 Hertz	0	1 Hz
62	Overload Early Warning Signal	(LCD) 70 - 150 %	100	1%

Frequency bias and frequency gain are operational only when the frequency setting method is analog (Function Code 16 = 01 or 02). See **WARNING in Section 1.*

***Function Code 51 sets width.*

****Factory recommendation is non-operational for most applications due to interference with torque boost operation.*

Function Code	Function	Display, Setting, Range	Factory Setting	Min. Unit
63*	X1, X2, and X3 Terminal Function	**00:** Multi-Step Frequency Op. **01:** INC/DEC Frequency Control **02:** DC Brake Operation	00	---
64	FWD/REV Command Hold	**00:** 2-Wire Operation **01:** 3-Wire Operation **02:** Switches Timer Oper. Mode	01	---
65	LV, OL, and FAR TerminalOutput Code	**00:** Output per FC 66, 67, 68 **01:** Binary speed step signal	00	---
66	LV Terminal Function (Signal Output)	**00:** Undervoltage **01:** Overvoltage	00	---
67	OL Terminal Function (Signal Output)	**00:** Overload Early Warning **01:** Current Limit Monitoring **02:** Undervoltage	00	---
68	FAR Terminal Function (Signal Output)	**00:** Freq. Equivalence Detection **01:** Setting Frequency Detection **02:** Not Running Signal	00	---
69	RUN Terminal Function (Signal Output)	**00:** Run signal **01:** Pattern Step Complete Signal **02:** Pattern Complete Signal	00	---
70	FM Terminal Function (Signal Output)	**00:** Frequency Meter **01:** Output Current Meter	00	---
71	Inverter Number Entry For Link Operation	**00:** For Master Record **01:** Slave #1 (Auxiliary #1) *Thru* **15:** Slave #15 (Auxiliary #15)	15	---
72	Number of Inverters Linked	**00:** Total Number of Inverters *Thru* **15:** Connected (master and slave)	00	
73	Link Mode Selection	**00:** Link Non-Operational **01:** Frequency Setting **02:** Monitoring Signal **03:** Link Operational	00	---
74	RUN Command Input In Link Operation	**00:** RUN/STOP via Keypad Panel **01:** RUN/STOP via Control Terms.	00	---
75	Slave Inverter Number Designation (To change Function Codes in slave from master keypad)	**00:** Inverter # to be changed *Thru* **15:** **16:** Selects All Inverters	00	---
76	Sets Gain On Automatic Current Regulation	**00:** Based On Acel/Decel Rate **01:** Quickest Response *Thru* **99:** Slowest Response	00	1

**For High Level Function Codes (63 - 82), see Table 10, "Function Code Descriptions" in Section 6.*

Function Code	Function	Display, Setting, Range	Factory Setting	Min. Unit
77-79	Optional - For Possible Future Use			
80*	Number of Bits for Frequency Setting	**00:** 4-Bit Binary Input **01:** 8-Bit Binary Input **02:** 12-Bit Binary Input	00	---
81*	Monitoring Signal Output Code In Link Operation	**00:** Set Output Data via Function Code for Each Inverter	00	---
82*	Monitoring Mode In Link	**00:** Operation Command Mode **01:** Function Code Setting Mode	00	---

**These Function Codes are present only on inverters with Serial Numbers 29 _ _ _ and prior.*

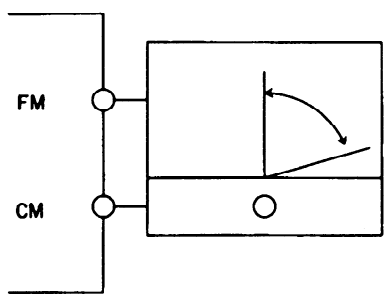
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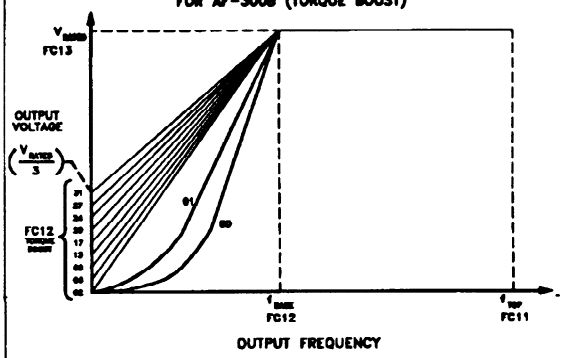
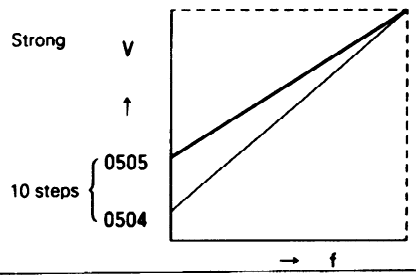
6. FUNCTION CODE DESCRIPTIONS

TABLE 10: FUNCTION CODE DESCRIPTIONS

Initial Settings

Function Code	Display	Function Explanation	Factory Setting
00 ■ DGTL MNTR GRHC MNTR MTR SOUND FM CALIBR LED Digital Monitor Selection	0000 0001 0002 0003 0004	**The default display of the four digit, seven segment LED display for power-up and for leaving the program mode can be changed as follows:** Set frequency [Hz] (during STOP), output frequency [Hz] (during RUN) displayed (Illustration 1). 60.00 FREQUENCY CURRENT 1. Output current [A] display (virtual value) (Illustration 2). 12.3 FREQUENCY CURRENT 2. Output voltage [V] display (virtual value) (Illustration 3). 400 FREQUENCY CURRENT 3. Synchronous rotation speed (rpm) display (Illustration 4). 1800 FREQUENCY CURRENT 4. Line speed (m/min) display (Illustration 5). 98.76 FREQUENCY CURRENT 5. For each display mode, normally the display can be changed by using the SHIFT key. This can be done while running. 60.00 → 00.02 → 12.3 → 400 → 1800 → 98.76 → For frequency display, one digit shift to the right for verification is possible by using the SHIFT key. (At this time the digit on the left will not be displayed) 60.002 → 60.002	00.00

Function Code	Display	Function Explanation	Factory Setting
01 DGTL MNTR ■ GRHC MNTR MTR SOUND FM CALIBR Graphics Monitor Selection	0100 0101 0102	The LCD (liquid Crystal) display can be changed to the following 3 modes. Output frequency [%], output current [%] Graph display (Illustration 1). 60.00 FREQUENCY ■■■■■■■■■■ CURRENT ■■■■■■■■■■ 1. Control terminal strip monitor 1 (Illustration 2). 60.00 ■FWD ■RT1 ■REV ■RT2 ■HLD 2. Control terminal strip monitor 2 (Illustration 3). 60.00 ■LV ■X1 ■OL ■X2 ■FAR ■X3 ■RUN 3. ① With function 11 the output frequency is a % of the set maximum frequency. (10% per block) ② The output current is a % of the inverter rated current. (10% per block). ③ Control terminal strip monitor On is indicated by ■. No indication = OFF.	0100
02 DGTL MNTR GRHC MNTR ■ MTR SOUND FM CALIBR Motor Noise Reduction	0200 ~ 0205	The sound quality of the sound produced by the motor can be changed. } Select from 6 types depending on the operating conditions	0203
03 DGTL MNTR GRHC MNTR MTR SOUND ■ FM CALIBR FM terminal Output Calibration	0300 ~ 0399	This function regulates the frequency indication meter voltage level output from the FM terminal. Approx. 6.5 V ↓ Approx. 10.5 V } Within this range regulation can be 1/100 resolution 	0385

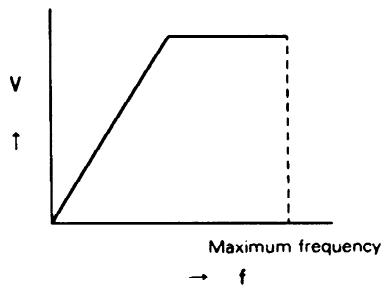
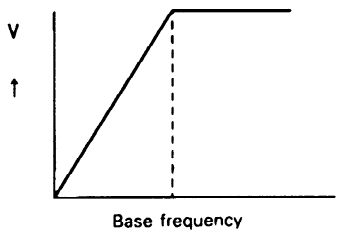
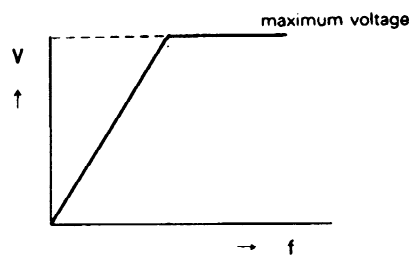
Function Code	Display	Function Explanation	Factory Setting
04 ■ AUTO TRQ TRQ BOOST TRQ FINE AUTO ACC Automatic torque boost control	0400 0401	Automatically regulates the output voltage to correspond to the operating load conditions. Nonoperate : operates at set torque boost value (05) Operate : auto torque boost, operates at set torque boost value (05) until output currect exceeds 100% inverter nameplate rating, then it is increased automatically.	0400
05 AUTO TRQ ■ TRQ BOOST TRQ FINE AUTO ACC Torque Boost	0500 0501 0502 Weak ↓ 0531 Strong	Setting can be made from 32 types, depending on type of load, motor characteristics, etc. For reduction torque load use (fan, pump, etc.) OUTPUT VOLTAGE VS. OUTPUT FREQUENCY FOR AF-300B (TORQUE BOOST)  **CAUTION: Set to lowest value that will start motor with maximum load, otherwise motor damage may result. Function Code 58 must be set to 00.**	0502
06 AUTO TRQ TRQ BOOST ■ TRQ FINE AUTO ACC Fine Adjustment Of Torque Boost	0600 ↓ 0609	For torque boost value set at 05 the torque boost value can be fine adjusted. (10 Steps) [Example] For torque boost value set at 05  	0600
07 AUTO TRQ TRQ BOOST TRQ FINE ■ AUTO ACC Automatic Accel/Decel Control	0700 0701	Automatically determines the acceleration/deceleration time corresponding to the load characteristics. The minimum times are 3 sec. for acceleration, and 8 sec. for deceleration. (0⟷ 60Hz). Nonoperate acceleration/deceleration time is set by Function Code 08 and 09 (or 23 through 28). Operate (auto acceleration/deceleration operation)	0700

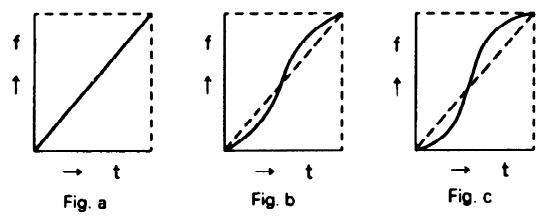
Function Code	Display	Function Explanation	Factory Setting
		① For heavy inertia load. ② Does not function when momentary power failure restart [2001], or current limit [5701] is selected. ③ Does not function for S curve acceleration/deceleration [1701] [1702]	
[08] Hz/RPM/M/min ■ ACCEL 1 DECEL 1 DATA PRTC Acceleration Time 1	[08] Hz/RPM/M/min F TOP 6.00 S t	Setting is possible within the 0.01S to 3,600S range to correspond to load characteristics. Is selected when both RT1-CM and RT2-CM are OFF.	[08] Hz/RPM/M/min F TOP 6.00 S t . * For 10 HP or less; 12 sec. for 15 HP or greater.
[09] Hz/RPM/M/min ACCEL 1 ■ DECEL 1 DATA PRTC Deceleration Time 1	[09] Hz/RPM/M/min F TOP 6.00 S t		[09] Hz/RPM/M/min F TOP 6.00 S t .

Setting time	Setting Resolution
0.01 ~ 9.99 s	By 0.01s
10.00 ~ 99.90 s	By 0.1s
100.0 ~ 999.0 s	By 1s
1000 ~ 3600 s	By 10s

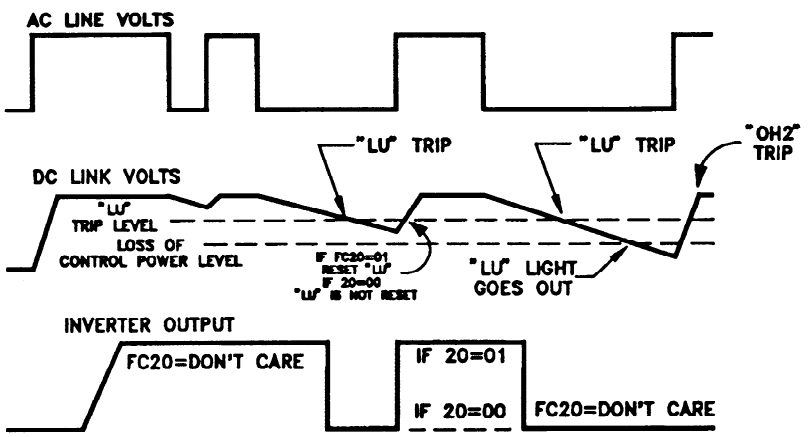
[10] Hz/RPM/M/min ACCEL 1 DECEL 1 ■ DATA PRTC Data Protection	[1000] [1001]	For data that does not need to be changed, the set data can be locked. No protect Protect ① ∧ ∨ keys + STOP key changes data. Data is stored with the SET key. ② Change of data other than [10] is not possible.	[1000]

Basic Functions

Function Code	Display	Function Explanation	Factory Setting
11 ■ MAX Hz BASE Hz RATED V MTR POLES Maximum Frequency	1100 1101 1102 1103 1104	Sets maximum value for output frequency 50Hzmax 60Hzmax 100Hzmax 120Hzmax  Between 0 to 400Hz, the maximum frequency can be set with 1Hz resolution. Damage may result if commonly used motors, etc. which are designed for low speed use, are operated at maximum frequency. Operate motors at a frequency setting conforming to the motor characteristics.	1101
12 MAX Hz ■ BASE Hz RATED V MTR POLES Base Frequency	1200 1201 1202	Sets the base frequency. (frequency for specified torque characteristics and specified output characteristic divergent point) 50Hz 60Hz  Between 0 to 400Hz, the base frequency can be set with 1Hz resolution. ① Operate at a setting conforming to the motor characteristics. ② A setting exceeding function code 11 is not possible.	1201
13:03 MAX Hz BASE Hz ■ RATED V MTR POLES Rated Output Voltage	1300 1301 1302 1303	Sets the maximum value for the output voltage. 200V (400V) 220V (440V) 230V (460V)  O—230V (O—460V) When set at 0V voltage relative to the AC supply voltage is output. Output of voltage exceeding the power supply voltage is not possible.	1302 (0V)

Function Code	Display	Function Explanation	Factory Setting
14 MAX Hz BASE Hz RATED V ■ MTR POLES Number of Motor Poles	1402 1404 1406 1408 1410 1412	The number of motor poles is set for the display of synchronous rpm. 2 pole conversion (3600 rpm). 4 pole conversion (1800 rpm). 6 pole conversion (1200 rpm) 8 pole conversion (900 rpm) 10 pole conversion (720 rpm) 12 pole conversion (600 rpm) [Example] Display when 4 pole motor is operated at 60Hz. 1800 ■RPM	1404
15 ■ OPR COMND Hz COMND ACC PTN H TRQ BRK Operation Command	1500 1501 1502	Selection can be made from the following 3 types. Keypad operation mode (RUN STOP keys) Terminal strip mode (FWD, REV, HLD terminals) (Remote Operation) Link mode (group operation: see Function Codes 71 through 82)	1500
16 OPR COMND ■ Hz COMND ACC PTN H TRQ BRK Frequency Command	1600 1601 1602	Selection can be made from the following 3 types. Digital setting (∧ ∨ keys) (Keypad Operation) Analog setting (DC 0 to 10V) (Speed Pot, Remote) Analog setting (DC 0 to 10V) + (DC 4 to 20mA) (Speed Pot + Process Follower) Even when 1502 link operation is selected, monitor of setting frequency is possible.	1600
17 OPR COMND Hz COMND ■ ACC PTN H TRQ BRK Accel/Decel Pattern	1700 1701 1702	Selection can be made from the following 3 types. Linear acceleration/deceleration (Fig. a) Weak S curve acceleration/deceleration (Fig. b) Strong S curve acceleration/deceleration (Fig. c)  Fig. a Fig. b Fig. c Does not function when 0701 auto acceleration/ deceleration is selected.	1700

Function Code	Display	Function Explanation	Factory Setting
18 OPR COMND Hz COMND ACC PTN ■ H TRQ BRK Normal/High Torque Dynamic Brake	1800 1801	Selection can be made from the following two types of regenerative braking methods: -- Standard brake torque via internal DB resistor. -- High brake torque via external DB resistor (opt.). For inverters with serial numbers starting with 2X and greater: -- For standard braking via internal DB resistor, braking operates below 66 Hz output frequency. -- For high braking via external DB resistor, braking operates throughout full output frequency range with max brake ON duty at 65 Hz/output frequency. For loads requiring an instantaneous stop, set high brake mode and connect optional external DB resistor.	1800
19 ■ PTN OPR RESTART MNTR COEF FUNC BLK Pattern Operation	1900 1901	Selects between operate/nonoperate for pattern operation set by function codes 29 to 42 Nonoperate (standard multistep frequency operation) Operate (pattern operation) See function codes 29 to 42 for pattern operation details.	1900
20 PTN OPR ■ RESTART MNTR COEF FUNC BLK Restart After Instantaneous Power Failure	2000 2001	Selects restart mode for instantaneous power failure and restoration. Nonoperate (No operation command: inverter stop With operation command: under voltage trip) Operate (Picks up the free running motor ① For 2001, operates only until LU light goes out. The LU light times for each unit is listed in the table below. (sec). **HP 0.5 1 2 3 5 7.5 10 15 20 25 30** **230 V 1.4 1.4 2.0 3.5 3.5 6.0 10 14 20 20 24** **460 V --- 0.9 1.0 1.2 2.0 2.2 3.0 3.4 4.3 5.1 5.8** Requirements for Restart operation: 1. Remote operation - Function Code 15 = 01 2. FWD-CM or REV-CM connection present when power is restored.	2000



Function Code	Display	Function Explanation	Factory Setting
21 PTN OPR RESTART ■ MNTR COEF FUNC BLK Coefficient for Line Speed	F coefficient Hz × 0.000 ↓ F coefficient Hz × 200	Coefficient K is for display of m/min Display value = output frequency × K Display output frequency (syncronous) × 0.00 ↓ Displays output frequency (syncronous) × 200 0.01 resolution is possible when K = 0.00 to 200. If the value for output frequency x K exceeds 9999. 9999 is displayed. [Example] K = 200 at output 100HZ 100Hz x K = 20,000 → display 9999	F coefficient Hz x 0.01
22 PTN OPR RESTART MNTR COEF ■ FUNC BLK Function Blocks Used	2200 2201 2202	The function code block display range for function codes can be defined. Up to basic function display 00 → 22 Up to standard function display 00 → 62 Function display (00 → 79	2200

Standard Functions

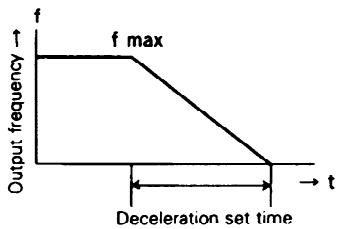
Function Code	Display	Function Explanation	Factory Setting
23 ■ ACCEL 2 ACCEL 3 ACCEL 4 Acceleration Time 2	23 F TOP 10.00 S t	Setting between the 0.01S to 3600S range is possible to correspond to the load characteristics. 0.01 to 9.99S : By 0.01S 10.00 to 99.90S: By 0.1S 100.0 to 999.0S: By 1S 1000 to 3600S: By 10S Selection of acceleration times 1 through 4 is made via RT1 and RT2 terminal combinations.	23 F TOP 10.00 S t
24 ACCEL 2 ■ ACCEL 3 ACCEL 4 Acceleration Time 3	24 F TOP 15.00 S t	✱ The setting time equals the time required to reach the set maximum frequency from zero. (0 → max. set freq.) Indicates time change (sec.) from 0 up to (11 setting)	24 F TOP 15.00 S t
25 ACCEL 2 ACCEL 3 ■ ACCEL 4 Acceleration Time 4	25 F TOP 3.00 S t	f, fmax, output frequency, t, acceleration set time	25 F TOP 3.00 S t

Function Code	8	23	24	25
Multiple Accel. Setting	ACCEL. 1	ACCEL. 2	ACCEL. 3	ACCEL. 4
RT1 - CM	OFF	ON	OFF	ON
RT2 - CM	OFF	OFF	ON	ON

Function Code	Display	Function Explanation	Factory Setting
26 ■ DECEL 2 DECEL 3 DECEL 4 Deceleration Time 2	26 F TOP 10.00 S t	Setting between the 0.01S to 3600S range is possible to correspond to the load characteristics. 0.01 to 9.99S : By 0.01S 10.00 to 99.90S : By 0.1S 100.0 to 999.0S : By 1S 1000 to 3600S : By 10S Selection of deceleration times 1 thru 4 is made via RT1 and RT2 terminal combinations.	26 F TOP 10.00 S t
27 DECEL 2 ■ DECEL 3 DECEL 4 Deceleration Time 3	27 F TOP 15.00 S t		27 F TOP 15.00 S t
28 DECEL 2 DECEL 3 ■ DECEL 4 Deceleration Time 4	28 F TOP 3.00 S t		28 F TOP 3.00 S t

Function Code	9	26	27	28
Multiple Decel. Setting	Decel. 1	Decel. 2	Decel. 3	Decel. 4
RT1 - CM	OFF	ON	OFF	ON
RT2 - CM	OFF	OFF	ON	ON

* The setting time equals the time required to reach 0 from the maximum frequency. (max. frequency → 0)
Indicates time change (sec.) from (11 setting to 0).



Function Code	Display	Function Explanation	Factory Setting

Function Code

29
■ MULT SPD1
TIMER 1
MULT SPD2
TIMER 2
Multistep Speed Setting 1

31
MULT SPD1
TIMER 1
■ MULT SPD2
TIMER 2
Multistep Speed Setting 2

33
■ MULT SPD3
TIMER 3
MULT SPD 4
TIMER 4
Multistep Speed Setting 3

35
MULT SPD3
TIMER 3
■ MULT SPD 4
TIMER 4
Multistep Speed Setting 4

37
■ MULT SPD5
TIMER 5
MULT SPD6
TIMER 6
Multistep Speed Setting 5

39
MULT SPD5
TIMER 5
■ MULT SPD6
TIMER 6
Multistep Speed Setting 6

41
■ MULT SPD7
TIMER 7
Multistep Speed Setting 7

Display

29 — F 10.00 Hz
31 — F 20.00 Hz
33 — F 30.00 Hz
35 — F 40.00 Hz
37 — F 50.00 Hz
39 — F 60.00 Hz
41 — F 60.00 Hz

Function Explanation

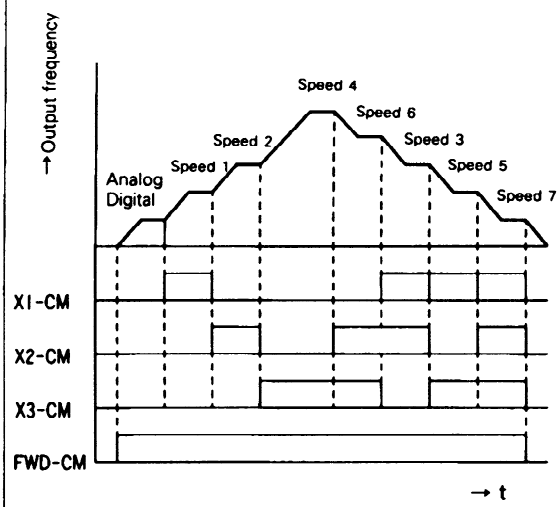
8 step multi-step frequency operation is possible through switching of the control terminals X1, X2, X3 (for setting) 1900

(Setting resolution)

Multi-step set value / Maximum frequency	0– 1.000Hz	1.000– 10.00Hz	10.00– 100.0Hz	100.0– Hz
60Hz	0.002Hz	0.01Hz	0.1Hz	—
300Hz	0.01Hz	0.01Hz	0.1Hz	1Hz
400Hz	0.02Hz	0.01Hz	0.1Hz	1Hz

(Relationship between the terminals and multi-step frequency 1 to 7)

Function code	16	29	31	33	35	37	39	41
Multi-step frequency	Analog Digital	Speed 1	Speed 2	Speed 3	Speed 4	Speed 5	Speed 6	Speed 7
X1-CM	OFF	ON	OFF	ON	OFF	ON	OFF	ON
X2-CM	OFF	OFF	ON	ON	OFF	OFF	ON	ON
X3-CM	OFF	OFF	OFF	OFF	ON	ON	ON	ON



① When X1-CM, X2-CM, X3-CM are all OFF, setting is made by digital (∧ ∨ keys), or analog (DC 0 to +10V, 4 to 20mA) which is selected by function 16

② A setting exceeding the maximum frequency set by 11 is not possible.

③ When 1901 is set, pattern operation is active.

④ When 6300 is set, multi-step frequency 1 thru 7 switching is possible via X1, X2, X3 terminals.

Factory Setting

29 — F 0.00 Hz
31 — F 0.00 Hz
33 — F 0.00 Hz
35 — F 0.00 Hz
37 — F 0.00 Hz
39 — F 0.00 Hz
41 — F 0.00 Hz

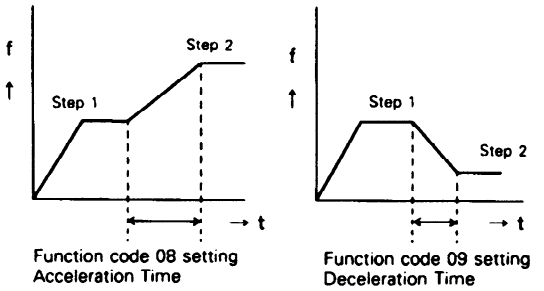
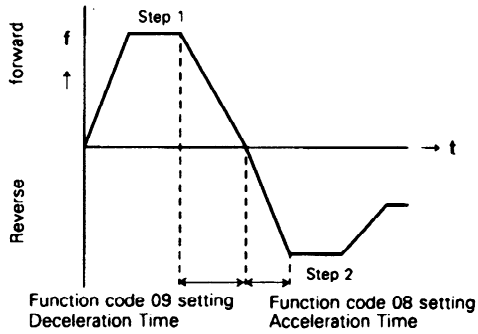
Function Code	Display	Function Explanation	Factory Setting
30	30:00	When 1901 is set, pattern operation is possible by multi-step frequency 1 thru 7 combination.	30:00
MULT SPD1 ■ TIMER 1 MULT SPD2 TIMER 2	F 0.01 S t	When in pattern operation, no frequency changeover via the X1, X2, X3 terminals is necessary.	F 5.00 S t
Timer 1		Pattern operation switches to stop frequencies 1 through 7 in numerical order only.	
32	32:00		32:03
MULT SPD1 TIMER 1 MULT SPD2 ■ TIMER 2	F 0.01 S t	→ Output frequency Step 1 Step 2 Step 3 Step 4 Step 5 Step 6 Step 7 Timer 1 Timer 2 Timer 3 Timer 4 Timer 5 Timer 6 Timer 7 1 2 3 4 5 6 7 → t	F 5.00 S t
Timer 2			
34	42:00		34:02
MULT SPD3 ■ TIMER 3 MULT SPD 4 TIMER 4	F 0.01 S t		F 5.00 S t
Timer 3			
36			36:01
MULT SPD3 TIMER 3 MULT SPD 4 ■ TIMER 4			F 5.00 S t
Timer 4			
38			38:02
MULT SPD5 ■ TIMER 5 MULT SPD6 TIMER 6			F 5.00 S t
Timer 5			
40			40:00
MULT SPD5 TIMER 5 MULT SPD6 ■ TIMER 6			F 5.00 S t
Timer 6			
42			42:01
MULT SPD7 ■ TIMER 7			F 5.00 S t
Timer 7			

[Rotation direction and acceleration/deceleration time] Function codes

30 32 34 36 38 40 42

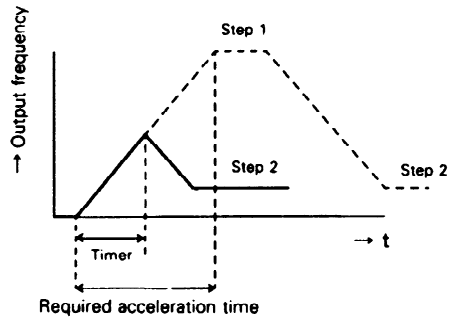
By using the chart below to see the code, rotation direction and acceleration/deceleration time can be determined.

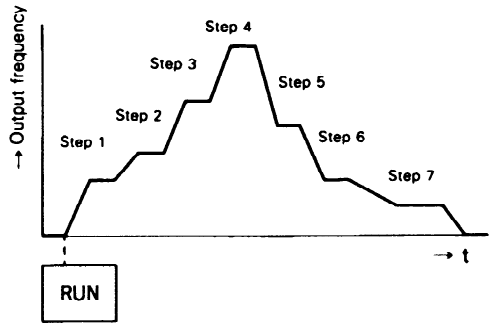
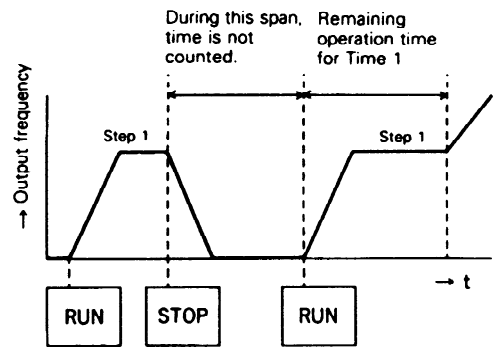
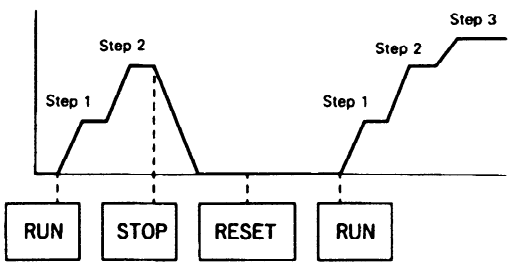
Data Code	Rotation Direction	Acceleration Time · Deceleration Time		
00	Forward (FWD)	ACCEL 1 08	09	DECEL 1
01		ACCEL 2 23	26	DECEL 2
02		ACCEL 3 24	27	DECEL 3
03		ACCEL 4 25	28	DECEL 4
04	Reverse (REV)	ACCEL 1 08	09	DECEL 1
05		ACCEL 2 23	26	DECEL 2
06		ACCEL 3 24	27	DECEL 3
07		ACCEL 4 25	28	DECEL 4

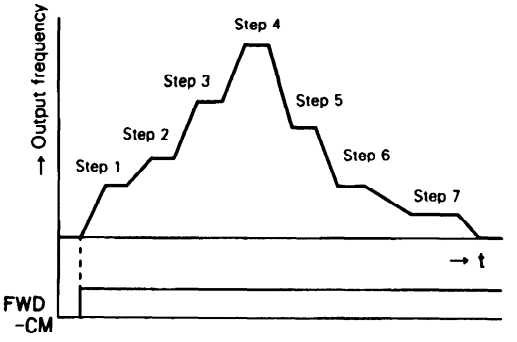
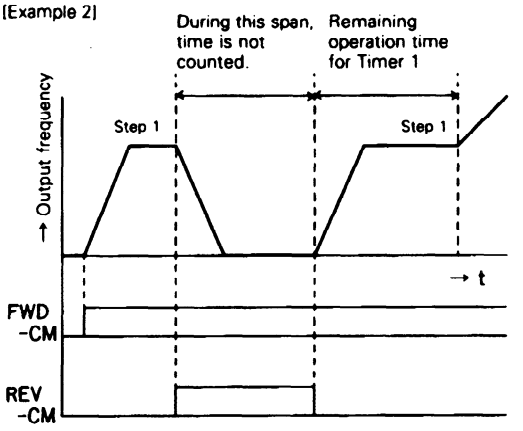
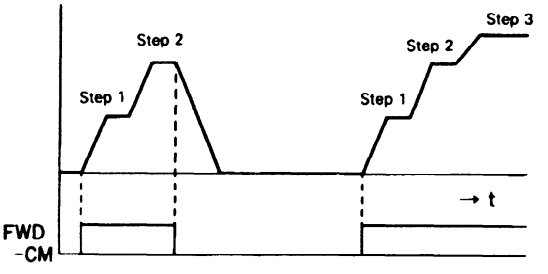
Function Code	Display	Function Explanation	Factory Setting
		When the next frequency in the process is higher than the present frequency, the acceleration time will be automatically selected. If the next frequency is lower than the present frequency, the deceleration time will be automatically selected. [Example 1] when data code is 3200  [Example 2] when data code is 3204  [Timer Setting] After setting the rotation direction and acceleration/deceleration time, set the timer.	

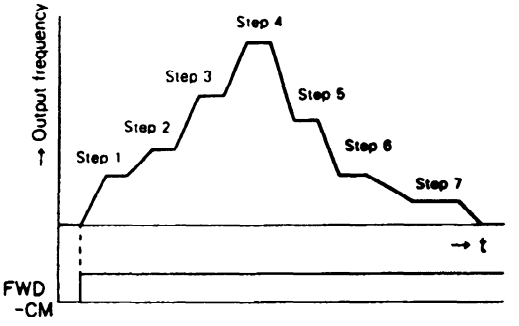
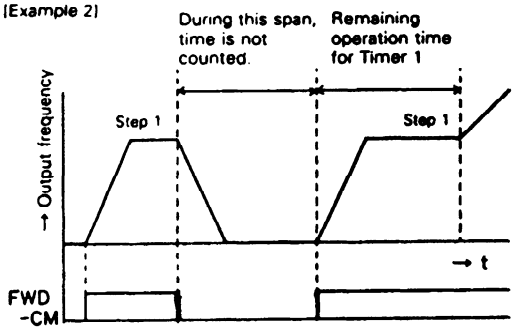
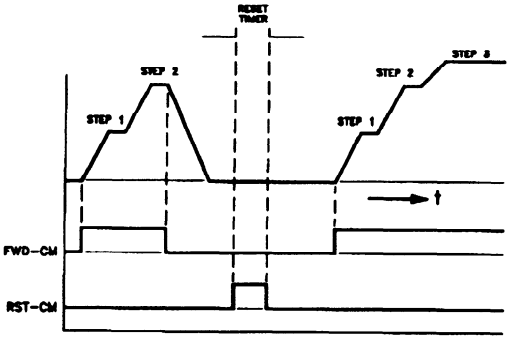
Timer Setting	Setting Resolution
0.01~9.99sec	0.01sec
10.00~99.90sec	0.1sec
100.0~999.0sec	1sec
1000~3600sec	10sec

The time set on the timer includes the time required for acceleration/deceleration. Therefore, if the time set on the timer is less than the time required for the acceleration.deceleration conditions, it will proceed to the next process before the set step frequency is reached.

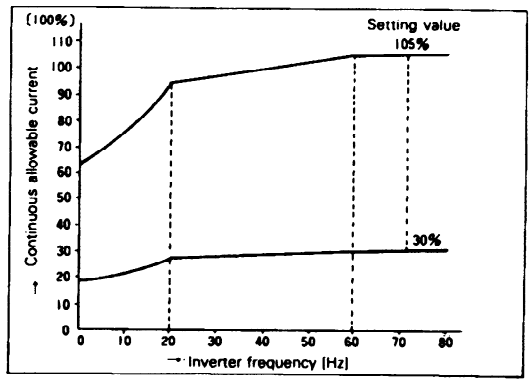
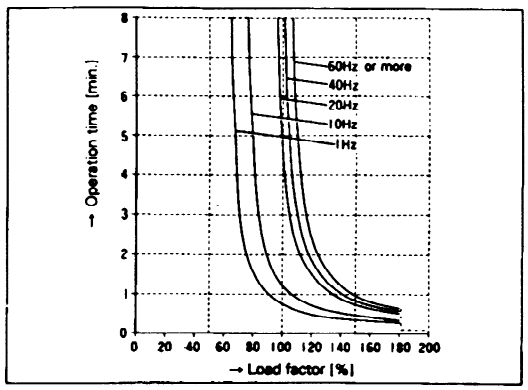


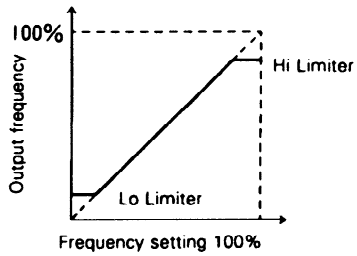
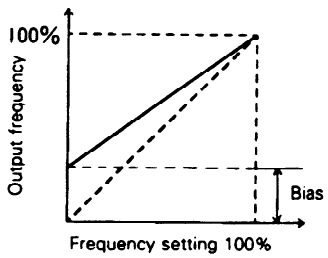
Function Code	Display	Function Explanation	Factory Setting
		(Pattern operation method) ① Keypad Panel Operation The key functions are changed as follows: RUN key : Start operation STOP key: Temporary stop of inverter operation (pause) * RESET key : Timer reset (The next RUN input will start operation from Step 1) * RESET key only functions when the inverter is stopped (or paused). [Example 1]  [Example 2]  [Example 3]  	

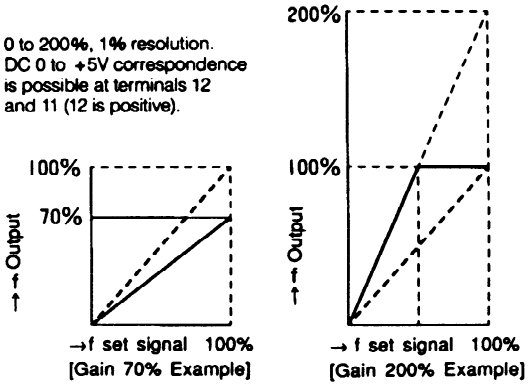
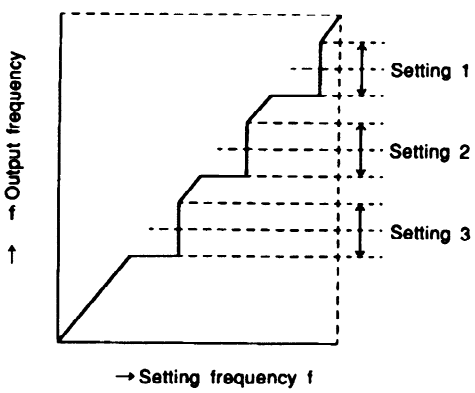
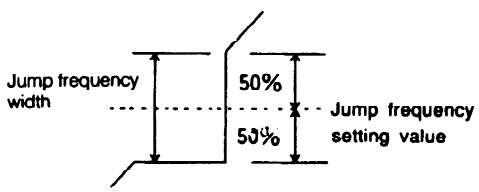
Function Code	Display	Function Explanation	Factory Setting
Pattern Operation in Remote Mode for inverters with serial numbers starting with 29 and earlier.		② Control Terminal (Remote) Operation The terminal functions are changed as follows: FWD-CM = ON : Start operation FWD-CM = OFF : Stop timer operation (resets timer) REV-CM = ON : Temporarily stop timer (pause) [Example 1]  [Example 2]  [Example 3]  	
		① During pattern operation, if RT1-CM, RT2-CM terminals are ON/OFF, operation will switch to acceleration, deceleration times set by 23 to 28 ② During pattern operation, if X1-CM, X2-CM, X3-CM terminals are ON/OFF, operation will switch to multi-step frequencies 1 to 7 set by 29 31 33 35 37 39 41 however, the timer is not stopped by operation of RT1, RT2, X1, X2, or X3 terminals.	

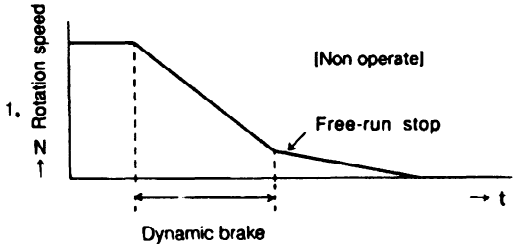
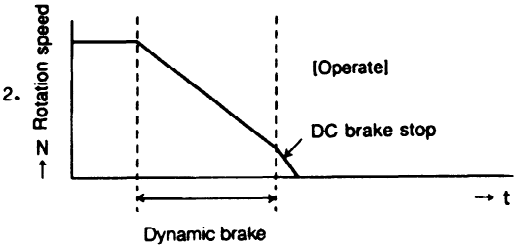
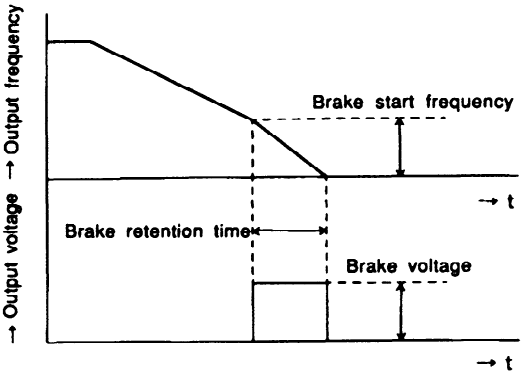
Function Code	Display	Function Explanation	Factory Setting
Pattern Operation in Remote Mode for inverters with serial numbers starting with 2X and greater.		**Control Terminal (Remote) Operation** **The terminal functions are changed as follows:** **FWD-CM = ON : Start operation** **FWD-CM = OFF : Temporary stop of inverter operation (pause)** **RST-CM = Momentary ON : Timer Reset (the next FWD-CM = ON input will start operation from step 1.** [Example 1]  [Example 2]  	
		① During pattern operation, if RT1-CM, RT2-CM terminals are ON/OFF, operation will switch to acceleration, deceleration times set by 23 to 28 ② During pattern operation, if X1-CM, X2-CM, X3-CM terminals are ON/OFF, operation will switch to multi-step frequencies 1 to 7 set by 29 31 33 35 37 39 41 however, the timer is not stopped by operation of RT1, RT2, X1, X2, or X3 terminals.	

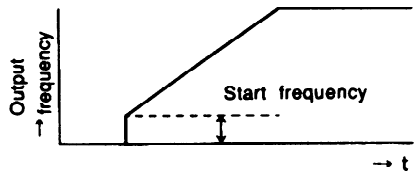
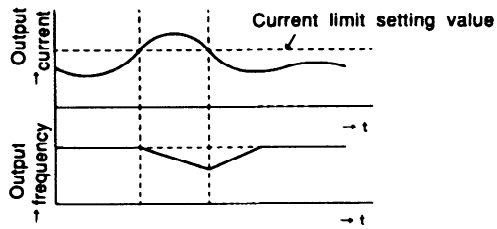
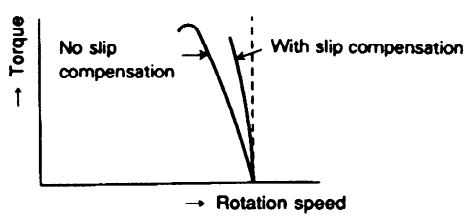
Function Code	Display	Function Explanation	Factory Setting
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Function Code	Display	Function Explanation	Factory Setting
43: ■ ERCTRN OL H LIMITER L LIMITER FREQ BIAS Electronic Thermal Overload Relay	4300 43:01 t 90%	Commonly used motor (4 pole) over-heating protection is possible. (all frequency range protection possible) **See Caution** Nonoperate Operate (electronic thermal protection operation) Setting range 30 to 105%, 1% resolution. **Setting Value [%] = K x Motor Rated Current x 100 / Inverter Rated Current [Output]** **Load Factor [%] = K x Running Current x 10000 / Setting Value [%] x Inverter Rated Current** K = 1.0 (when 50Hz is base frequency) K = 1.1 (when 60Hz is base frequency) Continuous allowable current characteristics  Electronic Thermal Operation Characteristics  **CAUTION: The electronic thermal overload relay meets the requirements of 4-Pole standard motors only. When any of the following conditions exist, use conventional overload relays instead of the electronic overload built into the inverter.** -- Motors other than 4-Pole type -- Special Motors (any non-standard) -- For a group operation where two or more motors are run using a single inverter. -- Application has frequent starting/stopping. -- Operation is at 10 Hz or less.	43:01 t 90%

Function Code	Display	Function Explanation	Factory Setting
44 ERCTRN OL ■ H LIMITER L LIMITER FREQ BIAS High Limiter	44 F 100 % F.SET	The output frequency for the upper limit as well as the lower limit are set at a % of the maximum frequency (Function Code 11). 0 to 100%, 1% resolution	44 F 100 % F.SET
45 ERCTRN OL H LIMITER ■ L LIMITER FREQ BIAS Low limiter	45 F 0 % F.SET	0 to 100%, 1% resolution 	45 F 0 % F.SET
46 ERCTRN OL H LIMITER L LIMITER ■ FREQ BIAS Bias Frequency	46 F 0 % F.SET	Outputs a biased frequency corresponding to the frequency setting Set as a % of the maximum frequency 0 to 100%, 1% resolution.  NOTE: Frequency Bias is operative only when frequency setting method is analog (Function Code 16 = 01 or 02). See WARNING in Section 1. WARNING: Bias is operational when Function Code 16 = 01 or 02. When the inverter does not have run command, the reference frequency flashes on the display. When bias is operational and the speed ref. is zero, the display flashes zero. When a run command is given, the motor runs at the bias setting, even though the display was flashing zero.	46 F 0 % F.SET

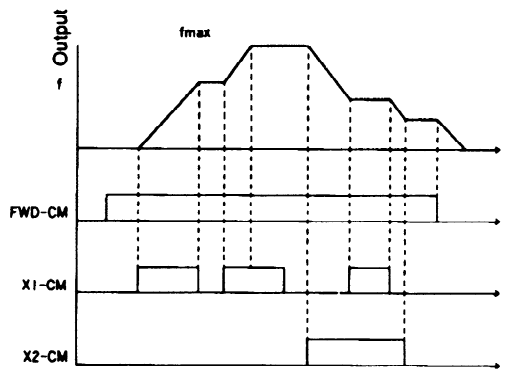
Function Code	Display	Function Explanation	Factory Setting
47 ■ FREQ GAIN JUMP Hz 1 JUMP Hz 2 JUMP Hz 3 Gain For Frequency Setting Signal	47 F ↕ 100 % F.SET	A rationed frequency corresponding to the frequency setting is output. It is set as a % of the maximum frequency. (Higher than the maximum frequency is not output) 0 to 200%, 1% resolution. DC 0 to +5V correspondence is possible at terminals 12 and 11 (12 is positive).  Frequency gain is operative only when frequency setting method is analog 1601 、 1602	47 F ↕ 100 % F.SET
48 FREQ GAIN ■ JUMP Hz 1 JUMP Hz 2 JUMP Hz 3 Jump Frequency 1	48 F 0 Hz F.SET	The frequency jumps to prevent the load mechanical resonance point and the inverter output frequency from overlapping. (3 location setting possible) 0 to 400Hz, 1Hz resolution.	48 F 0 Hz F.SET
49 FREQ GAIN JUMP Hz 1 ■ JUMP Hz 2 JUMP Hz 3 Jump Frequency 2	49 F 0 Hz F.SET		49 F 0 Hz F.SET
50 FREQ GAIN JUMP Hz 1 JUMP Hz 2 ■ JUMP Hz 3 Jump Frequency 3	50 F 0 Hz F.SET	A 1, 2, 3, 4, or 5Hz jump width is possible.	50 F 0 Hz F.SET
51 ■ J HYSTR DC BRAKE DC BRK Hz DC BRK U Jump Frequency Range	51 F ↔ 0 Hz F.SET	 **NOTE: Function Codes 48, 49, and 50 set the Jump Frequency Points, while Function Code 51 sets the width of the jump.**	51 F ↔ 0 Hz F.SET

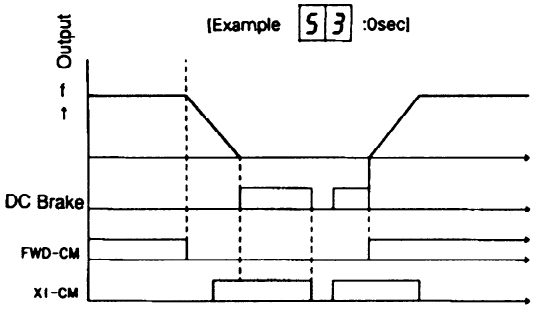
Function Code	Display	Function Explanation	Factory Setting
S2 J HYSTR ■ DC BRAKE DC BRK Hz DC BRK U DC Brake	S200 S201	Switches current brake between Operate / Nonoperate. Nonoperate (only dynamic brake) (Illustration 1).  Operate (DC brake after dynamic brake) (Illustration 2). 	S200
S3 J HYSTR DC BRAKE ■ DC BRK Hz DC BRK U DC Brake Starting Frequency	S400 ~ S415	Setting for the DC brake starting frequency 0 to 60Hz, 1Hz resolution (except at 0Hz braking is from 0.2 Hz) Week ↓ Setting can be made from 16 different DC brake strength settings. Strong	S3 F 0 Hz t S400
S4 J HYSTR DC BRAKE DC BRK Hz ■ DC BRK U DC Brake Voltage		Brake strength changes according to the motor characteristics	
S5 ■ DC BRK T START Hz I LIMITER SLIP COMP DC Braking Time		A 0.01 to 30 sec. time span setting is possible (try to match load inertia). Setting Range: 0.01~9.99sec — Setting Resolution: 0.01sec Setting Range: 10.00~30.00sec — Setting Resolution: 0.1sec 	S5 F 0.10 S t

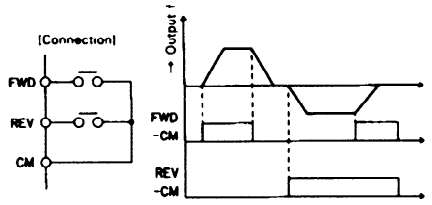
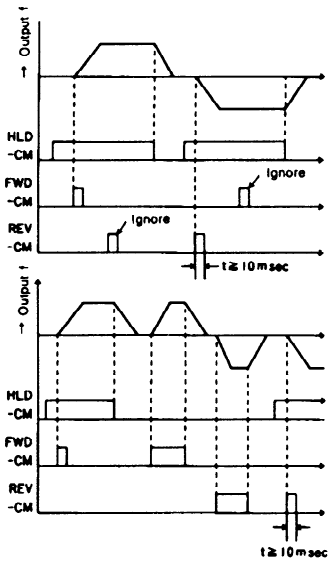
Function Code	Display	Function Explanation	Factory Setting
56 DC BRK T ■ START Hz I LIMITER SLIP COMP Starting Frequency	56 1 Hz	The start frequency setting can be from 0.2Hz to 60Hz, 1Hz resolution. 0Hz (0.2Hz start) ↓ 60Hz 	56 1 Hz
57 DC BRK T START Hz ■ I LIMITER SLIP COMP Current Limiter	57 150 % 5700 5701	When the output current reaches the current limit level, the output frequency is automatically lowered to avoid over-current trip. **NOTE: If I Limiter is set to Operate Mode and OC1, OC2, or OC3 trips still occur, try adjusting Function Code 76.** Nonoperate Operate (operates at 30 to 150% of the inverter rating, 1% setting resolution.  ① Does not operate for deceleration. ② When using this function, the automatic acceleration/deceleration function does not operate.	5701 (150%)
58 DC BRK T START Hz I LIMITER ■ SLIP COMP Slip Compensation Control	5800 5801	Automatically judges the relative size of the load by the output current fluctuations, and regulates the inverter frequency for consistent rpm operation. (controls speed fluctuations to 1/3 of normal slip) Nonoperate (no slip compensation) Operate (with slip compensation) **NOTE: Factory recommended setting is non-operational for most applications due to interference with torque boost operation.**  Operates only during constant speed operation.	5800

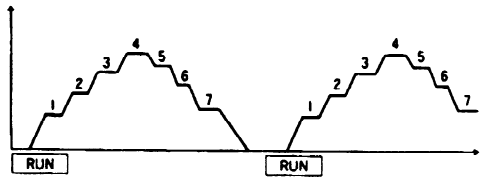
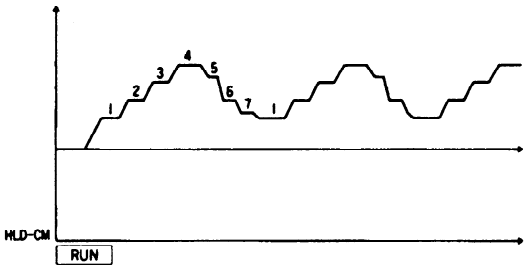
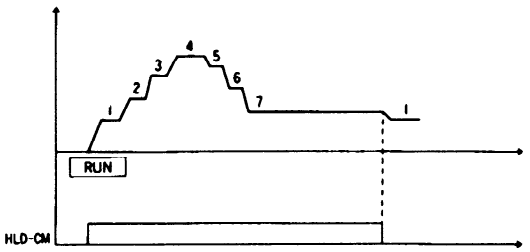
Function Code	Display	Function Explanation	Factory Setting
59 ■ FDT Hz FDT HYSTR RUN FINSH OL WARN **Frequency Detection Threshold**	59 F 60 Hz FDT	**The FAR-CM terminal ON signal is output when the output frequency goes above the frequency detection threshold.** The OFF level is set by 60 0 to 400Hz, 1Hz resolution.	59 F 60 Hz FDT
60 FDT Hz ■ FDT HYSTR RUN FINSH OL WARN FDT And FAR Signal Hysteresis	60 F 10 Hz FDT	0 to 300Hz, 1 Hz resolution. → Output frequency; FDT signal output start frequency; Hysteresis width; → t; FAR-CM; → t When 6801 is set, the FDT signal is operative.	60 F 10 Hz FDT
61 FDT Hz FDT HYSTR ■ RUN FINSH OL WARN RUN Signal Finishing Frequency	61 F 0 Hz RUN	The RUN-CM terminal ON signal is output when the output frequency goes above the RUN signal finishing frequency. 0 to 400Hz, 1Hz resolution. → Output frequency; RUN signal output frequency; → t; RUN-CM; → t When 6900 is set, the RUN signal is operative.	61 F 0 Hz RUN
62 FDT Hz FDT HYSTR RUN FINSH ■ OL WARN Overload Early Warning Signal	62 I 100 % OL	The OL-CM terminal ON signal is output when the output current exceeds the overload early warning signal value for over 10 sec. 70%to 150%, 1% resolution (100% corresponds to inverter rating) →Output current; Warning level; OL-CM; 10 Second; 1 Second → t When 6700 is set, the OL signal is operative.	62 I 100 % OL

High Level Functions

Function Code	Display	Function Explanation	Factory Setting
63 Hz rpm A m/min ■ X1-X2-X3 HOLD FUNC LV-OL-FAR X1, X2, and X3 Terminal Function	6300	The terminal X1, X2, X3 functions can be changed to the following 3 types via the data setting. 8 step multi-step frequency operation is possible through switching X1, X2, X3. See function codes 29 31 33 35 37 39 41 for details on multistep frequency operation.	6300
	6301	During operation, with RUN key or FWD, REV terminals: When X1-CM is ON: output frequency increase When X1-CM is OFF: output frequency fixed When X2-CM is ON: output frequency decrease When X2-CM is OFF: output frequency fixed The up/down variable speed is determined by the acceleration/deceleration time setting value. ① X3-CM terminal ON/OFF is ignored. ② When both X1-CM and X2-CM are ON at the same time, the frequency at that time is fixed. ③ Hi Limiter And LO Limiter take precedence (44 , 45) ④ Always at 0Hz for operation start ⑤ Operative only when frequency setting is by digital method 1600 ⑥ When there is no operation command (RUN , FWD, REV), X1, X2 input is ignored. ⑦ ∧ ∨ key input is ignored. 	

Function Code	Display	Function Explanation	Factory Setting
	6302	When DC brake is selected by 5201: with X1-CM = ON: DC brake during stop with X1-CM = OFF: brake reset  ① The strength of the DC brake is set by 54 ② Operation command take precedence. (RUN , FWD, REV) (While X1-CM is ON, if FWD-CM is ON: DC brake reset) ③ X2-CM, X3-CM terminal ON/OFF is ignored. ④ As DC current continues to flow to the motor during the X1-CM - On period, be careful of temperature rise, (When long periods of DC braking are required, measure the temperature continuously). ⑤ DC brake frequency (Function Code 53) and DC brake time (Function Code 55) are ignored. CAUTION: Motor may overheat if DC brake is left turned ON for too long.	

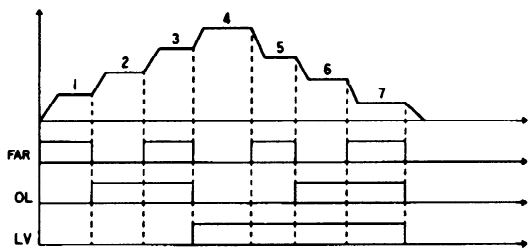
Function Code	Display	Function Explanation	Factory Setting
64 Hz A V X1-X2-X3 ■ HOLD FUNC LV-OL-FAR FWD/REV Command Hold (3-Wire Control)	6400	Depending on the data selection, the HLD terminal function can be changed to the following 3 types. 2 wire operation (Maintained Contacts) When terminal strip (remote) operation (1501) is selected, the following operation results.  When both FWD-CM and REV-CM are ON at the same time, deceleration stop results. HLD terminal input is ignored.	6401
	6401	3 wire operation (Momentary Contacts) When Terminal strip (remote) operation (1501) is selected, the following operation results.  ① When HLD-CM is ON : and FWD-CM is ON for over 10 msec. : Forward HLD-CM is OFF : Stop ② When HLD-CM is on and REV-CM is ON for over 10 msec. : Reverse HLD-CM is OFF : Stop ③ During the period when HLD-CM is ON, either the FWD or REV signal, whichever is input first, takes precedence. ④ When HLD-CM is OFF, if FWD and REV signal are input, 2 wire operation will result. ⑤ Operation will not start if FWD, REV signal t (time) is less than 10 msec.	

Function Code	Display	Function Explanation	Factory Setting
	6402	Switches operation mode. When 6400 6401 with 15 = 01, 19 = 01, and 64 = 01, HLD is ignored.  When 6402 with HLD-CM in OPEN, returns to the first step after 1 cycle of operation  With HLD-CM in ON, maintains the 7th step level after 1 cycle of operation.  ① Only operative when 1901 pattern operation is selected. With 1900, normal 2 wire (Maintained Contacts) operation results (same as 6400).	

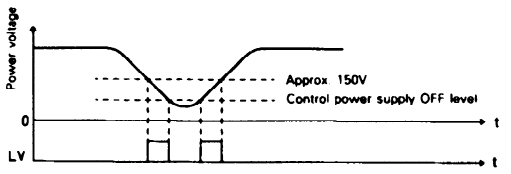
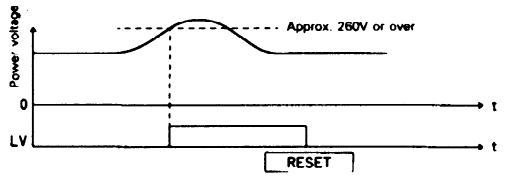
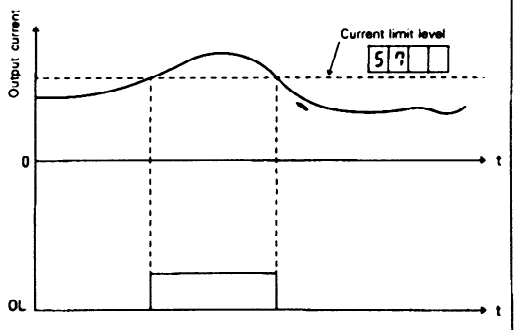
Function Code	Display	Function Explanation	Factory Setting
65		The signal data output from LV, OL, FAR terminals can be changed.	6500
X1-X2-X3 HOLD FUNC ■ LV-OL-FAR	6500	LV terminal outputs the signal selected by 66	
		OL terminal outputs the signal selected by 67	
LV, OL and FAR Terminal Output Code		FAR terminal outputs the signal selected by 68	
	6501	When 1901 pattern operation is selected, a binary signal equivalent to 1st speed → 7th speed is output from LV, OL, FAR terminals.	

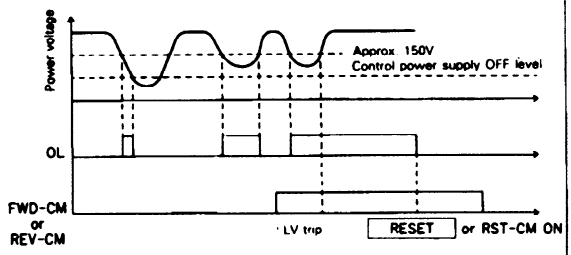
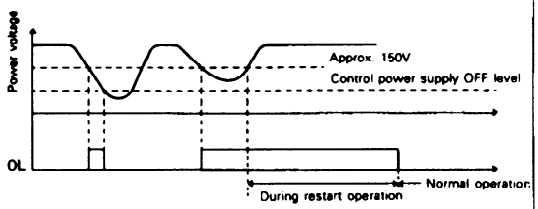
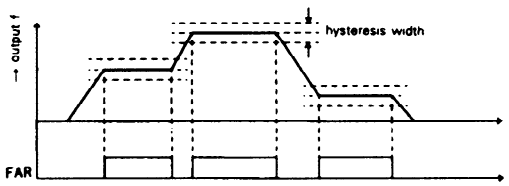
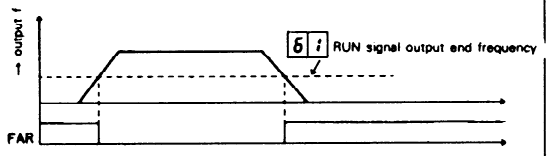
	1st Speed	2	3	4	5	6	7
FAR	1	0	1	0	1	0	1
OL	0	1	1	0	0	1	1
LV	0	0	0	1	1	1	1

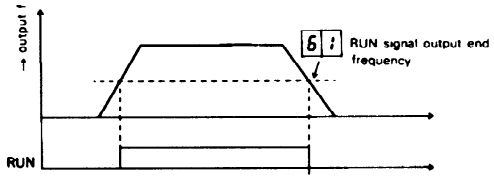
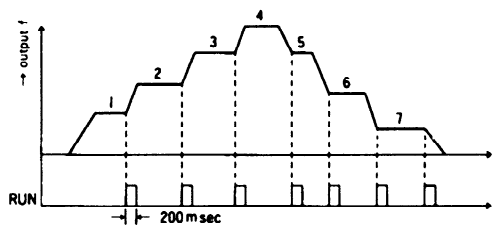
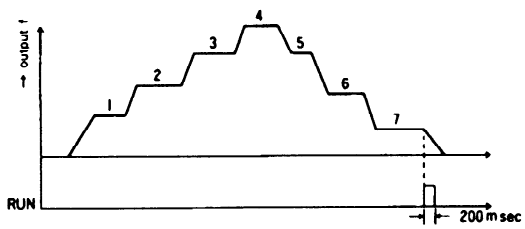
1 : ON 0 : OFF



With 1900, there is no output.

Function Code	Display	Function Explanation	Factory Setting
66 ■ LV FUNC OL FUNC FAR FUNC LV Terminal Function	6600	The function for terminal LV can be switched between the following 2 types. Only operative for 6500 Under-voltage signal (LU signal) is output from LV.  The LU signal is output from the time the power voltage drops below approx. 150V to the point where the control power no longer exists.	6600
	6601	The over-voltage signal (OU signal) is output from the LV terminal.  The OU signal, when activated for over-voltage protection, is output from the LV terminal. The OU signal is maintained until alarm reset is performed. (RESET key or RST-CM (momentary) = ON)	
67 LV FUNC ■ OL FUNC FAR FUNC OL Terminal Function	6700	The function for terminal OL can be switched between the following 3 types. Only operative for 6500 The inverter over-load early warning signal (OL signal) is output from the OL terminal. See 62 for signal details.	6700
	6701	The current-limiting monitoring signal (CL signal) is output from the OL terminal. 	

Function Code	Display	Function Explanation	Factory Setting
	6702	The under voltage signal (IP signal) is output from the OL terminal. Signal output when in 2000 AUTO restart nonoperate.  Signal output when in 2001 restarting. operate. 	
68 LV FUNC OL FUNC ■ FAR FUNC FAR Terminal Function	6800	The signal output from the FAR terminal can be selected from the following 3 types. The frequency equivalence detection signal (FAR signal) is output from the FAR terminal. The hysteresis width is set by 60 	6800
	6801	The setting frequency detection signal (FDT signal) is output at the FAR terminal. See 59 for signal details.	
	6802	During inverter operation "Not Run" signal (STOP signal) is output as the FAR terminal. The signal ON/OFF level is set by 61 The RUN signal's inverse is output. 	

Function Code	Display	Function Explanation	Factory Setting
69 ■ RUN FUNC FM FUNC Run Terminal Function	6900	The following 3 types of signal output at the RUN terminal can be selected. The inverter operation signal (RUN signal) is output at the RUN terminal. The signal ON/OFF level is set by 61 	6900
	6901	For 1901 pattern operation, the time-up signal (TP signal) is output by the RUN terminal. (outputs at time-up point of each timer)  No signal output when 1900	
	6902	For pattern operation 1901 , the cycle completed signal (TO signal) is output at the RUN terminal. (outputs at 7 step end point)  Also for the continuation of timer operation by the TO signal is output at 7 step end point. 6402	

Function Code	Display	Function Explanation	Factory Setting
70 Hz rpm ▲ ▼ M/min RUN FUNC ■ FM FUNC FM Terminal Function	7000 7001	**The following two types of output monitor signals from the FM terminal can be selected:** **Frequency Meter:** 0 to 10 V corresponds to 0 to top frequency At top frequency, output can be adjusted between 6.5 and 10.5 VDC via Function Code 03. **Output Current Monitor:** 0 to 10 V corresponds to 0 to 150% current (100% = Nameplate Output Current) Calibration adjustment is performed by Function Code 03.	7000

Function Code	Display	Function Explanation	Factory Setting
7 1 ■ NO. ENTRY TL UNITS Inverter Unit No. Entry For Link Operation (All Inverters)	7100 7101 7102 7103 7104 ⌇ 7115	Determines the function of the various inverters when performing link operation. Err4 will result when 2 identical numbers are recorded within the same group. for Master record Slave No.1 (Auxiliary No.1 for input terminal link) Slave No.2 (Auxiliary No.2 for input terminal link) Slave No.3 (Auxiliary No.3 for input terminal link) Slave No.4 ⌇ Slave No.15 Always record the inverter numbers consecutively from lowest to highest.	7115
72 NO. ENTRY ■ TL UNITS Number of Units Linked (Central Inverter)	7200 ⌇ 7215	**Needed only for Master setting:** **Record the total number of slave units connected -** **When the total number connected is greater than the recorded number.** — **The number of not linked inverters will appear** **When the total number connected is less than the recorded number.** — **Err 4**	7200
73 ■ LINK MODE INPUT SEL NO. SELECT Link Mode (All Inverters)	7300 7301 7302 7303	Link Mode Selection Link Nonoperate (After Err4 reset, automatically returns to 7300) → (for Mater) Frequency Setting → SET → 80XX Monitoring Signal → SET → 81XX Link Operation → SET → 82XX	7300

Function Code	Display	Function Explanation	Factory Setting
7 4		Setting for Master (required for link operation)	7 4 0 0
LINK MODE ■ INPUT SEL NO. SELECT	7 4 0 0	Selects interface for Master 1502 (link) operation control RUN/STOP via operation panel	
Run Command Input In Link Operation (Central Inverter)	7 4 0 1	RUN/STOP via terminal block	
7 5		Setting for Master (required for link operation)	7 5 0 0
LINK MODE INPUT SEL ■ NO. SELECT	7 5 0 0	Record command inverter no. (Individual Command)	
Inverter Unit No. (Central Inverter Only)	7 5 1 5		
	7 5 1 6	(Total Number Command)	
7 6	7 6 0 0	Current Limiter 1 Current Limitor operates during acceleration and constant speed running.	7 6 0 0
■ ACR GAIN OPTION 2 OPTION 3 OPTION 4	7 6 0 1	Current Limiter 2 Current Limiter operates during constand speed running. Gain is adjustable 7 6 0 1 High Response Speed	
Changeover of Current Limiter 1 and Current Limiter 2. Gain adjustment in Current Limiter 2.	7 6 9 9	7 6 9 9 Low Response Speed NOTE: Current Limiter enabled/disabled and Limit Level adjustment can be done through Function Code 57.	
8 0 8 1 8 2		These Function Codes may be accessable on certain early model inverters (serial numbers beginning with 29 and earlier), but are currently not used.	

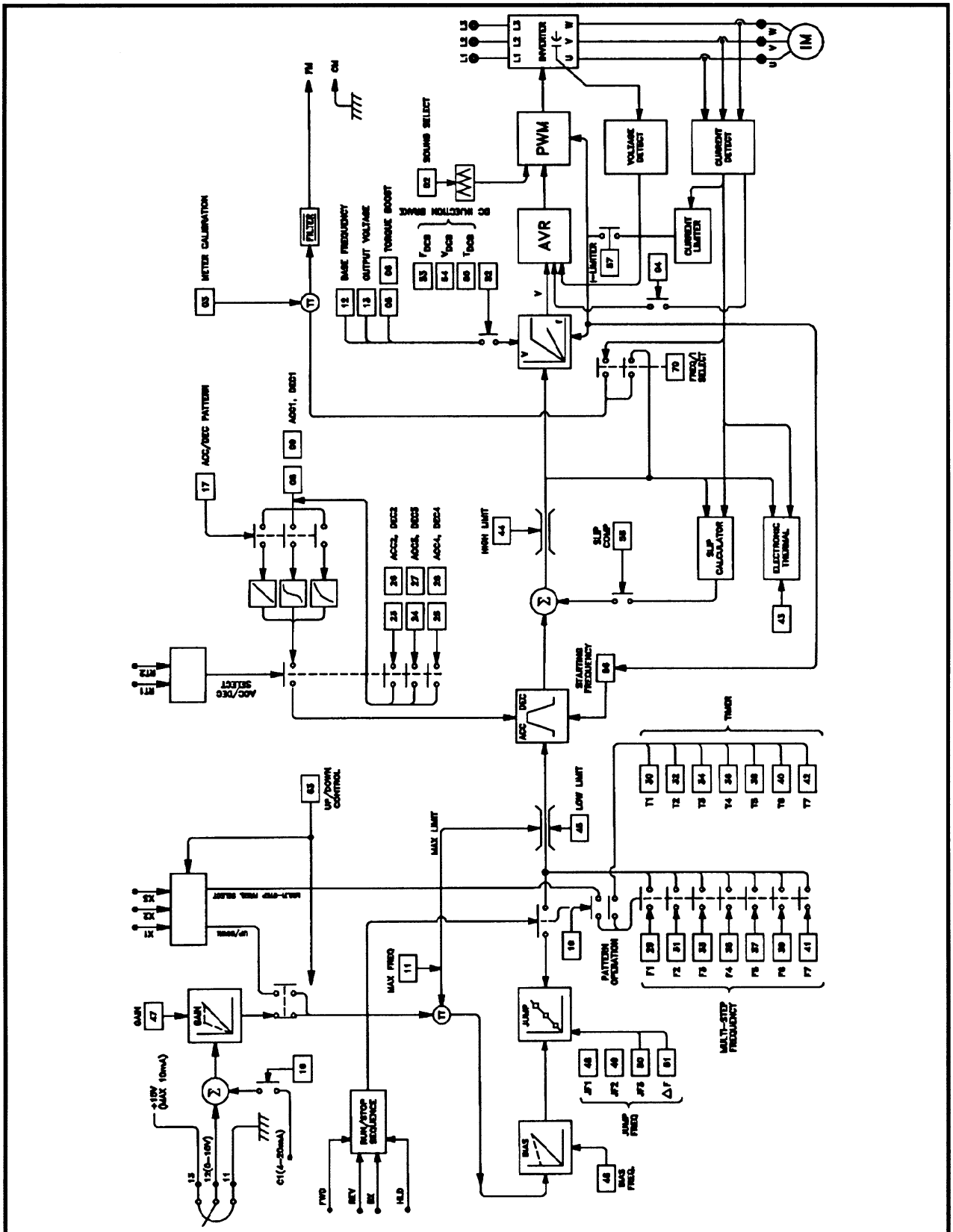


Figure 6-1. AF-300B FUNCTION CODE MAP

NOTES:

7. MAINTENANCE AND INSPECTION

So that the AF-300B inverter may give long periods of trouble-free operation and to prevent potential problems from occuring, the following items should be periodically inspected.

CAUTION: Do not conduct any inspections until after disconnecting the power supply and after the "CRG" light on the inverter has gone out.

MEGGER TEST

1. Disconnect all inverter terminals and never apply test voltage to the inverter when conducting an external circuit megger test.
2. Perform the test only on the main circuit as shown in Figure 7-1 when conducting a megger test on the inverter itself.
   -- Do not conduct a megger test on the control circuits.
   -- Short L1, L2, L3, U, V, W, P, N, DB, and P, then megger to ground.
3. Use a high resistance range type tester to conduct a continuity test on the control circuits and not a megger or a buzzer.

PERIODIC PARTS REPLACEMENT

The life of the inverter will vary according to the installation environment and the amount of running time. However, if continuous operation is within the allowable limits, the life of the DC Link capacitor is approximately five years and the life of the cooling fan is approximately three years. It is recommended that these parts be replaced before failure occurs.

INSPECTION ITEMS

Refer to the Inspection Items Chart in this section for inspection points and corrective action.

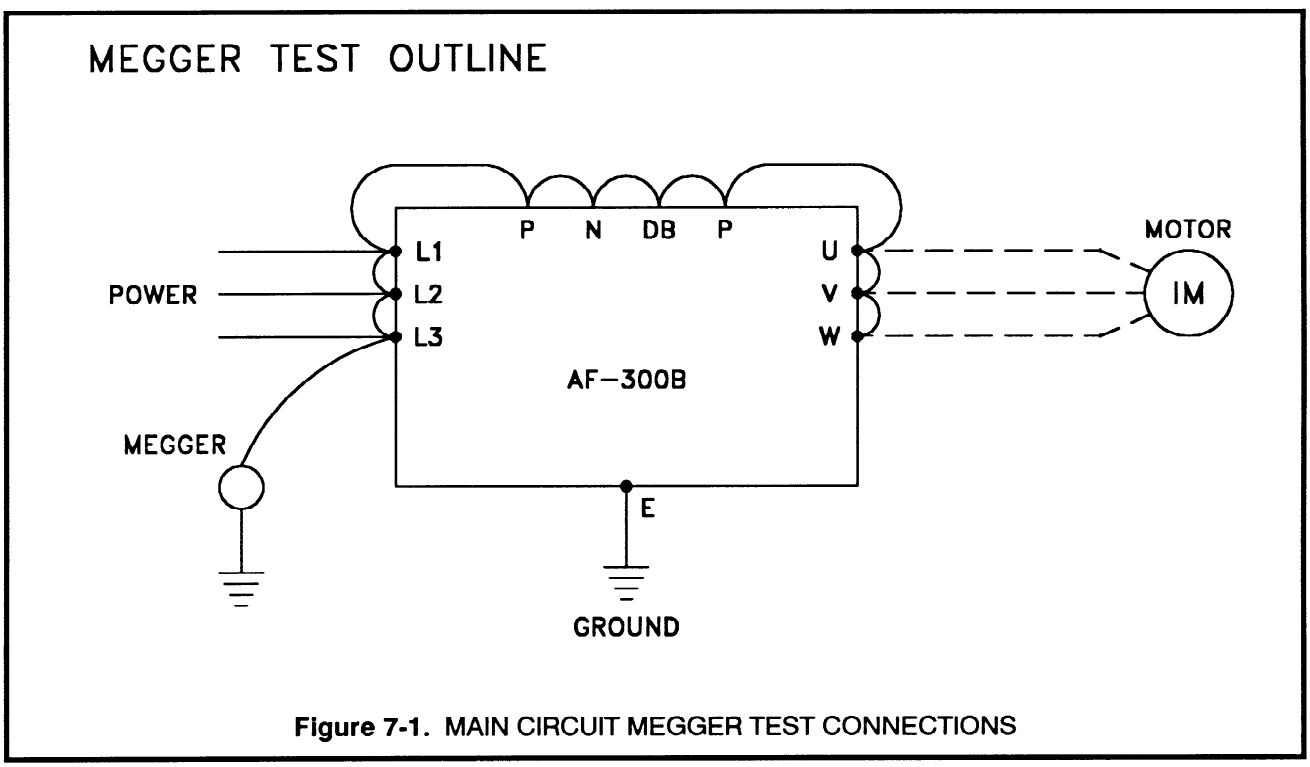


Figure 7-1. MAIN CIRCUIT MEGGER TEST CONNECTIONS

INSPECTION ITEMS CHART		
Item	**Inspection Criteria**	**Corrective Action**
Power Source Voltage	Within permissible limits (170 - 253 VAC) or (323 - 506) for 400-460 VAC inverters.	Adjust the power supply voltage
Ambient Temperature	Within permissible limits (-10° to +50° C) (+14° to +122°F)	Investigate cause and make corrections until environment is within permissible limits.
Ambient Humidity	Within permissible limits (20 - 90% RH) No dew condensation or freezing	Investigate cause and make corrections until environment is within permissible limits.
Vibration	Within permissible limit (0.6G or less)	Investigate cause and make adjustments until within permissible limits.
Noise	Abnormal Audio Noise from cooling fan, etc.	Contact the supplier where the inverter was purchased.
Odor	Smell or burning	Contact the supplier where the inverter was purchased.
Dust	Dust accumulation on cooling fins, cooling fan, or on the control board.	Clean and blow out with compressed air.
Screws/ Connectors	Check for any loosening	Retighten as needed.

MEASUREMENT POINTS AND METERS

Since the inverter's input/output voltage and current contain high frequencies, selection of the wrong measuring device can lead to gross miscalculations.

When using a CT (current-detection transformer) to measure the current, the amount of error will be large if the frequency is low. Because of this, always use a CT with as large a capacity as possible. See the following chart and Figure 7-2 for recommended measurement devices.

RECOMMENDED MEASUREMENT DEVICE CHART		
Item	**Simple Measurement**	**Precision Measurement**
Input Voltage	Tester	Moving-Iron type voltmeter
Input Current	Clamp Meter	Moving-Iron type ammeter
Input Power	--	Electrodynamometer type wattmeter
Output Voltage	Tester	Rectifier type voltmeter
Output Current	Clamp Meter	Moving-Iron type ammeter
Output Power	--	Electrodynamometer type wattmeter

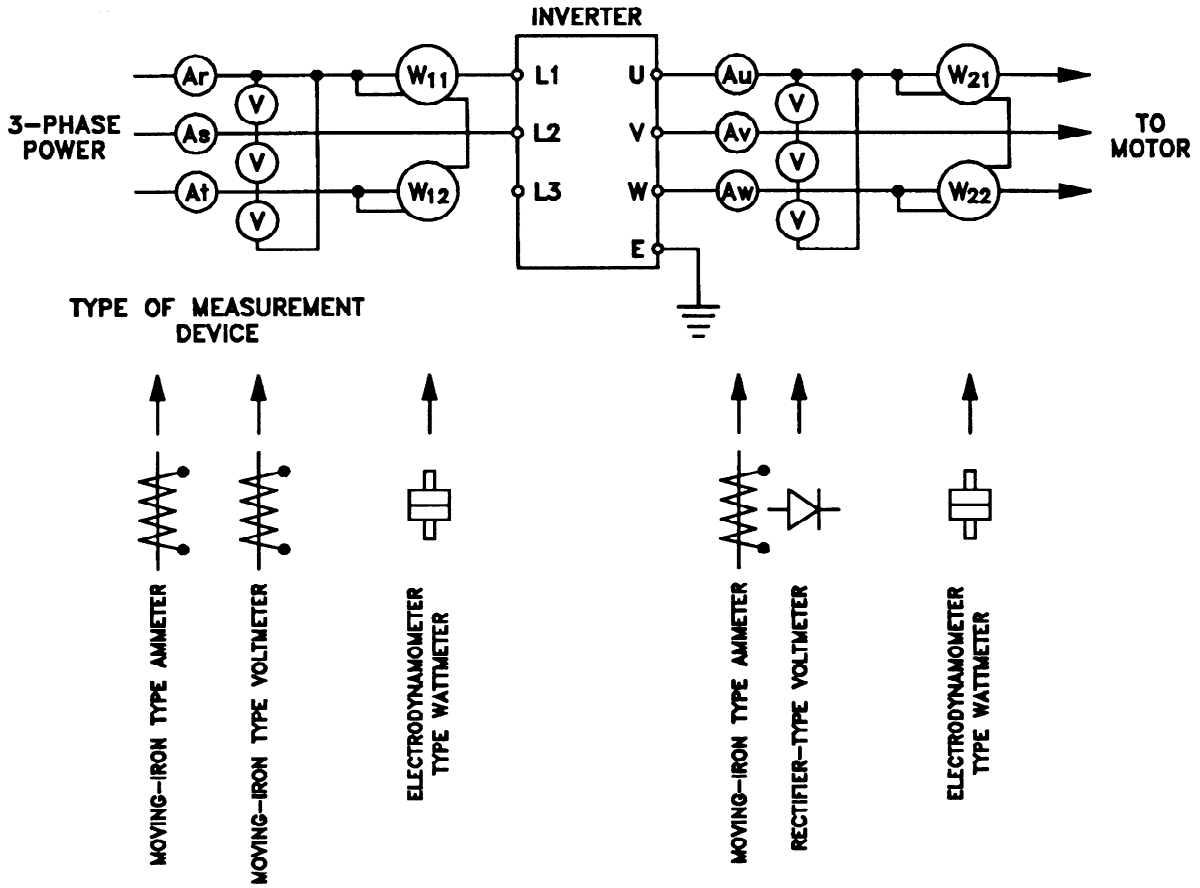


Figure 7-2. MEASUREMENT LOCATIONS AND DEVICES EXAMPLE

NOTES:

8. TROUBLESHOOTING

"OH2" TRIP ON POWER-UP

If the AF-300B inverter is configured to run from a remote operator station and a RUN command is present, upon power-up, the inverter will trip on **"OH2"**. This was the normal condition for factory configuration in the past. Inverters were factory configured for **remote** operation. Inverters are now factory configured for **keypad** operation.

-- The connection between **"FWD"** (or **"REV"**) and **"CM"** must be open during power-up when running from remote.

-- When the inverter is configured to run from the keypad, this connection will not cause the **"OH2"** trip on power-up.

If you have an inverter that is configured for **remote** operation and wish to run from the keypad, configure the inverter for **keypad** operation by performing the following procedure.

NOTE: *If you wish to run the inverter from a remote operator station, use the **UP** and **DOWN ARROW** keys to set **"1501"** in step 12 and **"1601"** in step 17, and do not reinstall the jumper in step 22.*

NOTE: *A 1K, 1/2 watt speed pot should be connected to terminals **"11"**, **"12"**, and **"13"** for the speed reference (only if Function Code 16 = 01). See Sections 4 and 6 for other possible reference signal choices.*

Keypad Operation Configuration Procedure

1. Remove AC power from the inverter and wait at least one minute for the DC link voltage to discharge.
2. Remove the terminal board access cover from the inverter.
3. Remove the wire jumper from between terminals **"FWD"** and **"CM"**. Do not throw this wire away, it will be reinstalled later.
4. Reinstall the terminal board access cover (removed in step 2) to prevent personal injury from electrical shock.
5. Apply AC power to the inverter and wait for the LED display to flash.
6. Press the **"PRG"** key on the keypad.
   -- The LED in the **"PRG"** key should light and the LED display should show **"00"**.
7. Press the **"SHIFT"** key.
   -- The LED display should show **"04"**.
8. Press the **"SHIFT"** key.
   -- The LED display should show **"08"**.
9. Press the **"SHIFT"** key.
   -- The LED display should show **"11"**.
10. Press the **"SHIFT"** key.
    -- The LED display should show **"15"**.
11. Press the **"SET"** key.
    -- The LED display will probably show **"1501"**.
12. Press the **DOWN ARROW** key until the display shows **"1500"**.
13. Press the **"SET"** key a couple times to be sure the data is entered.
14. Press the **"RESET"** key.
    -- The LED display should show **"15"**.
15. Press the **DOWN ARROW** key.
    -- The LED display should show **"16"**.
16. Press the **"SET"** key.
    -- The LED display will probably show **"1601"**.
17. Press the **DOWN ARROW** key until the display shows **"1600"**.
18. Press the **"SET"** key a couple times to be sure the data is entered.
19. Press the **"PRG"** key to exit the programming mode and wait 5 seconds.
20. Remove AC power from the inverter and wait at least one minute for the DC link voltage to discharge.
21. Remove the terminal board access cover from the inverter.
22. Reinstall the wire jumper between terminals **"FWD"** and **"CM"**.
23. Reinstall the terminal board access cover (removed in step 21) to prevent personal injury from electrical shock.
24. Apply AC power to the inverter and wait for the LED display to flash.

WARNING - MECHANICAL MOTION HAZARD: Be sure the motor and load are clear before pressing the "RUN" key because the motor will run.

25. Press the **"RUN"** key to activate the motor.
26. Use the **UP ARROW** and **DOWN ARROW** keys to control the speed of the motor.

NOTE: ***If momentary pushbuttons will be used,*** *connect the* ***"RUN"*** *pushbutton (normally open momentary contact) between terminals* ***"FWD"*** *and* ***"HLD"****, and the* ***"STOP"*** *pushbutton (normally closed momentary contact) between terminals* ***"HLD"*** *and* ***"CM"****. When wiring for reverse operation, refer to Section 4 of this Instruction Book.*

NOTE: ***If maintained contacts will be used,*** *connect the normally open contact between terminals* ***"FWD"*** *and* ***"CM"****. Make sure that the connections between* ***"FWD"*** *(or* ***"REV"****) and* ***"CM"*** *are open when power is applied to the inverter. When wiring for reverse operation, refer to Section 4 of this Instruction Book.*

TABLE 11: FAULT CONDITION DESCRIPTION AND OPERATION

Fault Cond.	Description	Disp.	Operation
Overcurrent Protection	Protects the inverter when the overcurrent flow momentarily reaches the specified protection level. OC1: During Acceleration OC2: During Deceleration OC3: During Constant Speed Oper.	OC1 OC2 OC3	Stops inverter output. Motor coast-to-stop. Fault Relay picks up. Fault signal is internally held until the fault trip is reset*.
Protection Against Momentary Power Failure Undervoltage Protection	For momentary power failure or undervoltage of greater than 15 milliseconds, the inverter is stopped. **NOTE:** *If the restart after instantaneous power failure mode is selected, operation will resume automatically if power is restored before control power voltage decays.*	LU (LV)	Stops inverter output.
Overvoltage Detection	Trips the inverter when the DC Link voltage reaches approximately 115%. (100% = 325 VDC for 230 Volt inverters or 650 VDC for 460 Volt inverters.) **NOTE:** *For protection against high AC line voltage, it is necessary to add a 3-phase contactor to disconnect power (see Figure 4-6 for wiring).*	OU (OV)	Stops inverter output. Motor coast-to-stop. Fault Relay picks up. Fault signal is internally held until the fault trip is reset*.
Inverter Overheat	Detects internal inverter overheating caused by overload operation, cooling fan failure, abnormally high ambient temperature, etc.	OH1	
External Trip	Occurs when connection between terminals THR and CM is opened.	OH2	

***Fault Signal Hold** - During a fault condition, the Fault Relay is picked-up. If power is removed from the inverter, the Fault Relay drops-out (to its normal state).*

TABLE 11: FAULT CONDITION DESCRIPTION AND OPERATION (Continued)

Fault Cond.	Description	Disp.	Operation
External Trip (continued)	Occurs when connection is made between terminals FWD or REV and CM during power-up when Function Code 15 = 01 (control terminal operation). Occurs when the inverter is running and the STOP key on the keypad is pressed when Function Code 15 = 01. (This is a safety feature to protect against having a RUN command from the control terminals and STOP command from the keypad at the same time.)	OH2	Stops inverter output. Motor coast-to-stop. Fault Relay picks up. Fault signal is internally held until the fault trip is reset*.
Electronic Thermal Overload	Performs motor overload protection when Function Code 43 is set to 01.	OL1	Stops inverter output. Motor coast-to-stop. Fault Relay picks-up. Fault signal is internally held until the fault trip is reset*.
Setting Error	Occurs when incompatible Function Codes are selected.	Err1	
Communication Error	Occurs when there is a continuous keypad panel communication error.	Err2	
DSP Error	Occurs when there is any malfunction of the internal DSP due to external electrical noise, abnormally high ambient temperatures, etc.	Err3	
Link Error	Occurs when there is a mismatch between the Function Code setting and the actual wiring during link operation.	Err4	

***Fault Signal Hold** - During a fault condition, the Fault Relay is picked-up. If power is removed from the inverter, the Fault Relay drops-out (to its normal state).*

NOTE: *Use the Keypad Panel **RESET** key or close the control terminals RST-CM to reset from a fault condition.*

NOTE: *The previous three faults are stored in the memory. This fault history information is displayed on the Graphic Display and is changed each time the **SHIFT** key is pressed (see Section 5 "Accessing Fault History" paragraph).*

TABLE 12: FAULT CONDITION DISPLAY AND CORRECTIVE ACTION

Display	Check For:	Corrective Action
OC1	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Output line short-circuited	Check output line insulation. Megger test the motor.
	Proper torque boost	Adjust as needed.
	Proper acceleration time	Lengthen the acceleration time. Check on operation of external output contactor if used. Eliminate sudden load changes. Try adjusting Function Code 76
	Other than above	Check transistors Increase inverter capacity (use larger HP inverter).
OC2	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Output line short-circuited	Check output line insulation. Megger test the motor.
	Improper deceleration time	Lengthen the deceleration time.
	Other than above	Connect DB braking resistor (optional). Increase inverter capacity (use larger HP inverter).
OC3	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Output line short-circuited	Check output line insulation. Megger test the motor.
	Sudden change in load	Eliminate sudden load changes. Try adjusting Function Code 76. Increase inverter capacity (use larger HP inverter).
	Other than above	Check for electrical noise. Check on operation of external output contactor if used.
OU	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Proper deceleration time	Lengthen the deceleration time.
	Other than above	Connect DB braking resistor (optional) or check operation of if already present.
OH1	Inverter ambient temperature not within permissible limits	Correct to proper temperature range.
	Cooling fan not operating (over 1.5 Kw)	Replace cooling fan.
	Load is over permissible limits	Reduce load. Increase inverter capacity (use larger HP inverter)
	Internal power supply overload	Check DC voltage between terminals 11 and 13 (should be 10 VDC; if not, correct the speed pot wiring).

NOTE: *Motor will coast-to-stop when fault condition is displayed. After correcting the cause of the problem, reset by pressing the RESET key on the Keypad Panel, or momentarily close contact between the RST-CM control circuit terminals (after the motor has stopped).*

NOTE: *"LU" is displayed when the AC supply is switched ON or OFF and does not indicate any abnormality.*

TABLE 12: FAULT CONDITION DISPLAY AND CORRECTIVE ACTION (continued)

Display	Check For:	Corrective Action
OH2	Proper setting of Function Code 15*	Open the connection between FWD or REV and CM, then press RESET key.
	Proper wiring between THR-CM	Rewire as needed.
	Thermal overload relay activated	Reduce load.
	Continuity check between external DB Braking Unit terminal 1-2	Correct ambient temp./lower braking duty cycle.
	Inverter ambient temperature not within permissible limits	Correct to proper temperature range.
LU	AC supply voltage not within permissible limits	Check inverter specs and adjust AC supply voltage.
	Circuit breaker is open	Close circuit breaker.
	Open AC supply phase	Correct input wiring and check AC supply fuses.
	Other than above	Check customer power supply capacity.
OL1	Electronic thermal overload set incorrectly	Check Function Code 43.
	Load is over permissible limits	Reduce load.
Err1	Correct Function Code selection	Verify Function Code.
Err2	Electrical noise source close to inverter	Take noise prevention measures.
Err3	Abnormal ambient temperature	Correct temperature range.
	Daughter board on Control card not seated properly	Check that board is installed properly.
	Keypad Faulty	Try replacement keypad
	Control Card Faulty	Replace Control Card.
Err4	Wiring correct	Confirm wiring and Function Code setting for link operation mode.

**Do not power-up with connection made between FWD or REV and CM when Function Code 15 = 01 (control terminal operation). Also, do not press STOP key on keypad when running from control terminals when Function Code 15 = 01.*

NOTE: *Motor will coast-to-stop when fault condition is displayed. After correcting the cause of the problem, reset by pressing the RESET key on the Keypad Panel or momentarily close contact between the RST-CM control circuit terminals (after the motor has stopped).*

NOTE: *"LU" is displayed when the AC supply is switched ON or OFF and does not indicate any abnormality.*

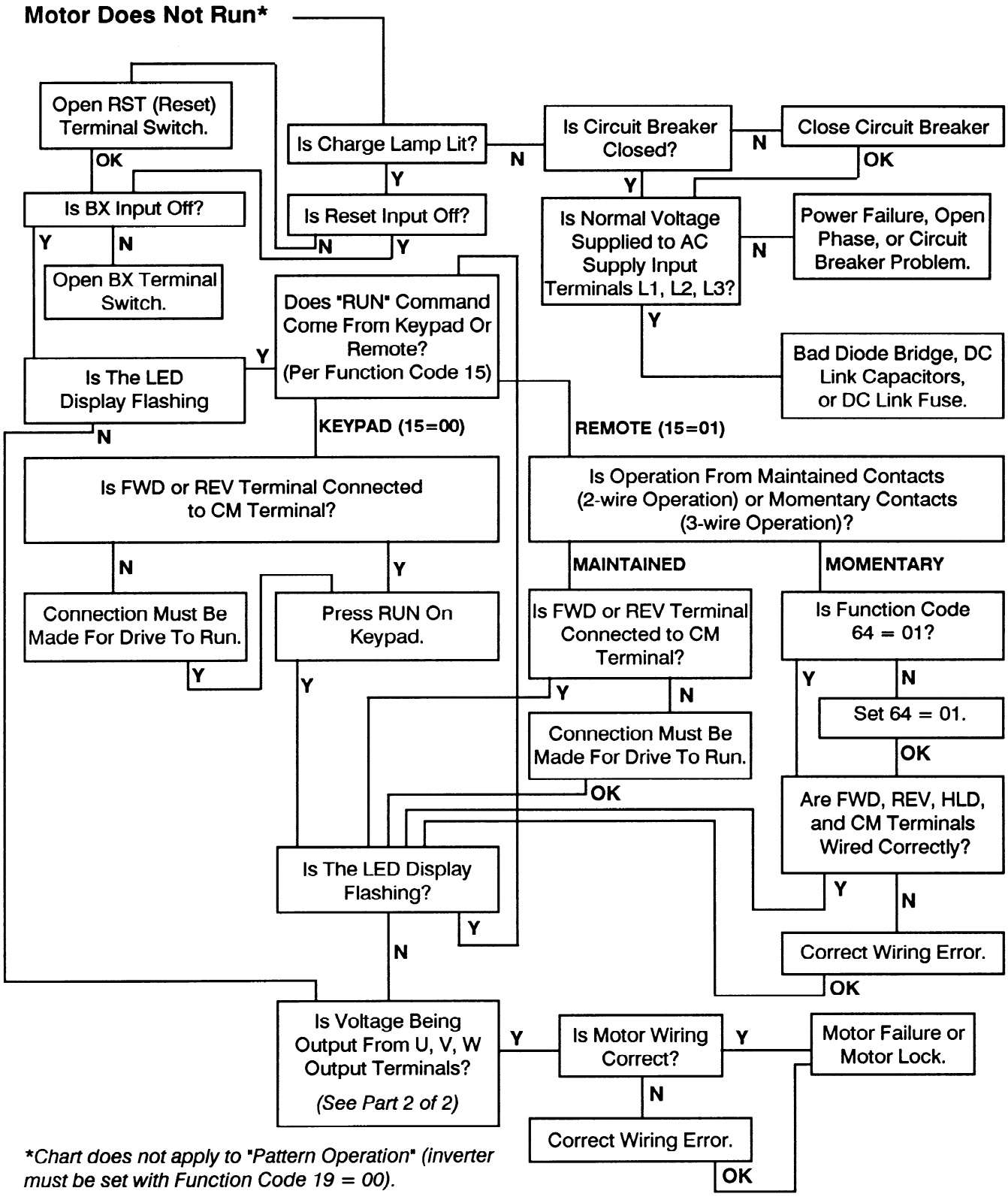


Figure 8-1. "MOTOR DOES NOT RUN" TROUBLESHOOTING CHART (Part 1 of 2)

Motor Does Not Run* (continued)

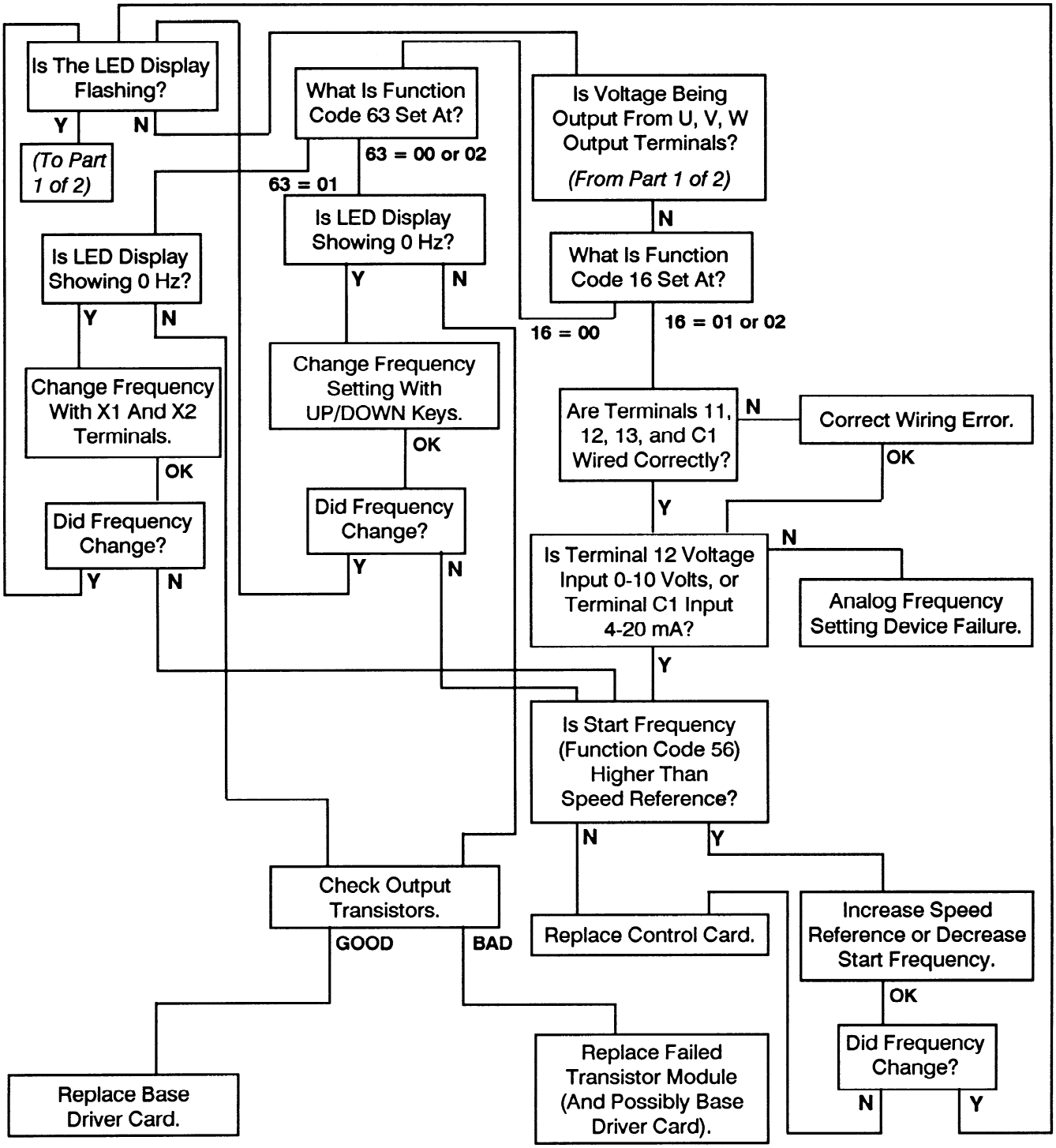


**Chart does not apply to "Pattern Operation" (inverter must be set with Function Code 19 = 00).*

Figure 8-1. "MOTOR DOES NOT RUN" TROUBLESHOOTING CHART (Part 2 of 2)

Motor Runs But Speed Does Not Change

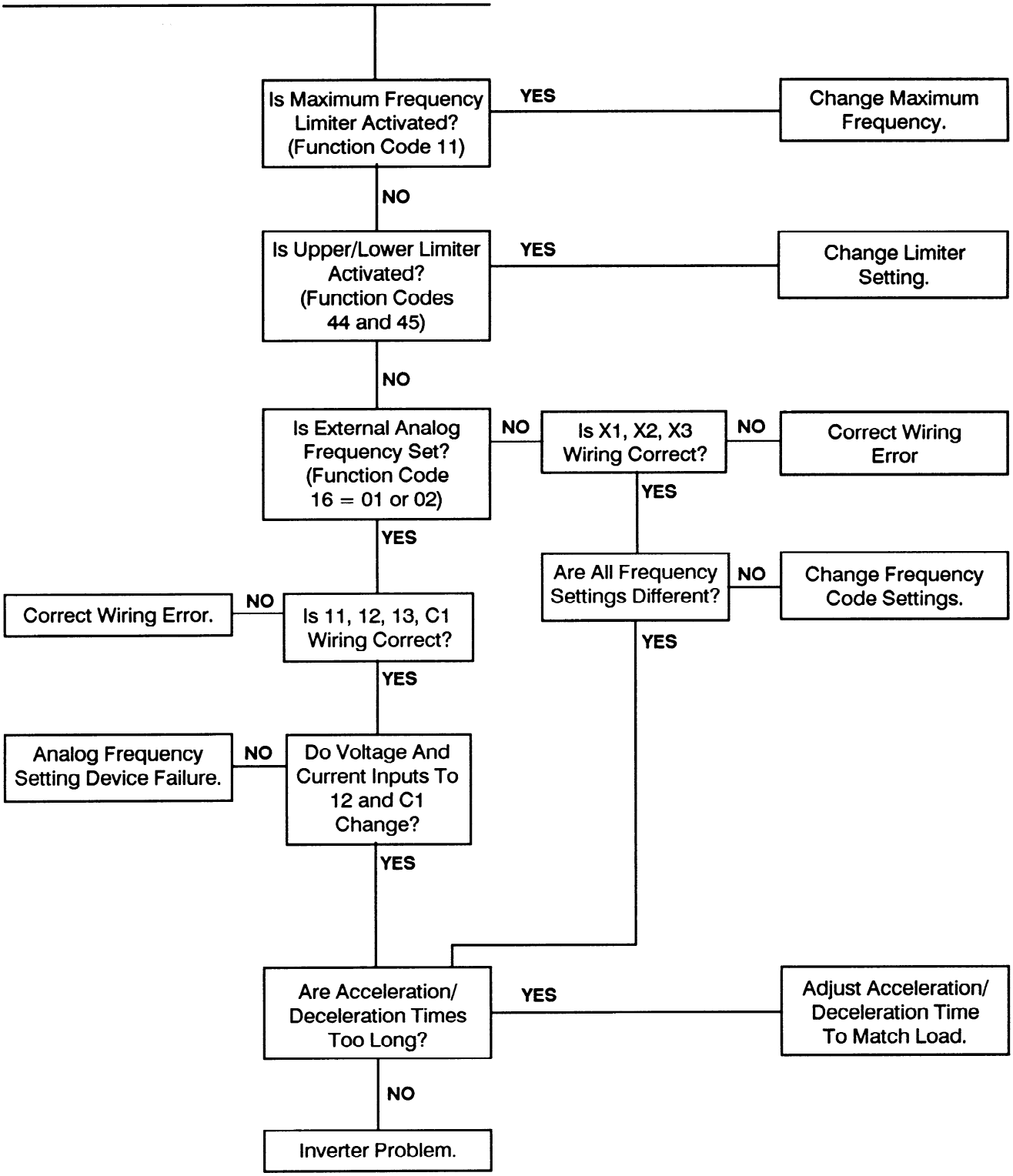


Figure 8-2. "MOTOR RUNS BUT SPEED DOES NOT CHANGE" TROUBLESHOOTING CHART

Motor Runs But Speed Does Not Change

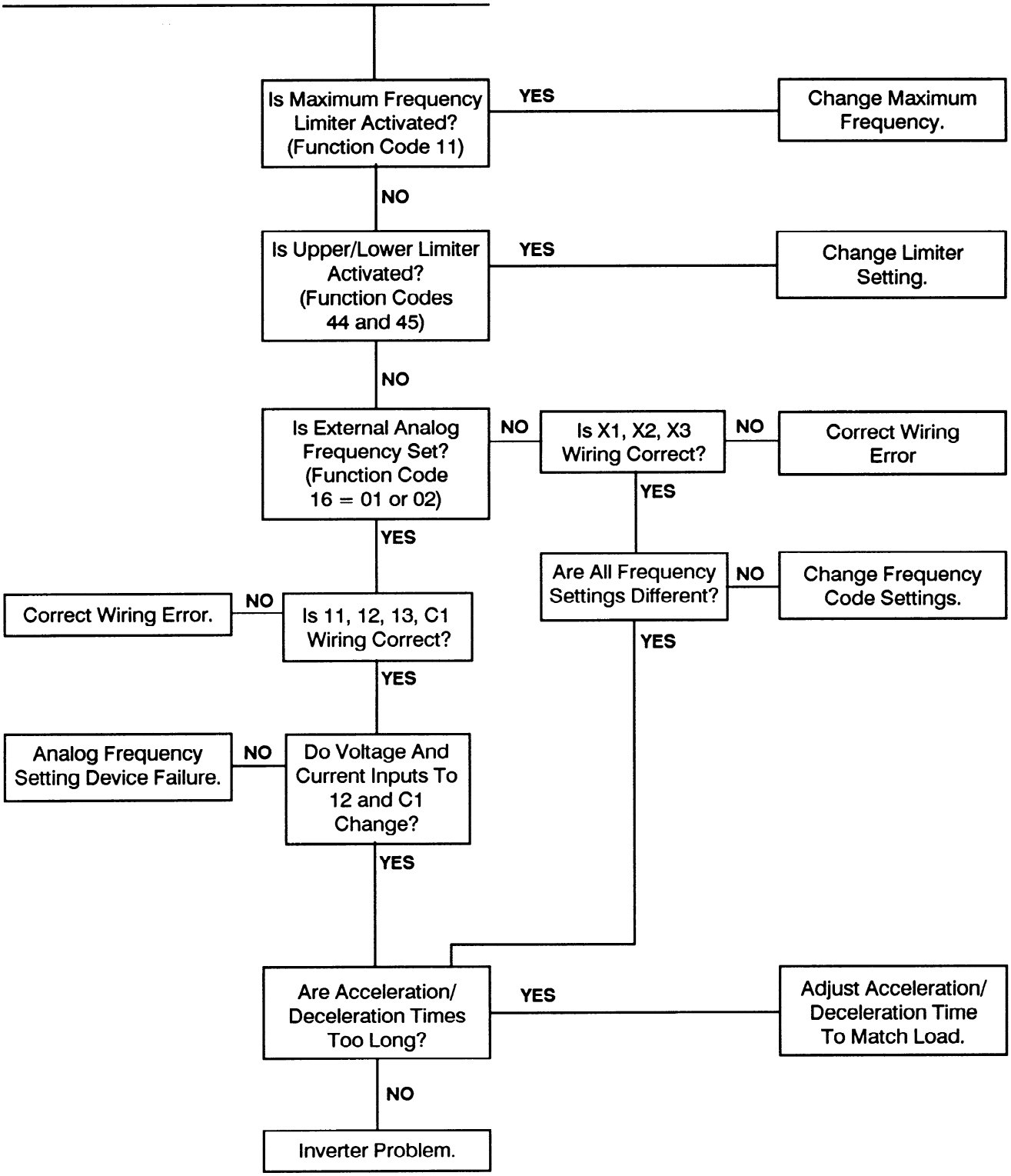


Figure 8-2. "MOTOR RUNS BUT SPEED DOES NOT CHANGE" TROUBLESHOOTING CHART

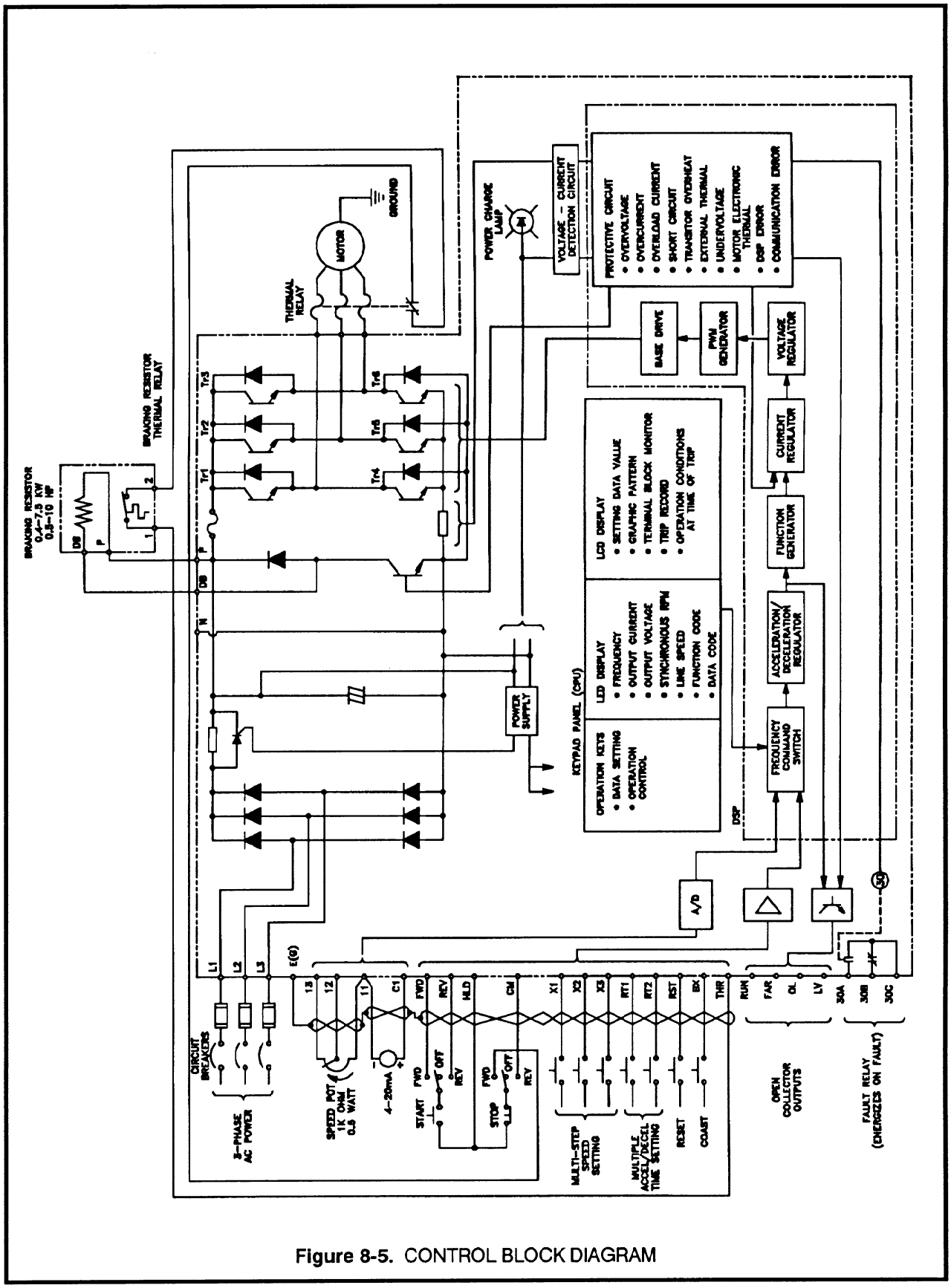


Figure 8-5. CONTROL BLOCK DIAGRAM

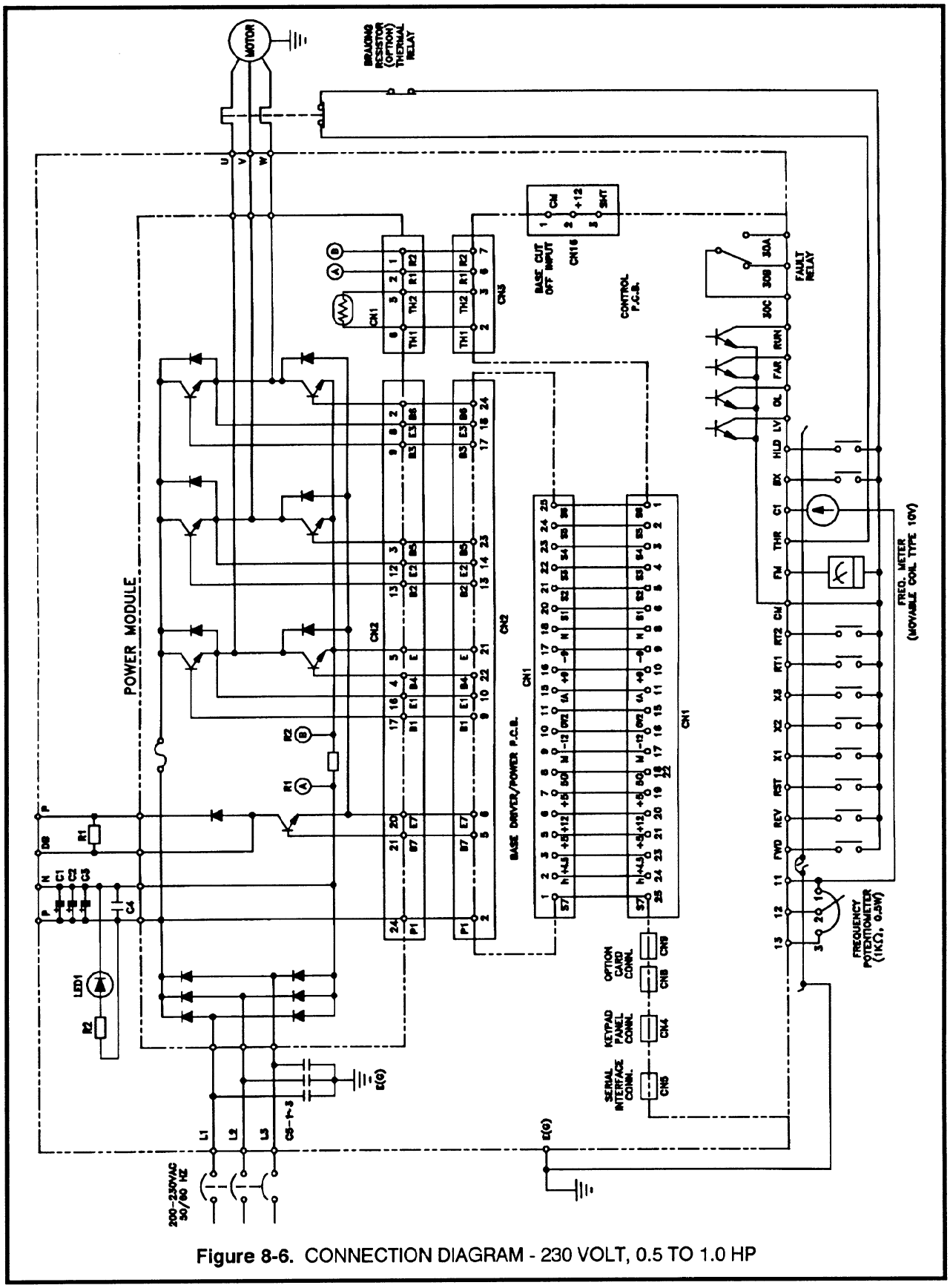


Figure 8-6. CONNECTION DIAGRAM - 230 VOLT, 0.5 TO 1.0 HP

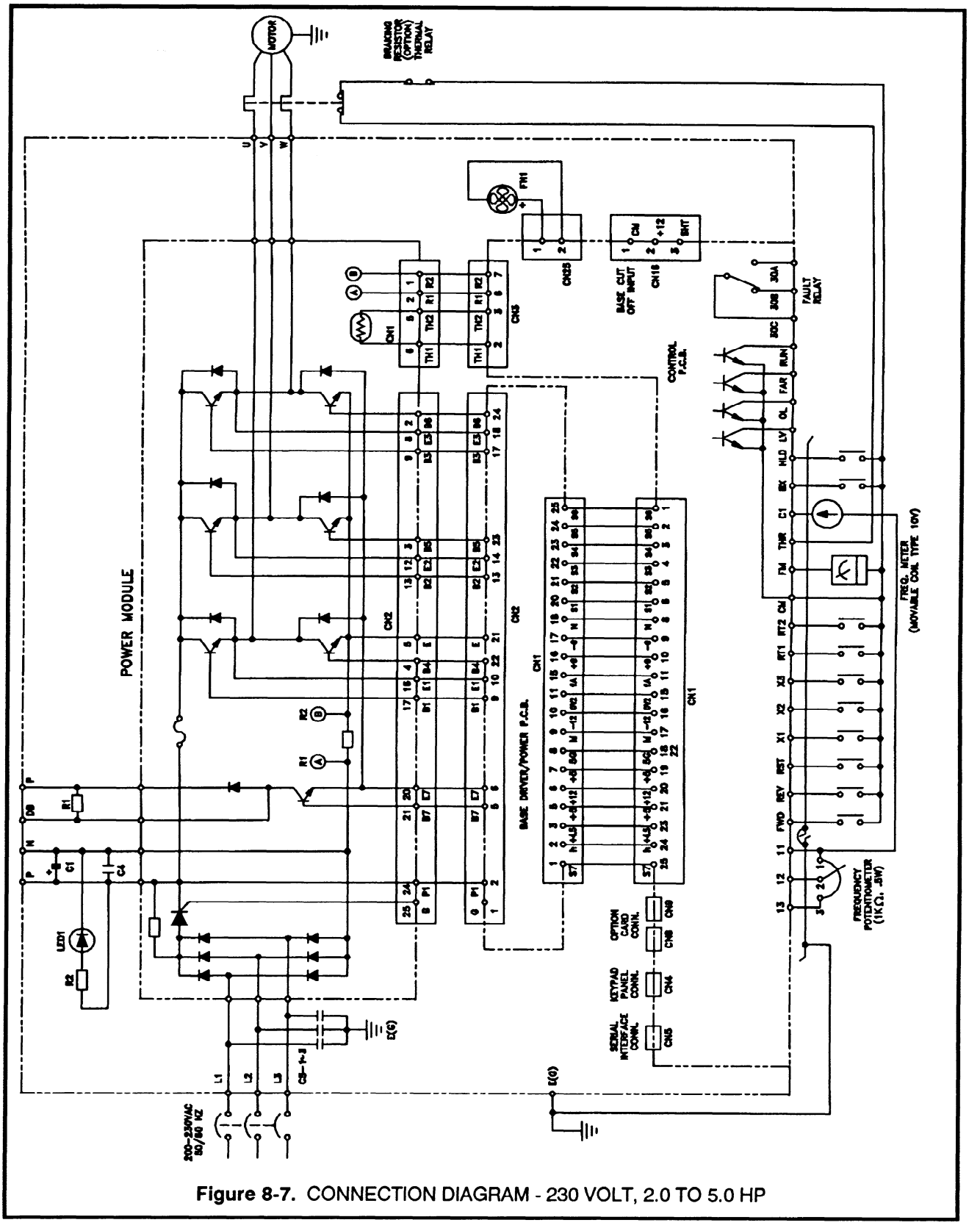


Figure 8-7. CONNECTION DIAGRAM - 230 VOLT, 2.0 TO 5.0 HP

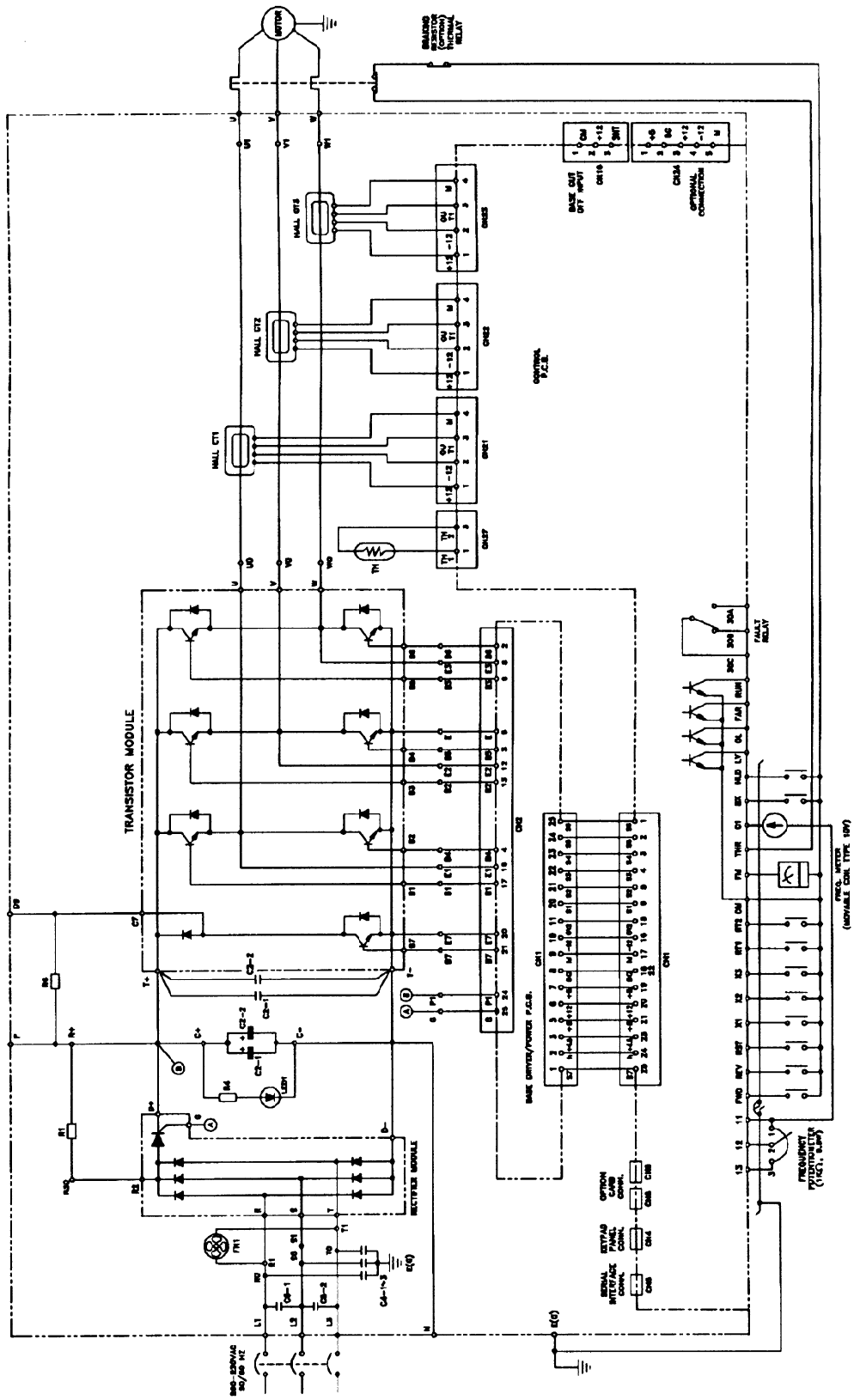


Figure 8-8. CONNECTION DIAGRAM - 230 VOLT, 7.5 TO 10.0 HP

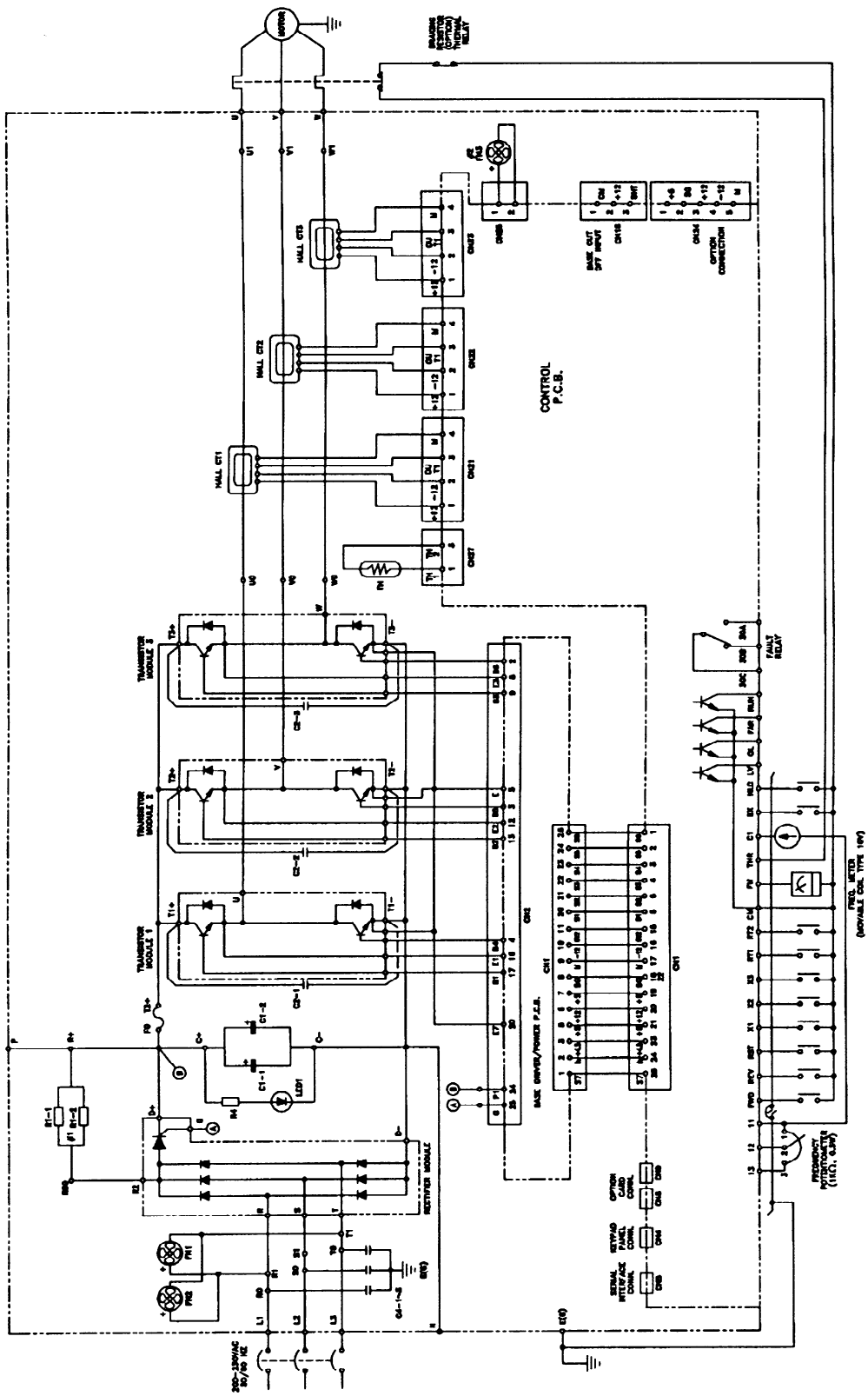


Figure 8-9. CONNECTION DIAGRAM - 230 VOLT, 15 TO 30 HP

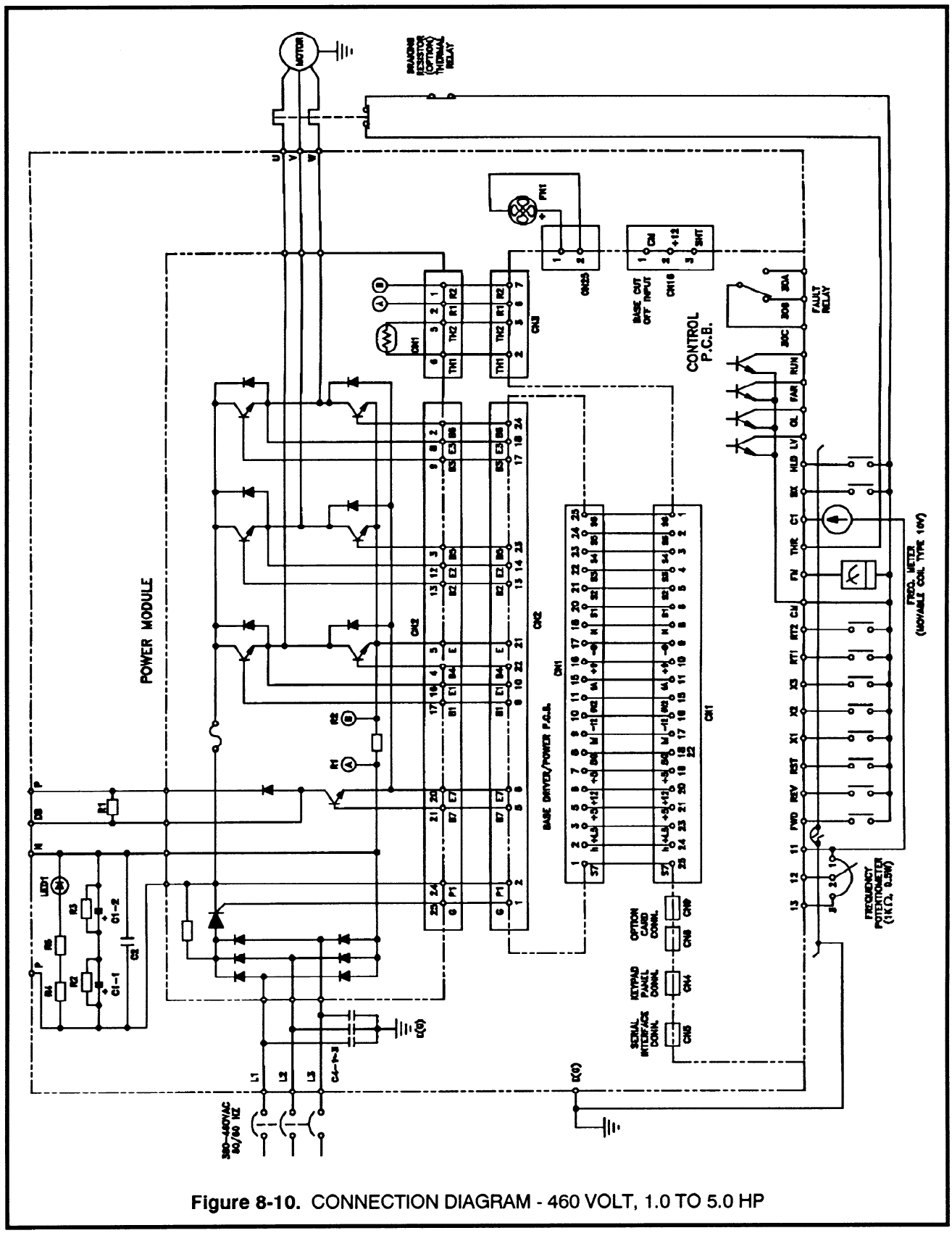


Figure 8-10. CONNECTION DIAGRAM - 460 VOLT, 1.0 TO 5.0 HP

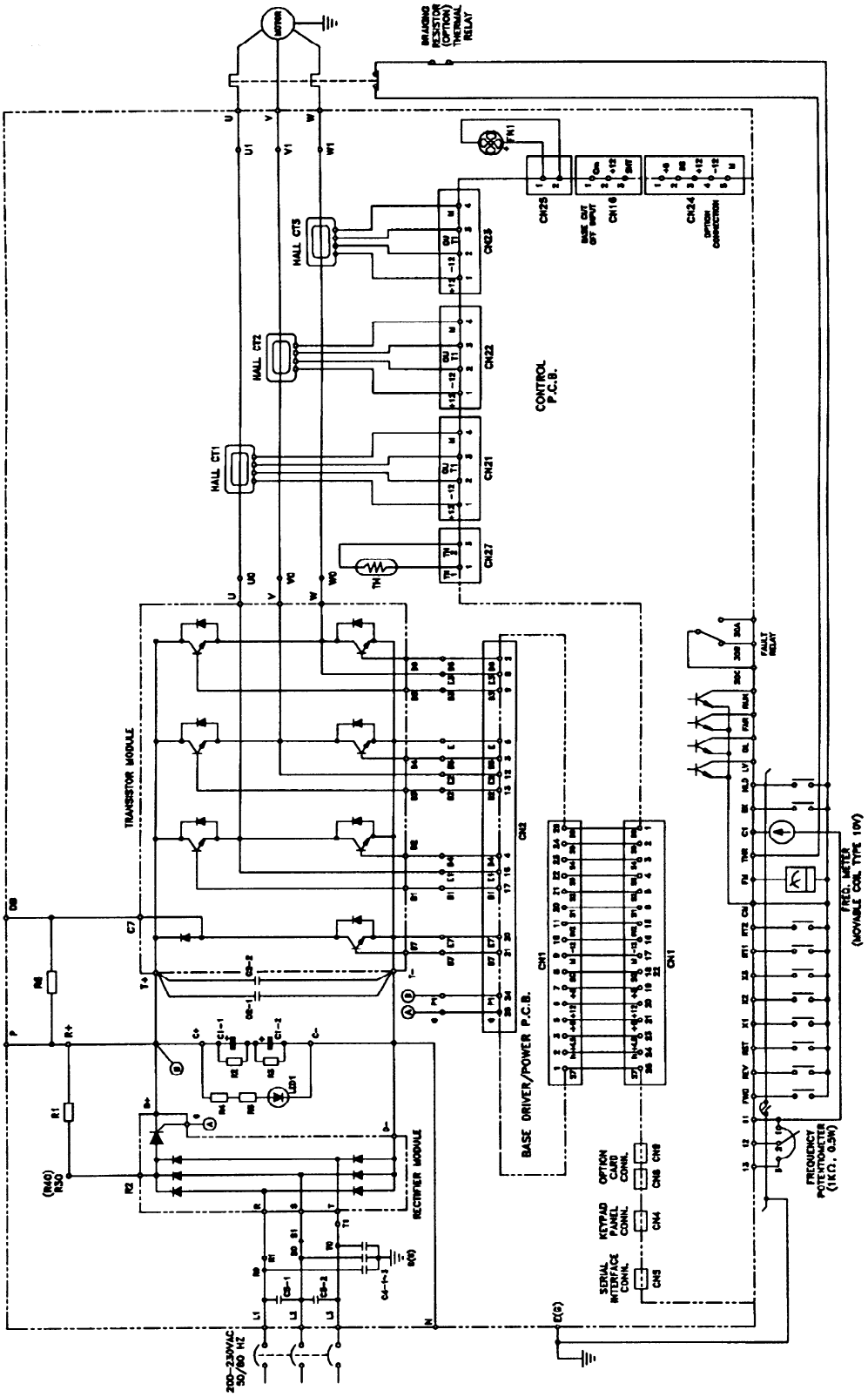


Figure 8-11. CONNECTION DIAGRAM - 460 VOLT, 7.5 TO 10.0 HP

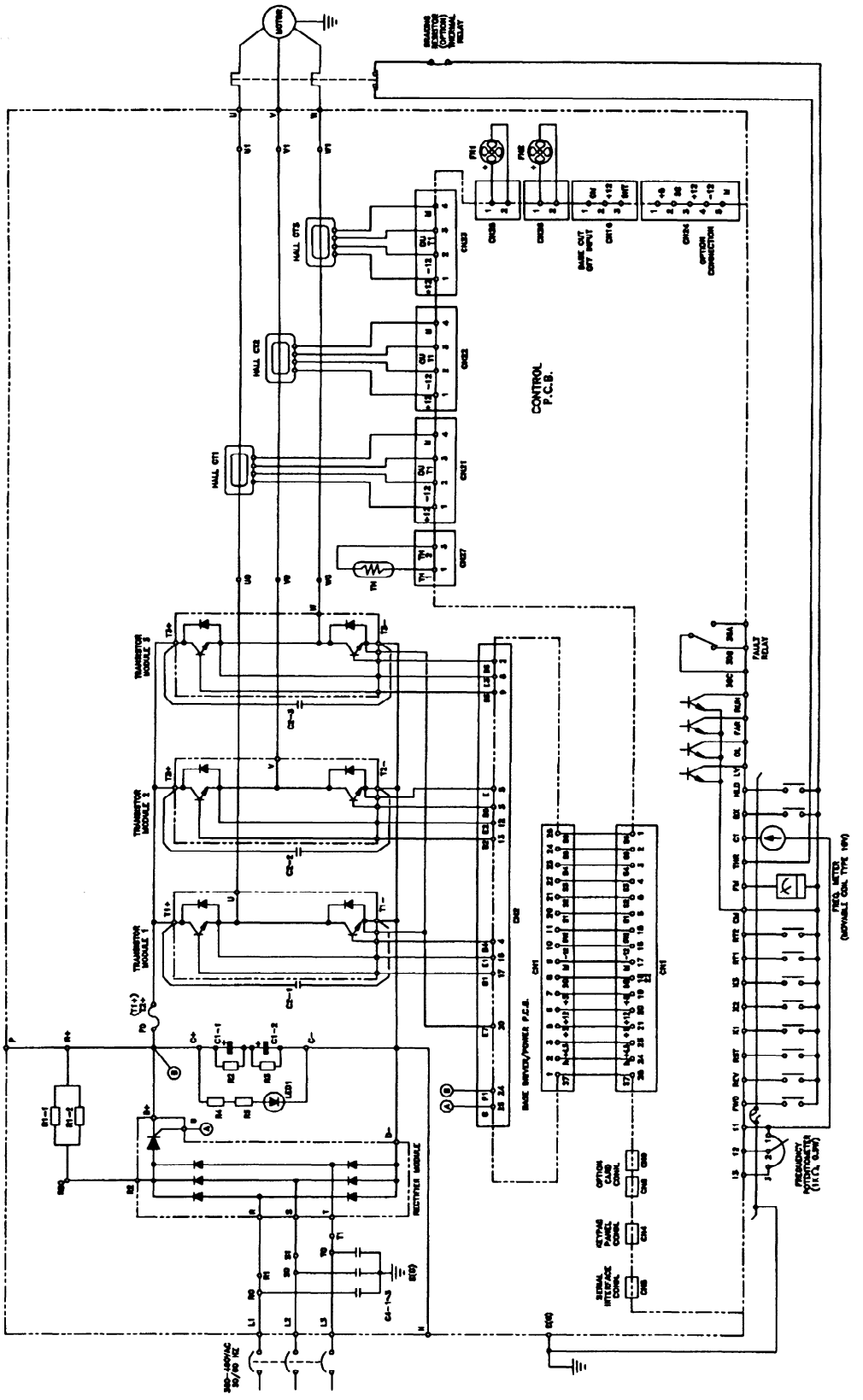


Figure 8-12. CONNECTION DIAGRAM - 460 VOLT, 15 TO 30 HP

NOTES:

9. WARRANTY PARTS AND SERVICE

The purpose of the following is to provide specific instructions to the user of the AF300B inverter regarding Warranty administration and how to obtain assistance on both in-warranty and out-of-warranty equipment.

For all warranty procedures, refer to Section 10 of this Instruction Book to identify the part or assembly.

If assistance is required to either determine warranty status, identify defective parts, or obtain name of local distributor, call:

GE Drive Systems
Salem, VA
(703) 387-7595
(8 AM to 5 PM, Mon. thru Fri.)

WARRANTY COVERAGE

The Warranty covers all major parts of the inverter such as the main printed circuit boards, transistor modules, etc. The warranty does not cover replacement of fuses or of the entire controller.

When the inverter is under warranty, information per the "In-Warranty-Failure Checklist" in this Instruction Book is required.

OUT-OF-WARRANTY PROCEDURES

When the defective part has been identified, contact your local Authorized AF-300B Distributor to order replacement parts.

Before calling the above number to determine warranty status, the inverter serial number will be required. This is located on the inverter nameplate. (If the inverter is still under warranty, further information will be required per the "In-Warranty Failure Checklist".)

MOTORS

Motor repairs on General Electric motors are generally handled by GE Small Motor Service Centers or GE Apparatus Service Shops. For specific instructions on your motor, call GE Drive Products Operation at the phone number listed previously and furnish complete nameplate data.

IN-WARRANTY FAILURE CHECKLIST

To assist with warranty troubleshooting, the following information is required. This data is needed to evaluate the cause in an effort to eliminate any further failures.

Model No.: _____

Serial No.: _____

Start-Up Date: _____

Failure Date: _____

Status When Failure Occurred (check one):

Power-Up _____ Running_____ Accel _____ Decel _____

Application Information (check Yes or No)

Input Transformer: Yes _____ No _____

If Yes: KVA _____

L1 Volts _____ L2 Volts _____ L3 Volts _____

Power Factor Correction Capacitors: Yes _____ No _____

If Yes: Microfarrad _____

IN-WARRANTY FAILURE CHECKLIST (continued)

Application Information (continued) (check Yes or No)

Other Equipment on Same Power Yes _____ No _____

If Yes, what?

Line Reactor on Input Yes _____ No _____

Input Starter Yes _____ No _____

Output Starter Yes _____ No _____

Motor Overloads Yes _____ No _____

Control Terminals Used (circle if used)

LV	OL	FAR	RUN	HOLD	RT1	RT2	THR	X1
X2	X3	30A	30B	30C	RST	BX	FWD	REV
C1	11	12	13					

Function Codes Different From Factory Settings

Function Code	Setting	Function Code	Setting

Failure Message (see Section 5)

Latest Fault _____ **Previous Faults:** **No Message** _____

Hz _____ 1. _____

A _____ 2. _____

V _____ 3. _____

After all of the Checklist information is acquired, contact the following number for assistance:

(703) 387-7595

When returning failed parts, reference the C_ _ _ _ _ _ _ that the replacement parts were received on and ship failed parts to:

GE Drive Systems
1501 Roanoke Boulevard
Salem, VA 24153
(Marked C_ _ _ _ _ _ _)

10. REPLACEMENT PARTS

Refer to the Following Tables when ordering parts per the procedures outlined in Section 9 of this Instruction Book. Tables are arranged per inverter horsepower and voltage (TABLE 13 and TABLE 15: 400-460 VAC Inverters and TABLE 14 and TABLE 16: 200-230 VAC Inverters).

TABLE 13: AF-300B (400-460 VAC) INVERTER RENEWAL PARTS

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter									
			1	2	3	5	7.5	10	15	20	25	30
Power Module	10 Amp, 1200 PIV	SA51714009	1	1								
Power Module	15 Amp, 1200 PIV	SA51714010			1							
Power Module	25 Amp, 1200 PIV	SA51714011				1						
Power Transistor	50 Amp, 1200 PIV	SA51714100					1	1				
Power Transistor	75 Amp, 1200 PIV	SA51714101							3			
Power Transistor	100 Amp, 1200 PIV	SA51714102								3		
Power Transistor	150 Amp, 1200 PIV	SA51714103									3	3
Rectifier Module	40 Amp	SA51714200					1	1				
Rectifier Module	50 Amp	SA51714201							1			
Rectifier Module	75 Amp	SA51714202								1		
Rectifier Module	100 Amp	SA51714203									1	1
Link Capacitor	330 uF, 400 VDC	SA51714300	2									
Link Capacitor	470 uF, 400 VDC	SA51714301		2								
Link Capacitor	680 uF, 400 VDC	SA51714302			2							
Link Capacitor	1000 uF, 400 VDC	SA51714321				2						
Link Capacitor	1500 uF, 400 VDC	SA51714304					2					
Link Capacitor	2200 uF, 400 VDC	SA51714305						2				
Link Capacitor	3300 uF, 400 VDC	SA51714306							2			
Link Capacitor	3900 uF, 400 VDC	SA51714307								2		
Link Capacitor	4700 uF, 400 VDC	SA51714308									2	
Link Capacitor	5400 uF, 400 VDC	SA51714309										2
Fan	24 VDC	SA51714400		1	1	1						
Fan	26 VDC	SA51714401					1	1				
Fan	26 VDC	SA51714402							2	2		
Fan	26 VDC	SA51714403									2	2
Base Driver Card		SA51714507	1	1	1	1						
Base Driver Card		SA51714508					1	1				

TABLE 13: AF-300B (400-460 VAC) INVERTER RENEWAL PARTS (continued)

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter									
			1	2	3	5	7.5	10	15	20	25	30
Base Driver Card		SA51714509							1	1	1	1
Main Control Card	Non UL Listed Inv.	SA51773900	1									
Main Control Card	Non UL Listed Inv.	SA51773901		1								
Main Control Card	Non UL Listed Inv.	SA51773902			1							
Main Control Card	Non UL Listed Inv.	SA51773903				1						
Main Control Card	Non UL Listed Inv.	SA51773904					1					
Main Control Card	Non UL Listed Inv.	SA51773905						1				
Main Control Card	Non UL Listed Inv.	SA51773906							1			
Main Control Card	Non UL Listed Inv.	SA51773907								1		
Main Control Card	Non UL Listed Inv.	SA51773908									1	
Main Control Card	Non UL Listed Inv.	SA51773909										1
Main Control Card	UL Listed Inverter	SA51773921	1									
Main Control Card	UL Listed Inverter	SA51773922		1								
Main Control Card	UL Listed Inverter	SA51773923			1							
Main Control Card	UL Listed Inverter	SA51773924				1						
Main Control Card	UL Listed Inverter	SA51773925					1					
Main Control Card	UL Listed Inverter	SA51773926						1				
Main Control Card	UL Listed Inverter	SA51773927							1			
Main Control Card	UL Listed Inverter	SA51773928								1		
Main Control Card	UL Listed Inverter	SA51773929									1	
Main Control Card	UL Listed Inverter	SA51773930										1
Key Pad	UL Listed Inverter	SA51714803	1	1	1	1	1	1	1	1	1	1
Key Pad	Non UL Listed Inv.	SA511312-20	1	1	1	1	1	1	1	1	1	1
Charge Resistor	30 Watt, 30 Ohm	SA51714700					1					
Charge Resistor	40 Watt, 30 Ohm	SA51714701						1	2	2		
Charge Resistor	40 Watt, 20 Ohm	SA51714702									2	2
DC Link Fuse	50 Amp	SA51714600							1			
DC Link Fuse	75 Amp	SA51714601								1	1	
DC Link Fuse	100 Amp	SA51714602										1
Discharge Resistor Card		SA51791001					1	1	1	1	1	1
Charge LED		SA51790900					1	1	1	1	1	1
AC Line Capacitor		SA51790800					1	1	1	1	1	1

TABLE 13: AF-300B (400-460 VAC) INVERTER RENEWAL PARTS (continued)

Description	Part Number	Inverter Horsepower and Quantity Per Inverter									
		1	2	3	5	7.5	10	15	20	25	30
Power Bus Printed Circuit Card	SA51790707					1	1				
Power Bus Printed Circuit Card	SA51790708							1	1		
Power Bus Printed Circuit Card	SA51790709									1	1
DC Link Filter Capacitor	SA51812500					1	1				
DC Link Filter Capacitor	SA51812501							1	1		
DC Link Filter Capacitor	SA51812502									1	1

NOTES:

TABLE 14: AF-300B (200-230 VAC) INVERTER RENEWAL PARTS

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter										
			0.5	1	2	3	5	7.5	10	15	20	25	30
Power Module	10 Amp	SA51714012	1										
Power Module	15 Amp	SA51714013		1									
Power Module	30 Amp	SA51714014			1	1							
Power Module	50 Amp	SA51714015					1						
Power Module	75 Amp	SA51714104						1					
Power Module	100 Amp	SA51714105							1				
Power Module		SA51714106								1	1		
Power Module		SA51714107										1	
Power Module		SA51714108											1
Link Capacitor	150 uF, 400 VDC	SA51714310	1										
Link Capacitor	220 uF, 400 VDC	SA51714311		1									
Link Capacitor	1000 uF, 400 VDC	SA51714312			1								
Link Capacitor	1500 uF, 400 VDC	SA51714313				1							
Link Capacitor	2200 uF, 400 VDC	SA51714314					1						
Link Capacitor	2700 uF, 400 VDC	SA51714315						1					
Link Capacitor	3900 uF, 400 VDC	SA51714316							1				
Link Capacitor	3300 uF, 400 VDC	SA51714317								1			
Link Capacitor	4700 uF, 400 VDC	SA51714318									1	1	
Link Capacitor	5400 uF, 400 VDC	SA51714309											1
Rectifier Module	75 Amp	SA51714204						1	1				
Rectifier Module		SA51714205								1	1		
Rectifier Module		SA51714206										1	1
Fan	24 VDC	SA51714404			1	1	1						
Fan	220 VAC	SA51714405						1	1				
Fan		SA51714407								2	2	2	2
Fan		SA51714408											1
Base Driver Card		SA51714510	1	1									
Base Driver Card		SA51714511			1	1	1						
Base Driver Card		SA51714512						1	1				
Base Driver Card		SA51714506								1	1	1	1
Keypad	UL Listed Inverters	SA51714803	1	1	1	1	1	1	1	1	1	1	1
Keypad	Non UL Listed Inv.	SA511312-20	1	1	1	1	1	1	1	1	1	1	1
Charge Resistor	30 Watt, 6.8 Ohm	SA51714703						1	1				
Charge Resistor		SA51714704								1	1	1	1
Charge LED		SA5179091						1	1	1	1	1	1

TABLE 14: AF-300B (200-230 VAC) INVERTER RENEWAL PARTS (cont.)

Description	Rating	Part Number	Inverter Horsepower and Quantity Per Inverter										
			0.5	1	2	3	5	7.5	10	15	20	25	30
Main Control Card	Non UL Listed Inv.	SA51003910	1										
Main Control Card	Non UL Listed Inv.	SA51003911		1									
Main Control Card	Non UL Listed Inv.	SA51003912			1								
Main Control Card	Non UL Listed Inv.	SA51003913				1							
Main Control Card	Non UL Listed Inv.	SA51003914					1						
Main Control Card	Non UL Listed Inv.	SA51003915						1					
Main Control Card	Non UL Listed Inv.	SA51003916							1				
Main Control Card	Non UL Listed Inv.*	SA51003917								1			
Main Control Card	Non UL Listed Inv.*	SA51003918									1		
Main Control Card	Non UL Listed Inv.*	SA51003919										1	
Main Control Card	Non UL Listed Inv.*	SA51003920											1
Main Control Card	UL Listed Inverter	SA51773931	1										
Main Control Card	UL Listed Inverter	SA51773932		1									
Main Control Card	UL Listed Inverter	SA51773933			1								
Main Control Card	UL Listed Inverter	SA51773934				1							
Main Control Card	UL Listed Inverter	SA51773935					1						
Main Control Card	UL Listed Inverter	SA51773936						1					
Main Control Card	UL Listed Inverter	SA51773937							1				
Main Control Card	UL Listed Inverter**	SA51773938								1			
Main Control Card	UL Listed Inverter**	SA51773939									1		
Main Control Card	UL Listed Inverter**	SA51773940										1	
Main Control Card	UL Listed Inverter**	SA51773941											1
AC Line Capacitor		SA51790800						1	1	1	1	1	1
Discharge Resistor Card		SA51791002								1	1	1	1
DC Link Filter Capacitor		SA51812503						1					
DC Link Filter Capacitor		SA51812504							1				
DC Link Filter Capacitor		SA51812505								1	1		
DC Link Filter Capacitor		SA51812506										1	1
Power Bus Printed Circuit Card		SA51790710						1					
Power Bus Printed Circuit Card		SA51790711							1				
Power Bus Printed Circuit Card		SA51790705								1	1		
Power Bus Printed Circuit Card		SA51790706										1	1

**Enclosed date of 10/92 and later.*

***Enclosed date of 9/92 and earlier.*

TABLE 15: AF-300B (400-460 VAC) AC LINE FUSES (FRS Type) AND FUSEHOLDERS*

Description	Rating In Amps	Part Number	Inverter Horsepower and Quantity Per Inverter									
			1	2	3	5	7.5	10	15	20	25	30
Fuse, 600 Volt	2.8	104X109BB028	3									
Fuse, 600 Volt	6.0	104X109BB030		3								
Fuse, 600 Volt	8.0	104X109BB007			3							
Fuse, 600 Volt	15	104X109BB009				3						
Fuse, 600 Volt	20	104X109BB010					3					
Fuse, 600 Volt	25	104X109BB011						3				
Fuse, 600 Volt	35	104X109BB013							3			
Fuse, 600 Volt	50	104X109BB015								3		
Fuse, 600 Volt	60	104X109BB016									3	
Fuse, 600 Volt	70	104X109BB017										3
3 Pole Fuseholder		104X111CA016	1	1	1	1	1	1				
3 Pole Fuseholder		104X111CA019							1	1	1	
3 Pole Fuseholder		104X111CA022										1

TABLE 16: AF-300B (200-230 VAC) AC LINE FUSES (FRN Type) AND FUSEHOLDERS*

Description	Rating In Amps	Part Number	Inverter Horsepower and Quantity Per Inverter										
			0.5	1	2	3	5	7.5	10	15	20	25	30
Fuse, 250 Volt	8.0	104X109BA006	3	3									
Fuse, 250 Volt	15	104X109BA008			3								
Fuse, 250 Volt	20	104X109BA009				3							
Fuse, 250 Volt	25	104X109BA010					3						
Fuse, 250 Volt	35	104X109BA011						3					
Fuse, 250 Volt	45	104X109BA013							3				
Fuse, 250 Volt	80	104X109BA015								3			
Fuse, 250 Volt	100	104X109BA016									3		
Fuse, 250 Volt	125	104X109BA017										3	3
3 Pole Fuseholder		104X111CA003	1	1	1	1	1						
3 Pole Fuseholder		104X111CA006						1	1				
3 Pole Fuseholder		104X111CA009								1	1		
3 Pole Fuseholder		104X111CA011										1	1

**Fuses and fuseholders are to be mounted external to the inverter.*

FUNCTION CODE SETTINGS: Factory and Field

Func. Code	Function	Display, Setting, Range	Min. Unit	Factory Setting	Field Setting
00	LED Digital Monitor Selection	00: Output Frequency (Hz) 01: Output Current (Amps) 02: Output Voltage (Volts) 03: Synchronous rpm (rpm) 04: Line Speed (m/min.)	---	00	
01	Graphics Monitor Selection	00: Hz Amp Monitor 01: Terminal Signal 1 02: Terminal Signal 2	---	00	
02	Motor Noise Reduction	00 to 05 (code)	---	03	
03	FM Terminal Output Level Calibration	00 to 99 (code, 6.5 to 10.5 VDC)	---	85	
04	Automatic Torque Boost Control	00: Non-Operational 01: Operational	---	00	
05	Torque Boost	00 to 31 (code)	---	02*	
06	Torque Boost Fine Adjustment	00 to 09 (code)	---	00	
07	Automatic Accel/Decel Control	00: Non-Operational 01: Operational	---	00	
08	Acceleration Time 1	(LCD) 0.01 to 3600 seconds	0.01	6/12**	
09	Deceleration Time 1	(LCD) 0.01 to 3600 seconds	0.01	6/12**	
10	Data Protection	00: Not Protected (changes) 01: Protected (no changes)	---	00***	
11	Maximum Frequency	00: 50 Hertz 01: 60 Hertz 02: 100 Hertz 03: 120 Hertz 04: Adjustable (0.2 - 400 Hertz)	1 Hz	01	
12	Base Frequency	00: 50 Hertz 01: 60 Hertz 02: Adjustable (0.2 - 400 Hertz)	1 Hz	01	
13	Rated Output Voltage	00: 200 (400) VAC 01: 220 (440) VAC 02: 230 (460) VAC 03: Adjustable (0 - 230, 460 VAC)	1V (2V)	02	
14	Number of Motor Poles	02:, 04:, 06:, 08:, 10:, 12:	---	04	
15	Operation Command	00: Keypad Panel Operation 01: Terminal Block Operation 02: Link Adaptor Operation	---	00	
16	Frequency Command	00: Digital (Keypad UP/DOWN key) 01: Analog (voltage, 0 - 10 VDC) 02: Analog (voltage & current, 0 - 10 VDC plus 4 - 20 mA)	---	00	
17	Accel/Decel Pattern	00: Linear 01: Weak "S" Curve 02: Strong "S" Curve	---	00	
18	Normal/High Torque Dynamic Brakes	00: Normal Brake 01: High Torque Brake	---	00	
19	Pattern Operation	00: Normal Operation 01: Pattern Operation	---	00	

**Factory setting is 13 for inverters of 3.7 KW (5 HP) or less and 8 for inverters of 5.5 KW (7.5 HP) or greater; it is recommended to change settings to 02 for all AF-300B inverters (see Section 6 for further explanation).*

***Factory setting for Accel Time 1 and Decel Time 1 is 6.0 for inverters of 7.5 KW (10 HP) or less; setting is 12.0 for inverters of 11 KW (15 HP) or greater.*

****Press and hold STOP, then use UP/DOWN arrows to change setting; press SET to store change and release STOP.*

Func. Code	Function	Display, Setting, Range	Min. Unit	Factory Setting	Field Setting
20	Restart After Instantaneous Power Failure	00: Non-Operational 01: Operational	—	00	
21	Coefficient for Line Speed	(LCD) 0.00 - 200	0.0	0.01	
22	Function Code Blocks Accessible	00: Basic Functions (00 - 22) 01: Basic & Standard Functions (00 - 62) 02: Basic, Standard, & High Level Functions (00 - 79) See Fig. 5-3	—	00	
23	Acceleration Time 2	(LCD) 0.01 - 3600 seconds	0.01	10.0	
24	Acceleration Time 3	(LCD) 0.01 - 3600 seconds	0.01	15.0	
25	Acceleration Time 4	(LCD) 0.01 - 3600 seconds	0.01	3.0	
26	Deceleration Time 2	(LCD) 0.01 - 3600 seconds	0.01	10.0	
27	Deceleration Time 3	(LCD) 0.01 - 3600 seconds	0.01	15.0	
28	Deceleration Time 4	(LCD) 0.01 - 3600 seconds	0.01	3.0	
29	Multistep Speed Setting 1	(LCD) 0.00 - 400 Hertz	0.002	0.0	
30	Timer 1	(LCD) 0.01 - 3600 seconds	0.01	5.0	
31	Multistep Speed Setting 2	(LCD) 0.01 - 400 Hertz	0.002	0.0	
32	Timer 2	(LCD) 0.01 - 3600 seconds	0.01	5.0	
33	Multistep Speed Setting 3	(LCD) 0.01 - 400 Hertz	0.002	0.0	
34	Timer 3	(LCD) 0.01 - 3600 seconds	0.01	5.0	
35	Multistep Speed Setting 4	(LCD) 0.01 - 400 Hertz	0.002	0.0	
36	Timer 4	(LCD) 0.01 - 3600 seconds	0.01	5.0	
37	Multistep Speed Setting 5	(LCD) 0.01 - 400 Hertz	0.002	0.0	
38	Timer 5	(LCD) 0.01 - 3600 seconds	0.01	5.0	
39	Multistep Speed Setting 6	(LCD) 0.01 - 400 Hertz	0.002	0.0	
40	Timer 6	(LCD) 0.01 - 3600 seconds	0.01	5.0	
41	Multistep Speed Setting 7	(LCD) 0.01 - 400 Hertz	0.002	0.0	
42	Timer 7	(LCD) 0.01 - 3600 seconds	0.01	5.0	
43	Electronic Thermal Overload Relay	00: Non-Operational 01: Operational (%)	1%	01 (LCD) 90%	
44	High Limiter	(LCD) 0 - 100 %	1%	100	
45	Low Limiter	(LCD) 0 - 100 %	1%	0	
46	Bias Frequency*	(LCD) 0 - 100 %	1%	0	
47	Frequency Setting Signal Gain*	(LCD) 0 - 200 %	1%	100	
48	Jump Frequency 1** (Freq. Rejection)	(LCD) 0 - 400 Hertz	1 Hz	0	
49	Jump Frequency 2** (Freq. Rejection)	(LCD) 0 - 400 Hertz	1 Hz	0	
50	Jump Frequency 3** (Freq. Rejection)	(LCD) 0 - 400 Hertz	1 Hz	0	
51	Jump Frequency Width	(LCD) 0 - 5 Hertz	1 Hz	0	
52	DC Brake	00: Non-Operational 01: Operational	—	00	
53	DC Brake Starting Frequency	(LCD) 0 - 60 Hertz	1 Hz	0	
54	DC Brake Voltage	0 to 15 (code)	1%	00	
55	DC Braking Time	(LCD) 0.01 - 30 seconds	0.01	0.10	
56	Starting Frequency	(LCD) 0 (0.2) - 60 Hertz	1 Hz	1	

Frequency bias and frequency gain are operational only when the frequency setting method is analog (Function Code 16 = 01 or 02). See **WARNING in Section 1.*

***Function Code 51 sets width.*

Func. Code	Function	Display, Setting, Range	Min. Unit	Factory Setting	Field Setting
57	Current Limiter	00: Non-Operational 01: Operational (30 - 150%)	1%	01 (LCD) 150%	
58	Slip Compensation Control	00: Non-Operational 01: Operational	---	00*	
59	Frequency Level Detection	(LCD) 0 - 400 Hertz	1 Hz	60	
60	FDT and FAR Signal Hysterisis	(LCD) 0 - 30 Hertz	1 Hz	10	
61	Run Signal Finishing Frequency	(LCD) 0 - 400 Hertz	1 Hz	0	
62	Overload Early Warning Signal	(LCD) 70 - 150 %	1%	100	
63	X1, X2, and X3 Terminal Function	00: Multi-Step Frequency Op. 01: INC/DEC Frequency Control 02: DC Brake Operation	---	00	
64	FWD/REV Command Hold	00: 2-Wire Operation 01: 3-Wire Operation 02: Switches Timer Oper. Mode	---	01	
65	LV, OL, and FAR Terminal Output Code	00: Output per FC 66, 67, 68 01: Binary speed step signal	---	00	
66	LV Terminal Function (Signal Output)	00: Undervoltage 01: Overvoltage	---	00	
67	OL Terminal Function (Signal Output)	00: Overload Early Warning 01: Current Limit Monitoring 02: Undervoltage	---	00	
68	FAR Terminal Function (Signal Output)	00: Freq. Equivalence Detection 01: Setting Frequency Detection 02: Not Running Signal	---	00	
69	RUN Terminal Function (Signal Output)	00: Run signal 01: Pattern Step Complete Signal 02: Pattern Complete Signal	---	00	
70	FM Terminal Function (Signal Output)	00: Frequency Meter 01: Output Current Meter	---	00	
71	Inverter Number Entry For Link Operation	00: For Master Record 01: Slave #1 (Auxiliary #1) *Thru* 15: Slave #15 (Auxiliary #15)	---	15	
72	Number of Inverters Linked	00: Total Number of Inverters *Thru* 15: Connected (master and slave)		00	
73	Link Mode Selection	00: Link Non-Operational 01: Frequency Setting 02: Monitoring Signal 03: Link Operational	---	00	
74	RUN Command Input In Link Operation	00: RUN/STOP via Keypad Panel 01: RUN/STOP via Control Terms.	---	00	
75	Slave Inverter Number Designation (For changing Function Codes in slave from master keypad)	00: Inverter # to be changed *Thru* 15: 16: Selects All Inverters	---	00	
76	Sets Gain On Automatic Current Regulation	00: Based On Acel/Decel Rate 01: Quickest Response *Thru* 99: Slowest Response	1	00	
77-79	Optional - For Future Use				
80	Number of Bits for Frequency Setting NOTE: *Function Codes 80-82 are on Ser. #29 and prior inverters only.*	00: 4-Bit Binary Input 01: 8-Bit Binary Input 02: 12-Bit Binary Input	---	00	
81	Monitoring Signal Output Code In Link Operation	00: Set Output Data via Function Code for Each Inverter	---	00	
82	Monitoring Mode In Link Operation	00: Operation Command Mode 01: Function Code Setting Mode	---	00	

**Factory recommendation is non-operational for most applications due to interference with torque boost operation*

NOTES:



GE Drive Systems

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